31 Suppose that a curve C is the graph of a polar equation $r = f(\theta)$. If $r' = dr/d\theta$ and $r'' = d^2r/d\theta^2$, show that the curvature K at $P(r, \theta)$ is given by

$$K = \frac{|2(r')^2 - rr'' + r^2|}{[(r')^2 + r^2]^{3/2}}.$$

(*Hint*: Use $x = r\cos\theta$ and $y = r\sin\theta$ to express C in parametric form).

In Exercises 32–34 find the curvature of the polar curve at $P(r, \theta)$ by means of the formula in Exercise 31.

32 The cardioid $r = a(1 - \cos\theta)$, where $0 < \theta < 2\pi$

33 The four-leafed rose $r = \sin 2\theta$, where $0 < \theta < \pi/2$

34 The spiral $r = e^{a\theta}$

35 Let $P(x, y)$ be a point on the graph of $y = f(x)$ at which $K \neq 0$. If (h, k) is the center of curvature for P show that

$$h = x - \frac{y'[1 + (y')^2]}{y''}, \quad k = y + \frac{[1 + (y')^2]}{y''}.$$

36–40 Use the formulas from Exercise 35 to find the center of curvature for P in Exercises 1–5.

41 A highway that coincides with the x-axis has an exit ramp at the origin O that follows along the curve $y = -x^3/27$ from O to the point $P(3, -1)$ and then along the arc of a circle as shown in the figure. Find the center of the circular arc that makes the curvature at $P(3, -1)$ continuous.

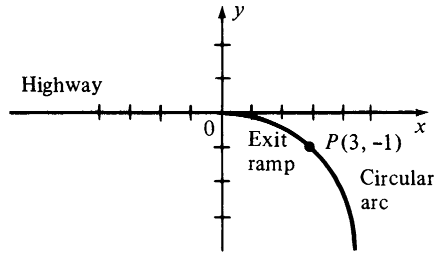


FIGURE FOR EXERCISE 41

42 Prove that straight lines and circles are the only *plane* curves that have a constant curvature. (Exercise 25 of Section 15.5 shows that this result is not true for space curves.)

43 Prove that if k is a nonnegative function that is nonzero and continuous on an interval $[0, a]$, then there is a plane curve C such that k represents the curvature of C as a function of arc length. (*Hint*: If s is in $[0, a]$, define

$$h(s) = \int_0^s k(t)\,dt,\ x = f(s) = \int_0^s \cos h(t)\,dt,$$

$$\text{and} \quad y = g(s) = \int_0^s \sin h(t)\,dt.)$$

44 Use Exercise 43 to rework Exercise 42.

15.5 Tangential and Normal Components of Acceleration

The results of the previous section may be applied to the motion of a particle. Let us suppose that at time t the particle is at the point $P(x, y, z)$ on a curve C, and that C is given parametrically by

$$x = f(t), \quad y = g(t), \quad z = h(t)$$

where f'', g'', and h'' exist. As usual, let $\mathbf{r}(t)$ represent the position vector $\overrightarrow{OP}$ and let s denote arc length measured along C. We shall also assume that s increases as t increases. The unit tangent vector $\mathbf{T}(s)$ introduced in the previous section may then be expressed as

$$\mathbf{T}(s) = \frac{1}{|\mathbf{r}'(t)|}\mathbf{r}'(t)$$

and hence

$$\mathbf{r}'(t) = |\mathbf{r}'(t)|\mathbf{T}(s) = \frac{ds}{dt}\mathbf{T}(s).$$

Differentiating with respect to t and using Exercises 39 and 40 of Section 15.2 gives us

$$\begin{aligned}\mathbf{r}''(t) &= \frac{d^2s}{dt^2}\,\mathbf{T}(s) + \frac{ds}{dt}\frac{d}{dt}\,\mathbf{T}(s)\\ &= \frac{d^2s}{dt^2}\,\mathbf{T}(s) + \frac{ds}{dt}\frac{ds}{dt}\,\mathbf{T}'(s).\end{aligned}$$

From the remark following Definition (15.14), we may write $\mathbf{T}'(s) = K\mathbf{N}(s)$ where K is the curvature of C and $\mathbf{N}(s)$ is the unit normal. Consequently,

$$\mathbf{r}''(t) = \frac{d^2s}{dt^2}\,\mathbf{T}(s) + \left(\frac{ds}{dt}\right)^2 K\mathbf{N}(s).$$

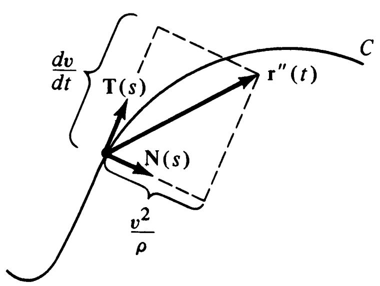


FIGURE 15.21

If we denote the speed ds/dt by v and write $K = 1/\rho$, where ρ is the radius of curvature of C, then the last formula for $\mathbf{r}''(t)$ may be written

$$\mathbf{r}''(t) = \frac{dv}{dt}\,\mathbf{T}(s) + \frac{v^2}{\rho}\,\mathbf{N}(s).$$

This formula expresses the acceleration $\mathbf{r}''(t)$ in terms of a **tangential component** dv/dt (the rate of change of speed with respect to time) and a **normal component** v^2/ρ. The sketch in Figure 15.21 illustrates one possibility that could occur.

The next theorem summarizes the preceding discussion.

Theorem (15.15)

Suppose the position of a point P on a curve C is given by

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

where t represents time. If the speed of P is $v = ds/dt$, then the acceleration of P is

$$\mathbf{r}''(t) = \frac{dv}{dt}\,\mathbf{T}(s) + \frac{v^2}{\rho}\,\mathbf{N}(s).$$

where ρ is the radius of curvature of C.

Note that the normal component v^2/ρ depends only on the speed and the curvature (or radius of curvature) of the curve C. If the speed or curvature is large (or the radius of curvature is small), then the normal component of acceleration is large. This result, obtained theoretically, proves the well-known fact that an automobile driver should slow down when attempting to negotiate a sharp turn.

Let us next find formulas for the tangential and normal components of acceleration which depend only on $\mathbf{r}(t)$. First, recall from the discussion at beginning of this section, that

$$\mathbf{r}'(t) = v\mathbf{T}(s).$$

Taking the dot product with $\mathbf{r}''(t)$ as given in Theorem (15.15) we obtain

$$\mathbf{r}'(t) \cdot \mathbf{r}''(t) = \frac{dv}{dt} v\mathbf{T}(s) \cdot \mathbf{T}(s) + \frac{v^3}{\rho} \mathbf{T}(s) \cdot \mathbf{N}(s).$$

Since $\mathbf{T}(s)$ has magnitude 1 and is orthogonal to $\mathbf{N}(s)$, the last equality reduces to

$$\mathbf{r}'(t) \cdot \mathbf{r}''(t) = \frac{dv}{dt} v = \frac{dv}{dt} |\mathbf{r}'(t)|.$$

If $|\mathbf{r}'(t)| \neq 0$, we obtain the following formula for the **tangential component of acceleration**:

(15.16)
$$\frac{dv}{dt} = \frac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{|\mathbf{r}'(t)|}$$

In like manner, if we take the cross product of $\mathbf{r}'(t)$ with $\mathbf{r}''(t)$, then

$$\begin{aligned}\mathbf{r}'(t) \times \mathbf{r}''(t) &= v\frac{dv}{dt} \mathbf{T}(s) \times \mathbf{T}(s) + \frac{v^3}{\rho} \mathbf{T}(s) \times \mathbf{N}(s) \\ &= \frac{v^3}{\rho} \mathbf{T}(s) \times \mathbf{N}(s).\end{aligned}$$

Since $\mathbf{T}(s)$ and $\mathbf{N}(s)$ are orthogonal unit vectors, $|\mathbf{T}(s) \times \mathbf{N}(s)| = 1$ and therefore

$$|\mathbf{r}'(t) \times \mathbf{r}''(t)| = \frac{v^3}{\rho}.$$

This gives us the formula for the **normal component of acceleration**:

(15.17)
$$\frac{v^2}{\rho} = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|}$$

There is an alternative way of finding the normal component of acceleration. To simplify the notation, denote $\mathbf{T}(s)$, $\mathbf{N}(s)$, and $\mathbf{r}''(t)$ in Theorem (15.15) by $\mathbf{T}$, $\mathbf{N}$, and $\mathbf{a}$, respectively. If the tangential and normal components of $\mathbf{a}$ are denoted by $a_{\mathbf{T}}$ and $a_{\mathbf{N}}$, then we may write

$$\mathbf{a} = a_{\mathbf{T}}\mathbf{T} + a_{\mathbf{N}}\mathbf{N}.$$

Since $\mathbf{T}$ and $\mathbf{N}$ are mutually orthogonal unit vectors, $\mathbf{T} \cdot \mathbf{N} = 0$, $\mathbf{T} \cdot \mathbf{T} = 1$ and $\mathbf{N} \cdot \mathbf{N} = 1$. Consequently,

$$|\mathbf{a}|^2 = \mathbf{a} \cdot \mathbf{a} = (a_{\mathbf{T}}\mathbf{T} + a_{\mathbf{N}}\mathbf{N}) \cdot (a_{\mathbf{T}}\mathbf{T} + a_{\mathbf{N}}\mathbf{N}) = a_{\mathbf{T}}^2 + a_{\mathbf{N}}^2.$$

Thus,

(15.18)
$$a_{\mathbf{N}} = \sqrt{|\mathbf{a}|^2 - a_{\mathbf{T}}^2}.$$

Formula (15.18) should be used if it is difficult to find the normal component by means of (15.17).

Example 1 The position of a particle at time t is (t, t^2, t^3). Find the tangential and normal components of acceleration at time t.

Solution We may write

$$\begin{aligned}
\mathbf{r}(t) &= t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}\\
\mathbf{r}'(t) &= \mathbf{i} + 2t\mathbf{j} + 3t^2\mathbf{k}\\
\mathbf{r}''(t) &= 2\mathbf{j} + 6t\mathbf{k}\\
|\mathbf{r}'(t)| &= (1 + 4t^2 + 9t^4)^{1/2}.
\end{aligned}$$

Using (15.16), the tangential component of acceleration is

$$a_{\mathbf{T}} = \frac{dv}{dt} = \frac{\mathbf{r}'(t)\cdot\mathbf{r}''(t)}{|\mathbf{r}'(t)|} = \frac{4t + 18t^3}{(1 + 4t^2 + 9t^4)^{1/2}}.$$

Let us find the normal component by means of (15.17). First,

$$\mathbf{r}'(t) \times \mathbf{r}''(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2t & 3t^2 \\ 0 & 2 & 6t \end{vmatrix} = 6t^2\mathbf{i} - 6t\mathbf{j} + 2\mathbf{k}.$$

Applying (15.17) gives us

$$a_{\mathbf{N}} = \frac{v^2}{\rho} = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|} = \frac{(36t^4 + 36t^2 + 4)^{1/2}}{(1 + 4t^2 + 9t^4)^{1/2}} = 2\left(\frac{9t^4 + 9t^2 + 1}{9t^4 + 4t^2 + 1}\right)^{1/2}.$$

After determining $a_{\mathbf{T}}$ we could have used (15.18) to find the normal component as follows

$$a_{\mathbf{N}} = \sqrt{(4 + 36t^2) - \frac{(4t + 18t^3)^2}{1 + 4t^2 + 9t^4}}.$$

This simplifies to the preceding expression for $a_{\mathbf{N}}$. ■

Example 2 If a point P moves around a circle of radius a with a constant speed v, find the tangential and normal components of acceleration.

Solution Since $v = ds/dt$ is a constant, the tangential component dv/dt is 0. By Example 3 of the previous section, the curvature of the circle is $1/a$. Hence the normal component of acceleration (see (15.17)) is v^2/a. This shows that the acceleration is a vector of constant magnitude v^2/a directed from P toward the center of the circle (see Example 2 in Section 15.3). ■

We may use (15.17) to obtain a formula for curvature of a space curve C. Specifically, if C is given parametrically by

$$x = f(t), \quad y = g(t), \quad z = h(t)$$

let

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}.$$

We may then regard C as the curve traced by the endpoint of $\mathbf{r}(t)$ as t varies. Since $v = |\mathbf{r}'(t)|$ and $K = 1/\rho$ we have, from (15.17),

$$\frac{|\mathbf{r}'(t)|^2}{(1/K)} = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|}.$$

Solving for K gives us the following result.

Theorem (15.19)

> Let a curve C be given by $x = f(t)$, $y = g(t)$, $z = h(t)$, where f'', g'', and h'' exist. The curvature K at the point $P(x, y, z)$ on C is
>
> $$K = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3}.$$

This formula may also be used for plane curves (see Exercise 24).

Example 3 Find the curvature of the twisted cubic $x = t$, $y = t^2$, $z = t^3$ at the point (x, y, z).

Solution If we let

$$\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$$

then the curve is the same as that considered in Example 1. Substituting the expressions obtained there for $\mathbf{r}'(t)$ and $\mathbf{r}'(t) \times \mathbf{r}''(t)$ into Theorem (15.19), we obtain

$$K = \frac{2(9t^4 + 9t^2 + 1)^{1/2}}{(1 + 4t^2 + 9t^4)^{3/2}}.$$ ■

15.5 Exercises

In Exercises 1–8, $\mathbf{r}(t)$ is the position vector of a particle at time t. Find the tangential and normal components of acceleration at time t.

1 $\mathbf{r}(t) = t^2\mathbf{i} + (3t + 2)\mathbf{j}$

2 $\mathbf{r}(t) = (2t^2 - 1)\mathbf{i} + 5t\mathbf{j}$

3 $\mathbf{r}(t) = 3t\mathbf{i} + t^3\mathbf{j} + 3t^2\mathbf{k}$

4 $\mathbf{r}(t) = 4t\mathbf{i} + t^2\mathbf{j} + 2t^2\mathbf{k}$

5 $\mathbf{r}(t) = t(\cos t\mathbf{i} + \sin t\mathbf{j})$

6 $\mathbf{r}(t) = \cosh t\mathbf{i} + \sinh t\mathbf{j}$

7 $\mathbf{r}(t) = 4\cos t\mathbf{i} + 9\sin t\mathbf{j} + t\mathbf{k}$

8 $\mathbf{r}(t) = e^t(\sin t\mathbf{i} + \cos t\mathbf{j} + \mathbf{k})$

9–16 Use (15.18) to find the curvature, at the point corresponding to t, of the curve traced by the endpoint of $\mathbf{r}(t)$ in each of Exercises 1–8.

17 A particle moves along the parabola $y = x^2$ such that the horizontal component of velocity is always 3. Find the tangential and normal components of acceleration at the point $P(1, 1)$. Sketch the path and illustrate the acceleration as a sum of two vectors as in Theorem (15.15).

18 Work Exercise 17 if the particle moves along the graph of $y = 2x^3 - x$.

19 Prove that if a particle moves along a curve C with a constant speed, then the accleration is always normal to C.

20 Use Theorem (15.19) to prove that if a particle moves through space with an acceleration that is always **0**, then the motion is rectilinear.

21 If a particle P moves along a curve C with a constant speed, show that the magnitude of the acceleration is directly proportional to the curvature of the curve; that is, $|\mathbf{a}| = cK$ for some real number c.

22 If, in Exercise 21, a second particle Q moves along C with a speed twice that of P, show that the magnitude of the acceleration is four times greater than that of P.

23 Show that if a particle moves along the graph of $y = f(x)$, where $a \leq x \leq b$, then the normal component of acceleration is 0 at a point of inflection. Illustrate this fact by using $\mathbf{r}(t) = t\mathbf{i} + t^3\mathbf{j}$.

24 If a plane curve is given parametrically by $x = f(t)$, $y = g(t)$, where f'' and g'' exist, use Theorem (15.19) to prove that the curvature at the point K is given by Theorem (15.13).

25 Show that the curvature at every point on the circular helix $x = a\cos t$, $y = a\sin t$, $z = bt$ where $a > 0$, is given by $K = a/(a^2 + b^2)$.

26 Find the curvature, at (x, y, z), of the elliptic helix that has parametric equations $x = a\cos t$, $y = b\sin t$, $z = ct$, where a, b, and c are positive.

15.6 Kepler's Laws

It is fitting to conclude this chapter with a display of the power and beauty of vector methods when applied to the derivation of three classical physical laws. The discussion in this section is not simple, for it is not a simple problem that we intend to consider. However, the reader should gain considerable insight and be justly rewarded by giving the material which follows a very careful reading.

After many years of analyzing an enormous amount of empirical data, the German astronomer Johannes Kepler (1571–1630) formulated three laws that describe the motion of planets about the sun. These laws may be stated as follows.

Kepler's Laws (15.20)

First Law. The orbit of each planet is an ellipse with the sun at one focus.

Second Law. The vector from the sun to a moving planet sweeps out area at a constant rate.

Third Law. If the time required for a planet to travel once around its elliptical orbit is T, and if the major axis of the ellipse is $2a$, then $T^2 = ka^3$ for some constant k.

Approximately 50 years later, Sir Isaac Newton (1642–1727) proved that Kepler's Laws were consequences of his Law of Universal Gravitation and Second Law of Motion. The achievements of both men were monumental, because these laws clarified all astronomical observations that had been made up to that time.

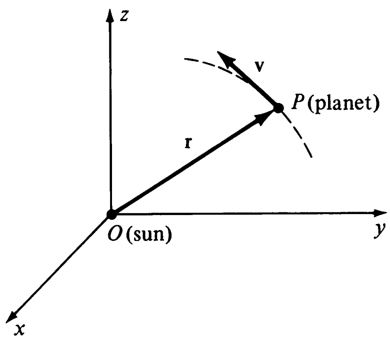


FIGURE 15.22

In this section, we shall prove Kepler's Laws by the use of vector techniques. Since the force of gravity which the sun exerts on a planet far exceeds that exerted by other celestial bodies, we shall neglect all other forces acting on a planet. From this point of view there are only two objects to consider: the sun and a planet revolving around it.

It is convenient to introduce a coordinate system with the center of mass of the sun at the origin, O, as illustrated in Figure 15.22. The point P represents the center of mass of the planet. To simplify the notation we shall denote the position vector of P by $\mathbf{r}$ instead of $\mathbf{r}(t)$, and use $\mathbf{v}$ and $\mathbf{a}$ to denote the velocity $\mathbf{r}'(t)$ and acceleration $\mathbf{r}''(t)$, respectively. Throughout our discussion we shall not distinguish between vectors represented by directed line segments and vectors represented by triples of real numbers.

Before proving Kepler's Laws, let us show that the motion of the planet takes place in one plane. If we let $r = |\mathbf{r}|$, then $\mathbf{u} = (1/r)\mathbf{r}$ is a unit vector having the same direction as $\mathbf{r}$. According to Newton's Law of Gravitation, the force $\mathbf{F}$ of gravitational attraction on the planet is given by

$$\mathbf{F} = -G\frac{Mm}{r^2}\mathbf{u}$$

where M is the mass of the sun, m is the mass of the planet, and G is the gravitational constant. Newton's Second Law of Motion states that

$$\mathbf{F} = m\mathbf{a}.$$

If we equate these two expressions for $\mathbf{F}$ and solve for $\mathbf{a}$ we obtain

$$\text{(a)} \qquad \mathbf{a} = -\frac{GM}{r^2}\mathbf{u}.$$

This shows that $\mathbf{a}$ is parallel to $\mathbf{r} = r\mathbf{u}$ and hence $\mathbf{r} \times \mathbf{a} = \mathbf{0}$. In addition, since $\mathbf{v} \times \mathbf{v} = \mathbf{0}$ we see that

$$\begin{aligned}\frac{d}{dt}(\mathbf{r} \times \mathbf{v}) &= \mathbf{r} \times \frac{d\mathbf{v}}{dt} + \frac{d\mathbf{r}}{dt} \times \mathbf{v} \\ &= \mathbf{r} \times \mathbf{a} + \mathbf{v} \times \mathbf{v} = \mathbf{0}.\end{aligned}$$

It follows that

$$\text{(b)} \qquad \mathbf{r} \times \mathbf{v} = \mathbf{c}$$

where $\mathbf{c}$ is a constant vector. The vector $\mathbf{c}$ will play an important role in the proofs of Kepler's Laws.

Since $\mathbf{r} \times \mathbf{v} = \mathbf{c}$, the vector $\mathbf{r}$ is orthogonal to $\mathbf{c}$ for every value of t. This implies that the curve traced by P lies in one plane, that is, *the orbit of the planet is a plane curve.*

Let us now prove Kepler's First Law. There is no loss of generality in assuming that the motion of the planet takes place in the xy-plane. In this case the vector $\mathbf{c}$ is perpendicular to the xy-plane, and we may assume that $\mathbf{c}$ has the same direction as the positive z-axis, as illustrated in Figure 15.23.

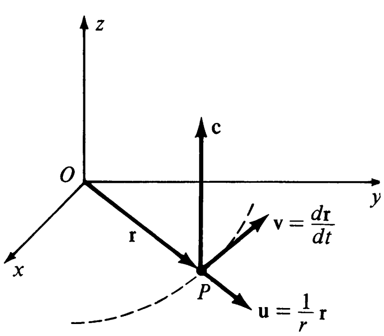


FIGURE 15.23

Since $\mathbf{r} = r\mathbf{u}$ we see that

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = r\frac{d\mathbf{u}}{dt} + \frac{dr}{dt}\mathbf{u}.$$

Substitution in $\mathbf{c} = \mathbf{r} \times \mathbf{v}$, and use of properties of vector products gives us

$$\begin{aligned}\mathbf{c} &= r\mathbf{u} \times \left(r\frac{d\mathbf{u}}{dt} + \frac{dr}{dt}\mathbf{u}\right)\\ &= r^2\left(\mathbf{u} \times \frac{d\mathbf{u}}{dt}\right) + r\frac{dr}{dt}(\mathbf{u} \times \mathbf{u}).\end{aligned}$$

Since $\mathbf{u} \times \mathbf{u} = \mathbf{0}$, this reduces to

(c) $$\mathbf{c} = r^2\left(\mathbf{u} \times \frac{d\mathbf{u}}{dt}\right).$$

Using (c) and (a) together with (ii) and (vi) of Theorem 14.36, we see that

$$\begin{aligned}\mathbf{a} \times \mathbf{c} &= \left(-\frac{GM}{r^2}\mathbf{u}\right) \times \left[r^2\left(\mathbf{u} \times \frac{d\mathbf{u}}{dt}\right)\right]\\ &= -GM\left[\mathbf{u} \times \left(\mathbf{u} \times \frac{d\mathbf{u}}{dt}\right)\right]\\ &= -GM\left[\left(\mathbf{u} \cdot \frac{d\mathbf{u}}{dt}\right)\mathbf{u} - (\mathbf{u} \cdot \mathbf{u})\frac{d\mathbf{u}}{dt}\right].\end{aligned}$$

Since $|\mathbf{u}| = 1$ it follows from Example 4 of Section 15.2 that $\mathbf{u} \cdot (d\mathbf{u}/dt) = 0$. In addition, $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2 = 1$, and hence the last formula for $\mathbf{a} \times \mathbf{c}$ reduces to

$$\mathbf{a} \times \mathbf{c} = GM\frac{d\mathbf{u}}{dt} = \frac{d}{dt}(GM\mathbf{u}).$$

We may also write

$$\mathbf{a} \times \mathbf{c} = \frac{d\mathbf{v}}{dt} \times \mathbf{c} = \frac{d}{dt}(\mathbf{v} \times \mathbf{c})$$

and, consequently,

$$\frac{d}{dt}(\mathbf{v} \times \mathbf{c}) = \frac{d}{dt}(GM\mathbf{u}).$$

Integrating both sides of this equation gives us

(d) $$\mathbf{v} \times \mathbf{c} = GM\mathbf{u} + \mathbf{b}$$

where $\mathbf{b}$ is a constant vector.

The vector $\mathbf{v} \times \mathbf{c}$ is orthogonal to $\mathbf{c}$ and, therefore, is in the xy-plane. Since $\mathbf{u}$ is also in the xy-plane it follows from (d) that $\mathbf{b}$ is in the xy-plane.

Up to this point, our proof has been independent of the positions of the x- and y-axes. Let us now choose a coordinate system such that the positive x-axis has the direction of the constant vector $\mathbf{b}$, as illustrated in Figure 15.24.

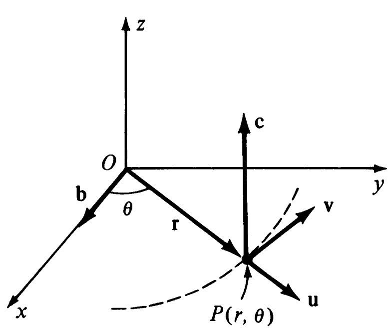


FIGURE 15.24

Let (r, θ) be polar coordinates for the point P, where $r = |\mathbf{r}|$. It follows that

$$\mathbf{u} \cdot \mathbf{b} = |\mathbf{u}|\,|\mathbf{b}| \cos\theta = b\cos\theta$$

where $b = |\mathbf{b}|$. If we let $c = |\mathbf{c}|$, then using (b) together with properties of the dot and vector products, and also (d),

$$\begin{aligned} c^2 = \mathbf{c}\cdot\mathbf{c} &= (\mathbf{r}\times\mathbf{v})\cdot\mathbf{c} = \mathbf{r}\cdot(\mathbf{v}\times\mathbf{c}) \\ &= (r\mathbf{u})\cdot(GM\mathbf{u} + \mathbf{b}) \\ &= rGM(\mathbf{u}\cdot\mathbf{u}) + r(\mathbf{u}\cdot\mathbf{b}) \\ &= rGM + rb\cos\theta. \end{aligned}$$

Solving the last equation for r gives us

$$r = \frac{c^2}{GM + b\cos\theta}.$$

Dividing numerator and denominator of this fraction by GM we obtain

(e) $$r = \frac{p}{1 + e\cos\theta}$$

where $p = c^2/GM$ and $e = b/GM$. From Theorem (13.8), the graph of this polar equation is a conic with eccentricity e and focus at the origin. Since the orbit is a closed curve, it follows that $0 < e < 1$ and that the conic is an ellipse. This completes the proof of Kepler's First Law.

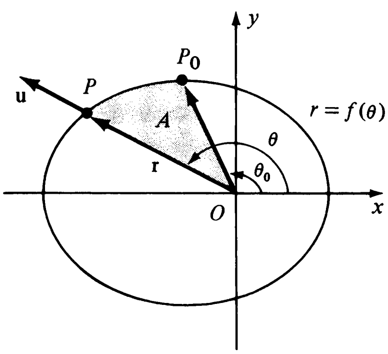


FIGURE 15.25

Let us next prove Kepler's Second Law. It may be assumed that the orbit of the planet is an ellipse in the xy-plane. Let $r = f(\theta)$ be a polar equation of the orbit, with the sun centered at the focus O. Let P_0 denote the position of the planet at time t_0, and P its position at any time $t \geq t_0$. As illustrated in Figure 15.25, θ_0 and θ will denote the angles measured from the positive x-axis to $\overrightarrow{OP_0}$ and $\overrightarrow{OP}$, respectively.

By Theorem (13.9), the area A swept out by $\overrightarrow{OP}$ in the time interval $[t_0, t]$ is given by

$$A = \int_{\theta_0}^{\theta} \frac{1}{2} r^2\, d\theta$$

and hence

$$\frac{dA}{d\theta} = \frac{d}{d\theta}\int_{\theta_0}^{\theta} \frac{1}{2} r^2\, d\theta = \frac{1}{2} r^2.$$

Using this fact and the Chain Rule gives us

(f) $$\frac{dA}{dt} = \frac{dA}{d\theta}\frac{d\theta}{dt} = \frac{1}{2} r^2 \frac{d\theta}{dt}.$$

Next we observe that since $\mathbf{r} = r\cos\theta\mathbf{i} + r\sin\theta\mathbf{j} + 0\mathbf{k}$, the unit vector $\mathbf{u} = (1/r)\mathbf{r}$ may be expressed in the form

$$\mathbf{u} = \cos\theta\mathbf{i} + \sin\theta\mathbf{j} + 0\mathbf{k}.$$

Consequently,

$$\frac{d\mathbf{u}}{dt} = -\sin\theta\frac{d\theta}{dt}\mathbf{i} + \cos\theta\frac{d\theta}{dt}\mathbf{j} + 0\mathbf{k}.$$

A direct calculation may be used to show that

$$\mathbf{u} \times \frac{d\mathbf{u}}{dt} = \frac{d\theta}{dt}\mathbf{k}.$$

If $\mathbf{c}$ is the vector obtained in the proof of Kepler's First Law, then by (c) and the last equation,

$$\mathbf{c} = r^2\left(\mathbf{u} \times \frac{d\mathbf{u}}{dt}\right) = r^2\frac{d\theta}{dt}\mathbf{k}$$

and hence

(g) $$c = |\mathbf{c}| = r^2\frac{d\theta}{dt}.$$

Combining (f) and (g) we see that

(h) $$\frac{dA}{dt} = \frac{1}{2}c$$

that is, the rate at which A is swept out by $\overrightarrow{OP}$ is a constant. This establishes Kepler's Second Law.

To prove Kepler's Third Law we shall retain the notation used in the proofs of the first two laws. In particular, we assume that a polar equation of the planetary orbit is given by

$$r = \frac{p}{1 + e\cos\theta}$$

where $p = c^2/GM$ and $e = b/GM$ (see (e)).

Let T denote the time required for the planet to make one complete revolution about the sun. By (h) the area swept out in the time interval $[0, T]$ is given by

$$A = \int_0^T \frac{dA}{dt}dt = \int_0^T \frac{1}{2}c\,dt = \frac{1}{2}cT.$$

This also equals the area of the plane region bounded by the ellipse. However, it was shown in Example 4 of Section 12.3 that $A = \pi ab$, where $2a$ and $2b$ are lengths of the major and minor axes, respectively, of the ellipse. Consequently,

$$\frac{1}{2}cT = \pi ab \quad \text{or} \quad T = \frac{2\pi ab}{c}.$$

From our work in Section 13.4, we know that

$$e = \frac{\sqrt{a^2 - b^2}}{a}$$

and hence

$$a^2e^2 = a^2 - b^2, \quad \text{or} \quad b^2 = a^2(1 - e^2).$$

Thus
$$T^2 = \frac{4\pi^2 a^2 b^2}{c^2} = \frac{4\pi^2 a^4(1 - e^2)}{c^2}.$$

It was shown in the proof of Theorem 13.7 (where we had $p = de$) that

$$a^2 = \frac{p^2}{(1 - e^2)^2}, \quad \text{or} \quad a = \frac{p}{1 - e^2}$$

and hence
$$T^2 = \frac{4\pi^2 a^4}{c^2}\left(\frac{p}{a}\right).$$

Since $p = c^2/GM$, this reduces to

$$T^2 = \frac{4\pi^2}{GM}a^3 = ka^3$$

where $k = 4\pi^2/GM$. This completes the proof.

In our proofs of Kepler's Laws, remember that we assumed that the only gravitational force acting on a planet was that of the sun. If forces exerted by other planets are taken into account, then irregularities in the elliptical orbits may occur. Indeed, it was because of observed irregularities in the motion of Uranus that the British astronomer J. Adams (1819–1892) and the French astronomer U. LeVerrier (1811–1877) were able to predict the orbit of an unknown planet that was causing the irregularities. Using their predictions, this planet, later named Neptune, was first observed with a telescope by the German astronomer J. Galle in 1846.

15.7 Review: Concepts

Define or discuss each of the following.

1 Vector-valued function

2 Limit of a vector-valued function

3 Continuity of a vector-valued function

4 Derivative of a vector-valued function

5 Differentiation formulas for vector-valued functions

6 Tangent vector to a curve

7 Integrals of vector-valued functions

8 Velocity and acceleration

9 Normal vector to a curve

10 Curvature of a plane curve

11 Circle of curvature

12 Radius of curvature

13 Curvature of a space curve

14 Tangential component of acceleration

15 Normal component of acceleration

16 Kepler's Laws

Review: Exercises

1 If $\mathbf{r}(t) = t^2\mathbf{i} + (4t^2 - t^4)\mathbf{j}$,

(a) sketch the curve determined by the components of $\mathbf{r}(t)$.

(b) find $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$.

(c) sketch vectors corresponding to $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$ if $t = 0$, $t = 1$, and $t = 2$.

2 The position of a particle moving in a plane is given by

$$\mathbf{r}(t) = (t - \sin t)\mathbf{i} + (1 - \cos t)\mathbf{j}.$$

Find its velocity, acceleration, and speed at time t. Sketch the path of the particle together with vectors corresponding to the velocity and acceleration for the following values of t.

(a) $t = 0$

(b) $t = \pi/4$

(c) $t = \pi/2$

(d) $t = 3\pi/4$

(e) $t = \pi$

(f) $t = 5\pi/4$

(g) $t = 3\pi/2$

(h) $t = 7\pi/4$

(i) $t = 2\pi$

3 If the curve C is given by $x = e^t \sin t$, $y = e^t \cos t$, $z = e^t$, where $0 \le t \le 1$, find

(a) a unit tangent vector to C at the point P corresponding to $t = 0$.

(b) the length of C.

4 The position of a particle at time t is given by

$$\mathbf{r}(t) = 3t\mathbf{i} + t^3\mathbf{j} + t^4\mathbf{k}.$$

(a) Find the velocity and acceleration at time t.

(b) Sketch the path of the particle together with vectors corresponding to the velocity and acceleration at $t = 1$.

(c) Find the speed at $t = 1$.

5 If the curve C is given by $x = 3t^2 + 1$, $y = 4t$, $z = e^{t-1}$, find an equation of the tangent line at the point $P(4, 4, 1)$.

6 If $\mathbf{u}(t) = t^2\mathbf{i} + 6t\mathbf{j} + t\mathbf{k}$ and $\mathbf{v}(t) = t\mathbf{i} - 5t\mathbf{j} + 4t^2\mathbf{k}$ find the values of t for which $\mathbf{u}(t)$ and $\mathbf{v}'(t)$ are orthogonal.

7 If $\mathbf{u}(t)$ and $\mathbf{v}(t)$ are as in Exercise 6, find $D_t[\mathbf{u}(t) \times \mathbf{v}(t)]$ and $D_t[\mathbf{u}(t) \cdot \mathbf{v}(t)]$.

8 Find $\displaystyle\int (\sin 3t\mathbf{i} + e^{-2t}\mathbf{j} + \cos t\mathbf{k})\,dt$.

9 Evaluate $\displaystyle\int_0^1 (4t\mathbf{i} + t^3\mathbf{j} - \mathbf{k})\,dt$.

10 Find $\mathbf{u}(t)$ if $\mathbf{u}'(t) = e^{-t}\mathbf{i} - 4\sin 2t\mathbf{j} + 3\sqrt{t}\mathbf{k}$ and $\mathbf{u}(0) = -\mathbf{i} + 2\mathbf{j}$.

Verify the identities in Exercises 11 and 12 without using components.

11 $D_t|\mathbf{u}(t)|^2 = 2\mathbf{u}(t) \cdot \mathbf{u}'(t)$

12 $D_t(\mathbf{u}(t) \cdot \mathbf{u}(t) \times \mathbf{u}''(t)) = \mathbf{u}(t) \cdot \mathbf{u}'(t) \times \mathbf{u}'''(t)$

In Exercises 13–15 find the curvature of the given curve at the point P.

13 $y = xe^x$; $P(0, 0)$

14 $x = 1/(1 + t)$, $y = 1/(1 - t)$; $P(\frac{2}{3}, 2)$

15 $x = 2t^2$, $y = t^4$, $z = 4t$; $P(x, y, z)$

16 Find the x-coordinates of the points on the graph of $y = x^3 - 3x$ at which the curvature is a maximum.

17 If C is the graph of $y = \cosh x$,

(a) find an equation of the circle of curvature for the point $P(0, 1)$.

(b) sketch C and the circle of curvature for P.

18 Given the limaçon $r = 2 + \sin\theta$, find the radius of curvature for the point $P(\pi, 2)$.

In Exercises 19 and 20 find the tangential and normal components of acceleration at time t if the position vector of a particle is as indicated.

19 $\mathbf{r}(t) = \sin 2t\mathbf{i} + \cos t\mathbf{j}$

20 $\mathbf{r}(t) = 3t\mathbf{i} + t^3\mathbf{j} + t\mathbf{k}$

16 Partial Differentiation

In this chapter the concept of derivative is generalized to functions of more than one variable. Applications include finding rates of change, extrema, and approximations by differentials.

16.1 Functions of Several Variables

The functions considered in previous chapters involved only one independent variable. As we have seen, such functions can be used to solve a variety of problems; however, there are numerous applications in which *several* independent variables occur. For example, the area of a rectangle depends on *two* quantities, length and width. If an object is located in space, then the temperature at a point P in the object may depend on *three* rectangular coordinates x, y, and z of P. If, in addition, the temperature changes with time t, then *four* variables x, y, z, and t are involved. As a final illustration, a manufacturer may find that the cost C of producing a certain item depends on material, labor, equipment, overhead, and maintenance charges. Thus C is influenced by *five* different variables.

In this section we shall define functions of several variables and discuss some of their properties. Recall that a function f is a correspondence that associates with each element of a set X a unique element of a set Y (see Definition (1.21)). If both X and Y are subsets of $\mathbb{R}$, then f is called a function of one (real) variable. If X is a subset of $\mathbb{R} \times \mathbb{R}$ and Y is a subset of $\mathbb{R}$, then f is called a *function of two (real) variables*. Another way of stating this is as follows.

Definition (16.1)

> Let D be a set of ordered pairs of real numbers. A **function f of two variables** is a correspondence that associates with each pair (x, y) in D a unique real number, denoted by $f(x, y)$. The set D is called the **domain** of f. The **range** of f consists of all real numbers $f(x, y)$, where (x, y) is in D.

In Chapter 1 the domain and range of a function of one variable were represented geometrically by points on two real lines (see Figure 1.26). For functions of *two* variables we may represent the domain D by points in an xy-plane and the range by points on a real line, say a w-axis. This is illustrated in Figure 16.1, where several curved arrows are drawn from ordered pairs in D to the corresponding numbers in the range. To obtain a physical interpretation of this situation we could imagine a flat metal plate having the shape of D. To each point (x, y) on the plate there corresponds a temperature $f(x, y)$ that can be recorded on a thermometer, represented by the w-axis. As another illustration, we could regard D as the surface of a lake and let $f(x, y)$ denote the depth of the water under the point (x, y).

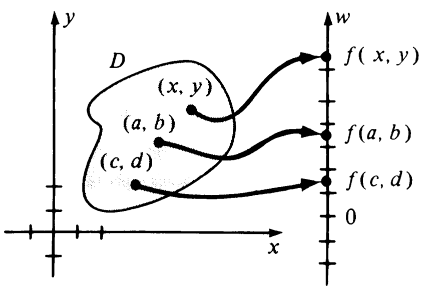


FIGURE 16.1

A function f of two variables is often defined by using an expression in x and y to specify $f(x, y)$. In this event, it is assumed that the domain is the set of all pairs (x, y) for which the given expression is meaningful, and we call f **a function of x and y**.

Example 1 If

$$f(x, y) = \frac{xy - 5}{2\sqrt{y - x^2}},$$

find the domain D of f. Illustrate D and the numbers $f(2, 5)$, $f(1, 2)$, and $f(-1, 2)$ in the manner of Figure 16.1.

Solution The domain D is the set of all pairs (x, y) such that $y - x^2$ is positive, that is, $y > x^2$. (Why?) Thus the graph of D is that part of the xy-plane that lies above the parabola $y = x^2$, as shown in Figure 16.2. By substitution,

$$f(2, 5) = \frac{(2)(5) - 5}{2\sqrt{5 - 4}} = \frac{5}{2}.$$

Similarly, $f(1, 2) = -\frac{3}{2}$ and $f(-1, 2) = -\frac{7}{2}$. These functional values are indicated on the w-axis in Figure 16.2. ■

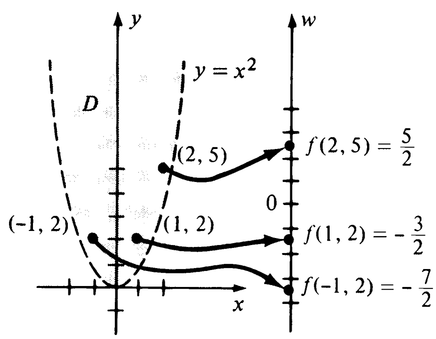


FIGURE 16.2

Formulas are sometimes used to define functions of two variables. For example, the formula $V = \pi r^2 h$ expresses the volume V of a right circular cylinder as a function of the altitude h and base radius r. The symbols r and h are referred to as **independent variables** and V is the **dependent variable**.

A function f of three (real) variables is defined as in (16.1), except that the domain D is a subset of $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$. In this case, with each ordered triple (x, y, z) in D there is associated a unique real number $f(x, y, z)$. If we represent

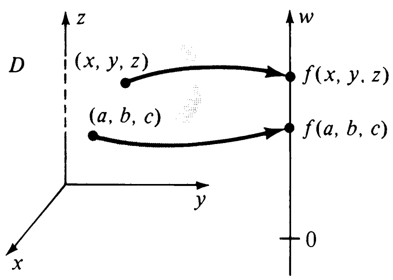


FIGURE 16.3

D by a region in a three-dimensional rectangular coordinate system, then as illustrated in Figure 16.3, to each point (x, y, z) in D there corresponds a unique point on the w-axis with coordinate $f(x, y, z)$. To obtain a physical illustration we could regard D as a solid object and, as before, let $f(x, y, z)$ be the temperature at (x, y, z).

As in the two-variable case, functions of three variables are often defined by means of expressions. For example,

$$f(x, y, z) = \frac{xe^y + \sqrt{z^2 + 1}}{xy \sin z}$$

determines a function of x, y, and z. Formulas such as $V = lwh$ for the volume of a rectangular parallelepiped of dimensions l, w, and h also illustrate how functions of three variables may arise.

Let us now return to a function f of two variables x and y. The **graph** of f is, by definition, the graph of the equation $z = f(x, y)$ in a three-dimensional rectangular coordinate system, and hence is usually a surface S of some type. If we represent D by a region in the xy-plane, then the pair (x, y) in D is represented by the point $(x, y, 0)$. Functional values $f(x, y)$ are the (directed) distances from the xy-plane to S as illustrated in Figure 16.4.

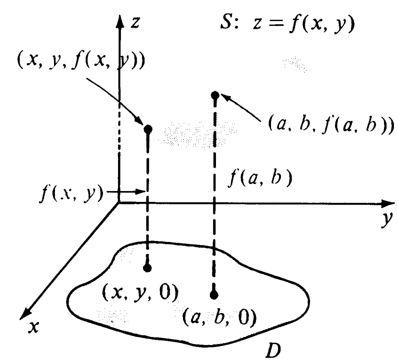


FIGURE 16.4

Example 2 Sketch the graph of f if $f(x, y) = 9 - x^2 - y^2$ and the domain is $D = \{(x, y): x^2 + y^2 \leq 9\}$.

Solution The domain D of f may be represented geometrically by the collection of all points within and on the circle $x^2 + y^2 = 9$ in the xy-plane. The graph of f is the portion of the graph of

$$z = 9 - x^2 - y^2$$

shown in Figure 16.5. For future reference we have sketched circular traces on the planes $z = k$ where $k = 0, 2, 4, 6$, and 8. ■

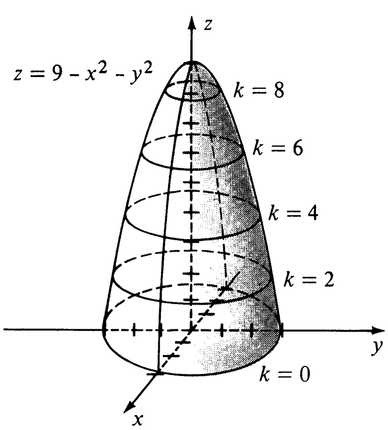


FIGURE 16.5 Circular traces are on the planes $z = k$.

There is another useful graphical technique for describing a function f of two variables. The method consists of sketching, in an xy-plane, the

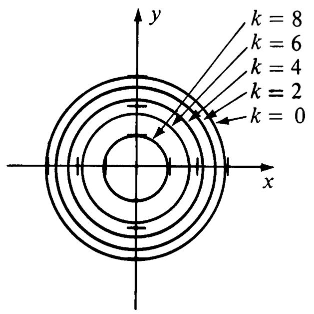


FIGURE 16.6
Level curves: $9 - x^2 - y^2 = k$.

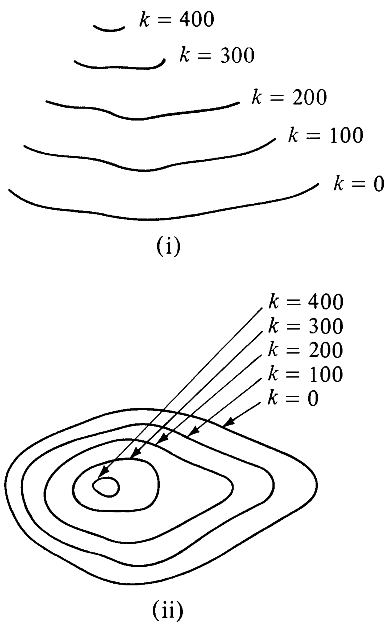


FIGURE 16.7

graphs of the equations $f(x, y) = k$ for various values of k. The graphs obtained in this way are called **level curves** of the function f. It is important to note that as a point (x, y) moves on a level curve, the values $f(x, y)$ of the function are constant.

Example 3 Sketch some level curves of the function f in Example 2.

Solution The level curves are graphs, in the xy-plane, of equations of the form $f(x, y) = k$, that is,

$$9 - x^2 - y^2 = k$$

or

$$x^2 + y^2 = 9 - k.$$

These are circles, provided $0 \leq k < 9$. In Figure 16.6 we have sketched the level curves corresponding to $k = 0, 2, 4, 6$, and 8. These level curves are the projections, in the xy-plane, of the circular traces shown in Figure 16.5. ■

If we sketch the level curves $f(x, y) = k$ for equispaced values of k, such as $k = 0, 2, 4, 6$, and 8 in Example 3, then the nearness of successive curves gives information about the steepness of the surface. In Figure 16.6 the level curves corresponding to $k = 0$ and $k = 2$ are closer than those corresponding to $k = 6$ and $k = 8$, which indicates that the surface shown in Figure 16.5 is steeper at points near the xy-plane than at points farther away.

Level curves are often used in making **topographic maps** of rough terrain. For example, suppose $f(x, y)$ denotes the altitude (in feet) at a point (x, y) of latitude x and longitude y. On the hill pictured in (i) of Figure 16.7 we have sketched curves (in three dimensions) corresponding to altitudes of $k = 0, 100, 200, 300$, and 400 feet. We may regard these curves as having been obtained by "slicing" the hill with planes parallel to the base. A person walking along one of these curves would always remain at the same altitude. The (two-dimensional) level curves corresponding to the same values of k are shown in (ii) of the figure. They represent the view obtained by looking down on the hill from an airplane.

Configurations of the type illustrated in (ii) of Figure 16.7 also occur on maps that indicate the depth of the water in a lake. One example is sketched in Figure 16.8, where $f(x, y)$ is the depth under the point (x, y).

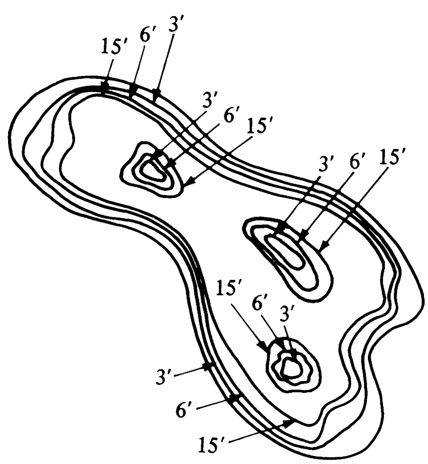


FIGURE 16.8 Depth of water in a lake

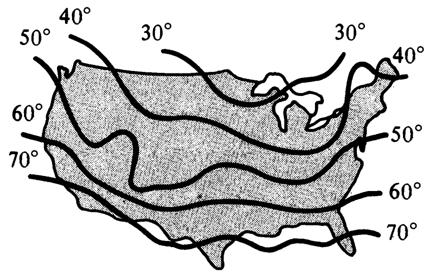


FIGURE 16.9 Isothermal curves

As another illustration of level curves, Figure 16.9 shows a weather map of the United States, where $f(x, y)$ denotes the high temperature at (x, y) during a certain day. The level curves are called **isothermal curves** to indicate that the temperature is constant on each curve. Another weather map could be drawn in which $f(x, y)$ represents the barametric pressure at (x, y). In this case, the level curves are called **isobars**.

If f is a function of three variables x, y, and z, then by definition the **level surfaces** of f are the graphs of $f(x, y, z) = k$, where k takes on suitable real values. If we let $k = w_0$, w_1, and w_2, there may result surfaces S_0, S_1, and S_2, respectively, as illustrated in Figure 16.10. As a point (x, y, z) moves along one of these surfaces, $f(x, y, z)$ does not change. In applications, if $f(x, y, z)$ is the temperature at (x, y, z), then the level surfaces are called **isothermal surfaces**. If $f(x, y, z)$ represents electrical potential, they are called **equipotential surfaces**, since the voltage does not change if (x, y, z) remains on such a surface.

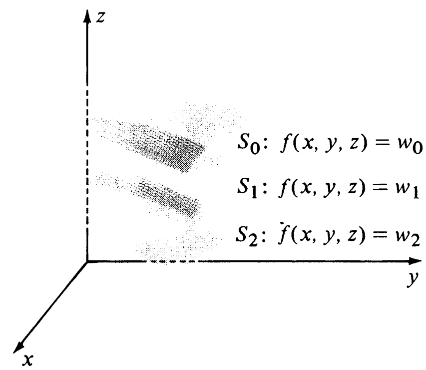


FIGURE 16.10 Level surfaces of $f(x, y, z)$

Example 4 If $f(x, y, z) = z - \sqrt{x^2 + y^2}$, sketch some level surfaces of f.

Solution The level surfaces are graphs of

$$z - \sqrt{x^2 + y^2} = k$$

or equivalently,

$$z = \sqrt{x^2 + y^2} + k$$

where k is any real number. For each k we obtain a right circular cone with its axis along the z-axis. The special cases $k = -1$, $k = 0$, $k = 1$, and $k = 2$ are sketched in Figure 16.11. ■

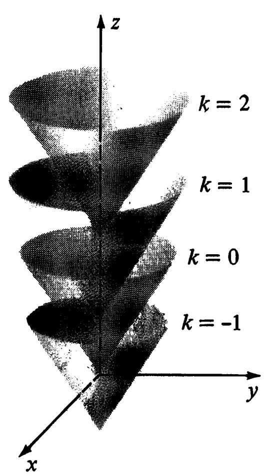
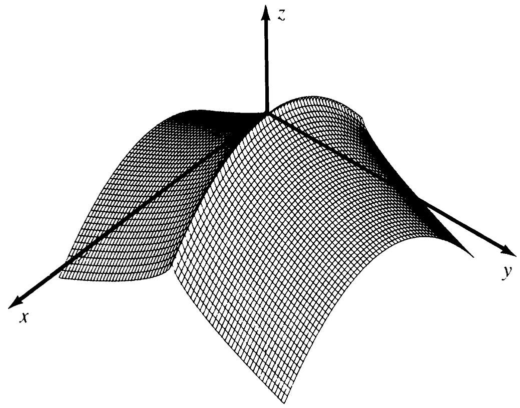
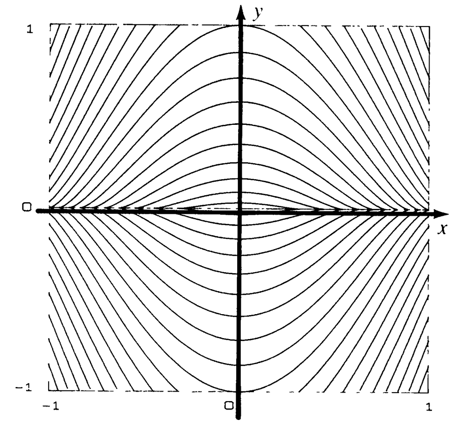


FIGURE 16.11
Level surfaces $z - \sqrt{x^2 + y^2} = k$

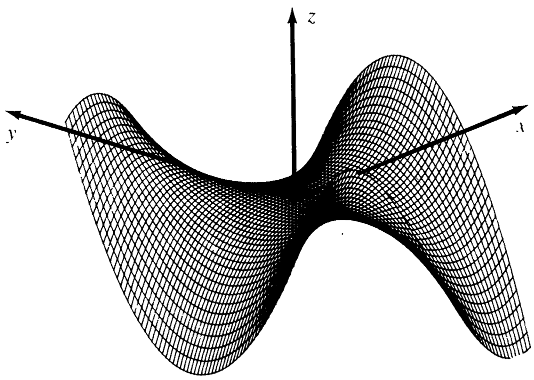
Computers can be programmed to exhibit level curves and traces of surfaces on various planes. It is also possible to represent surfaces from various perspectives. Sketches of this type are sometimes referred to as *computer graphics*, or *computer generated graphs*. Several illustrations obtained by means of a computer are shown in Figure 16.12. Students who are interested in computer programming should look for details of this technique in other books and courses.



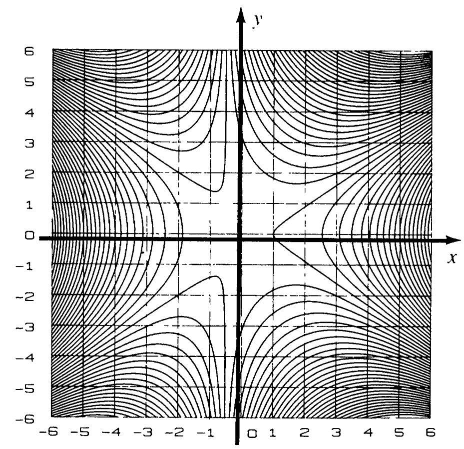
$z = -(x^2 + y^{2/3})$



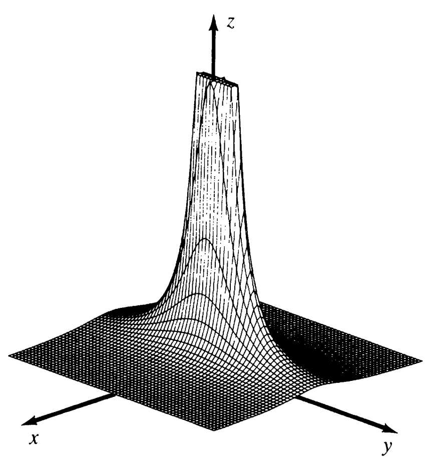
Level curves of $z = -(x^2 + y^{2/3})$



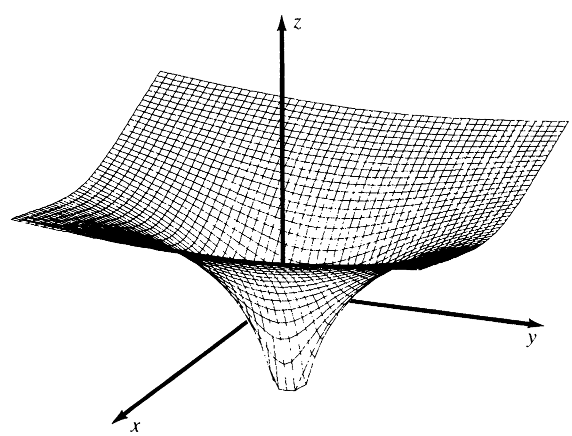
$z = x^3 - 2xy^2 - x^2 - y^2$



Level curves of $z = x^3 - 2xy^2 - x^2 - y^2$

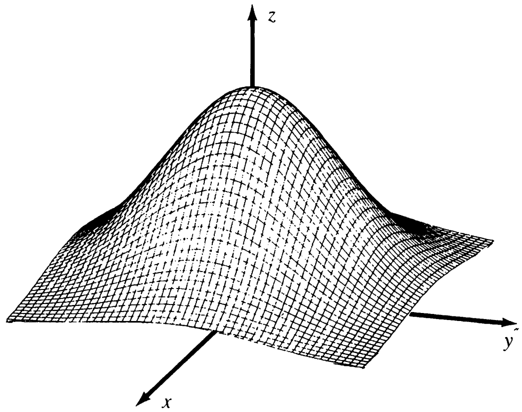


$z = 1/(9x^2 + y^2)$

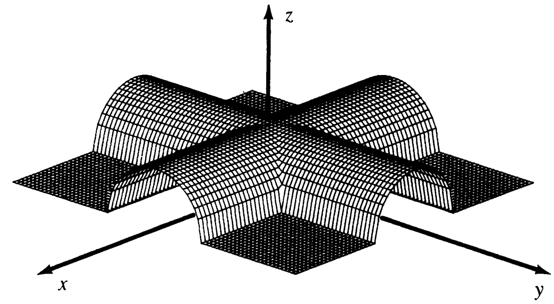


$z = \ln(2x^2 + y^2)$

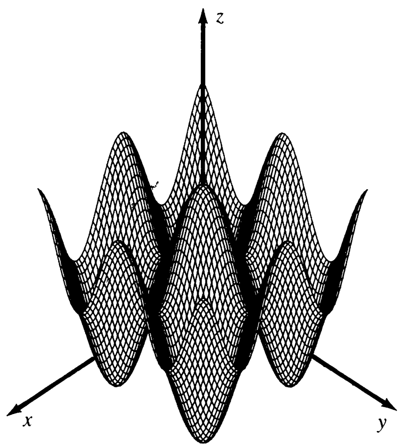
FIGURE 16.12



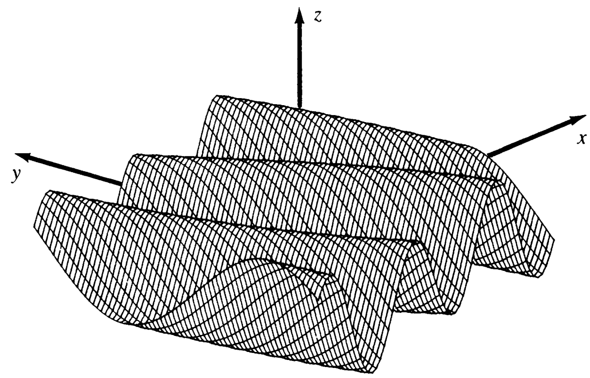
$z = 8e^{-(x^2+y^2)/4}$



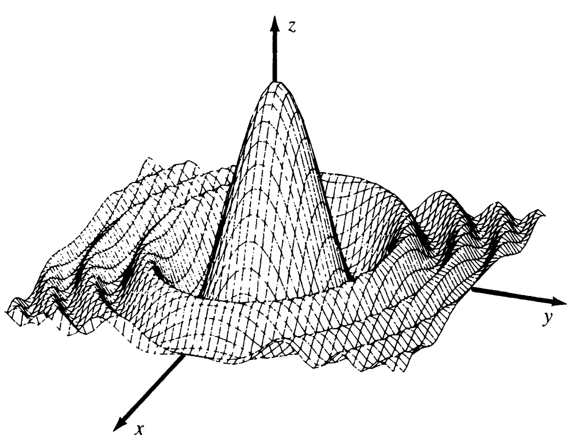
$$z = \begin{cases} \sqrt{1-y^2} & \text{if } |x| \geq 1 \text{ and } |y| \leq 1, \text{ or if } |x| \leq 1 \text{ and } |y| \leq |x| \\ \sqrt{1-x^2} & \text{if } |y| \geq 1 \text{ and } |x| \leq 1, \text{ or if } |y| \leq 1 \text{ and } |x| \leq |y| \\ 0 & \text{if } |x| \geq 1 \text{ and } |y| \geq 1 \end{cases}$$



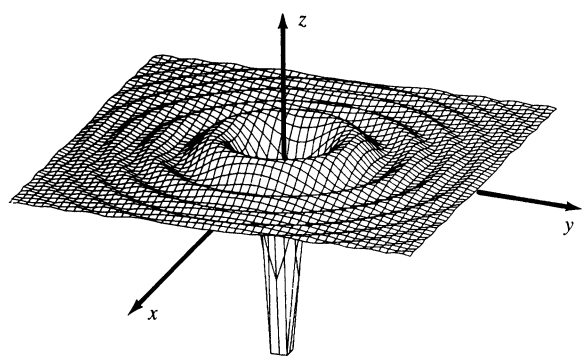
$z = \cos x + \cos y$



$z = \cos x + 3\cos(3x + y)$



$$z = \frac{\frac{1}{2}\cos(2x^2+y^2)}{1+2x^2+y^2}$$



$$z = \frac{1-2\cos(x^2+y^2)}{x^2+y^2}$$

16.1 Exercises

In Exercises 1–8 determine the domain of f and the value of f at the indicated points.

1 $f(x, y) = 2x - y^2$; $(-2, 5)$, $(5, -2)$, $(0, -2)$

2 $f(x, y) = (y + 2)/x$; $(3, 1)$, $(1, 3)$, $(2, 0)$

3 $f(u, v) = uv/(u - 2v)$; $(2, 3)$, $(-1, 4)$, $(0, 1)$

4 $f(r, s) = \sqrt{1 - r} - e^{r/s}$; $(1, 1)$, $(0, 4)$, $(-3, 3)$

5 $f(x, y, z) = \sqrt{25 - x^2 - y^2 - z^2}$; $(1, -2, 2)$, $(-3, 0, 2)$

6 $f(x, y, z) = 2 + \tan x + y \sin z$; $(\pi/4, 4, \pi/6)$, $(0, 0, 0)$

7 $f(r, s, v, p) = rs^2 \tan v + 4sv \ln p$; $(3, -1, \pi/4, e)$

8 $f(x, y, u, v, w) = w \ln (x - y) - xue^{v/w}$; $(2, 1, 3, 4, -1)$

In Exercises 9–12 find

$$\frac{f(x + h, y) - f(x, y)}{h} \quad \text{and} \quad \frac{f(x, y + h) - f(x, y)}{h}$$

where $h \neq 0$.

9 $f(x, y) = x^2 + y^2$

10 $f(x, y) = y^2 + 3x$

11 $f(x, y) = xy^2 + 3x$

12 $f(x, y) = xy^3 + 4x^2 - 2$

In Exercises 13–22 describe the graph of f.

13 $f(x, y) = \sqrt{1 - x^2 - y^2}$

14 $f(x, y) = x^2 + y^2 - 1$

15 $f(x, y) = 6 - 2x - 3y$

16 $f(x, y) = \sqrt{y^2 - 4x^2 - 16}$

17 $f(x, y) = 5$

18 $f(x, y) = 4 - x$

19 $f(x, y) = \sqrt{72 + 4x^2 - 9y^2}$

20 $f(x, y) = \sqrt{x^2 + 4y^2 + 25}$

21 $f(x, y) = 16y^2 - x^2$

22 $f(x, y) = -\sqrt{16 - 9x^2}$

In Exercises 23–30 sketch some level curves of f.

23 $f(x, y) = y^2 - x^2$

24 $f(x, y) = 3x - 2y$

25 $f(x, y) = x^2 - y$

26 $f(x, y) = xy$

27 $f(x, y) = y - \sin x$

28 $f(x, y) = 4x^2 + y^2$

29 $f(x, y) = x^2 + y^2 - 4x + 6y + 13$

30 $f(x, y) = e^x - y$

31 If $f(x, y) = y \arctan x$, find an equation of the level curve of f that passes through the point $P(1, 4)$.

32 If $f(x, y, z) = x^2 + 4y^2 - z^2$, find an equation of the level surface of f that passes through the point $P(2, -1, 3)$.

In Exercises 33–38 describe the level surfaces of f.

33 $f(x, y, z) = x^2 + y^2 + z^2$

34 $f(x, y, z) = z + x^2 + 4y^2$

35 $f(x, y, z) = x + 2y + 3z$

36 $f(x, y, z) = x^2 + y^2 - z^2$

37 $f(x, y, z) = x^2 + y^2$

38 $f(x, y, z) = z$

39 A flat metal plate is situated on an xy-plane such that the temperature T (in degrees Celsius) at the point (x, y) is inversely proportional to the distance from the origin.

(a) Describe the isothermal curves.

(b) If the temperature at the point $P(4, 3)$ is 40, find an equation of the isothermal curve where the temperature is 20.

40 If the voltage V at the point $P(x, y, z)$ is given by $V = 6/(x^2 + 4y^2 + 9z^2)^{1/2}$,

(a) describe the equipotential surfaces.

(b) find an equation of the equipotential surface where $V = 120$.

41 According to Newton's *Law of Universal Gravitation* if a particle of mass M is at the origin of a rectangular coordinate system, then the magnitude F of the force exerted on a particle of mass m located at the point (x, y, z) is given by

$$F = \frac{gMm}{x^2 + y^2 + z^2}$$

where g is a gravitational constant. How many independent variables are present? If M and m are constant, describe the level surfaces of the resulting function of x, y, and z. What is the physical significance of these level surfaces

42 According to the *Ideal Gas Law*, the pressure P, volume V, and temperature T of a confined gas are related by the formula $PV = kT$, where k is a constant. Express P as a function of V and T and describe the level curves associated with this function. What is the physical significance of these level curves?

43 Define, in a manner similar to (16.1), a function of (a) four variables; (b) five variables; (c) n variables, where n is any positive integer.

44 If f and g are functions of two variables define, in a manner that is analogous to the single variable case, the *sum* $f + g$, *difference* $f - g$, *product* fg and *quotient* f/g. What restrictions are required for the domains of these functions?

16.2 Limits and Continuity

If f is a function of two variables, it is often important to study the variation of $f(x, y)$ as (x, y) varies through the domain D of f. As a physical illustration, consider a thin metal plate which has the shape of the region D in Figure 16.1. To each point (x, y) on the plate there corresponds a temperature $f(x, y)$, which is recorded on a thermometer represented by the w-axis. As the point (x, y) moves on the plate the temperature may increase, decrease, or remain constant, and hence the point on the w-axis that corresponds to $f(x, y)$ will move in a positive direction, negative direction, or remain fixed, respectively. If the temperature $f(x, y)$ gets closer to a fixed value L as (x, y) gets closer and closer to a fixed point (a, b), we use the following notation.

(16.2) $$\lim_{(x, y)\to(a, b)} f(x, y) = L$$

This may be read as follows: *f has the limit L as (x, y) approaches (a, b)*. To make our remarks mathematically precise, let us employ a geometric device similar to that used for functions of one variable in Chapter 2. For any $\varepsilon > 0$, consider the open interval $(L - \varepsilon, L + \varepsilon)$ on the w-axis. If (16.2) is true, then as illustrated in Figure 16.13 there is a $\delta > 0$ such that for every point (x, y) inside the circle of radius δ with center (a, b), except possibly (a, b) itself, the number corresponding to $f(x, y)$ is in the interval $(L - \varepsilon, L + \varepsilon)$. This is equivalent to the following statement:

$$\text{If} \quad 0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta, \quad \text{then} \quad |f(x, y) - L| < \varepsilon.$$

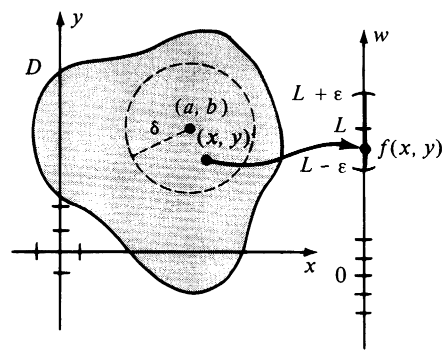


FIGURE 16.13

The preceding discussion is summarized in the next definition.

Definition (16.3)

Let a function f of two variables be defined throughout the interior of a circle with center (a, b), except possibly at (a, b) itself. The **limit of $f(x, y)$ as (x, y) approaches (a, b) is L**, written

$$\lim_{(x, y)\to(a, b)} f(x, y) = L$$

means that for every $\varepsilon > 0$ there corresponds a $\delta > 0$ such that

$$\text{if} \quad 0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta, \quad \text{then} \quad |f(x, y) - L| < \varepsilon.$$

If we consider the graph of f illustrated in Figure 16.4, then intuitively, Definition (16.3) means that as the point $(x, y, 0)$ approaches $(a, b, 0)$ on the xy-plane, the corresponding point $(x, y, f(x, y))$ on the graph S of f approaches (a, b, L) (which may or may not be on S). It can be shown that if the limit L exists, it is unique.

If f and g are functions of two variables, then $f + g, f - g, fg$, and f/g are defined in the usual way, and Theorem (2.8) concerning limits of sums, products, and quotients can be extended. For example, if f and g have limits as (x, y) approaches (a, b), then it can be proved that

$$\lim_{(x, y)\to(a, b)} [f(x, y) + (g(x, y)] = \lim_{(x, y)\to(a, b)} f(x, y) + \lim_{(x, y)\to(a, b)} g(x, y).$$

A function f of two variables is a **polynomial function** if $f(x, y)$ can be expressed as a sum of terms of the form $cx^m y^n$, where c is a real number and m and n are nonnegative integers. A **rational function** is a quotient of two polynomial functions. As in the single variable case, it can be shown that limits of such functions may be found by substituting for x and y. For example,

$$\begin{aligned}\lim_{(x, y)\to(2, -3)} (x^3 - 4xy^2 + 5y - 7) &= 2^3 - 4(2)(-3)^2 + 5(-3) - 7 \\ &= 8 - 72 - 15 - 7 = -86.\end{aligned}$$

The next example is an illustration of a function that has no limit as (x, y) approaches $(0, 0)$.

Example 1 Show that $\displaystyle\lim_{(x, y)\to(0, 0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Solution If $f(x, y) = (x^2 - y^2)/(x^2 + y^2)$, then f is defined everywhere except the origin $(0, 0)$. If we consider any point $(x, 0)$ on the x-axis, then $f(x, 0) = x^2/x^2 = 1$, provided $x \neq 0$. For points $(0, y)$ on the y-axis, we have $f(0, y) = -y^2/y^2 = -1$, if $y \neq 0$. Consequently, as illustrated in Figure 16.14,

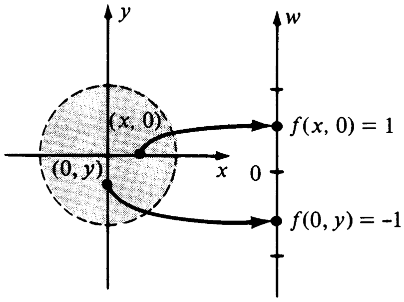


FIGURE 16.14

every circle with center (0, 0) contains points at which the value of f is 1 and points at which the value of f is -1. It follows that the limit does not exist, for if we take $\varepsilon = 1$, there is no open interval $(L - \varepsilon, L + \varepsilon)$ on the w-axis containing both 1 and -1, and hence it is impossible to find a $\delta > 0$ that satisfies the conditions of Definition (16.3). ■

For certain functions in Chapter 2 we proved that $\lim_{x\to a} f(x)$ did not exist by showing that $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ were not equal. When considering such one-sided limits we may regard the point on the x-axis with coordinate x as "approaching" the point with coordinate a either from the left or from the right, respectively. The analogous situation for functions of two variables is more complicated, since in a coordinate plane there are an infinite number of different curves (which we shall call **paths**) along which (x, y) can approach (a, b). However, if the limit in Definition (16.3) exists, then $f(x, y)$ must have the limit L regardless of the path taken. This illustrates the following important rule for investigating certain limits.

(16.4) If two different paths to a point $P(a, b)$ produce two different limiting values for f, then the limit of $f(x, y)$ as (x, y) approaches (a, b) does not exist.

Example 2 Rework Example 1 using (16.4).

Solution If, on the one hand, (x, y) approaches (0, 0) along the x-axis, then the y-coordinate is always zero and the expression $(x^2 - y^2)/(x^2 + y^2)$ reduces to x^2/x^2 or 1. Hence the limiting value is 1. On the other hand, if we let (x, y) approach (0, 0) along the y-axis, then the x-coordinate is 0 and $(x^2 - y^2)/(x^2 + y^2)$ reduces to $-y^2/y^2$ or -1. Since two different limits are obtained, the given limit does not exist by (16.4). We could, of course, have chosen other paths to the origin (0, 0). For example, if we let (x, y) approach (0, 0) along the line $y = 2x$, then

$$\frac{x^2 - y^2}{x^2 + y^2} = \frac{x^2 - 4x^2}{x^2 + 4x^2} = \frac{-3x^2}{5x^2} = -\frac{3}{5}$$

and hence the limiting value would be $-\frac{3}{5}$. ■

Example 3 If $f(x, y) = \dfrac{x^2 y}{x^4 + y^2}$ show that $\displaystyle\lim_{(x, y)\to(0, 0)} f(x, y)$ does not exist.

Solution If we let (x, y) approach (0, 0) along any line $y = mx$ that passes through the origin, we see that if $m \neq 0$,

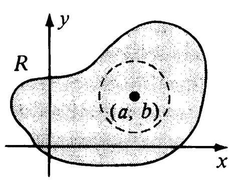
$$\lim_{(x, y)\to(0, 0)} \frac{x^2 y}{x^4 + y^2} = \lim_{(x, y)\to(0, 0)} \frac{x^2(mx)}{x^4 + (mx)^2} = \lim_{(x, y)\to(0, 0)} \frac{mx^3}{x^4 + m^2x^2}$$

$$= \lim_{(x, y)\to(0, 0)} \frac{mx}{x^2 + m^2} = \frac{0}{0 + m^2} = 0.$$

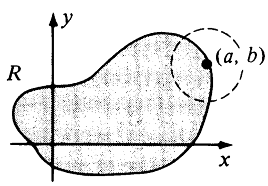
Because of this fact it is tempting to conclude that $f(x, y)$ has the limit 0 as (x, y) approaches (0, 0). However, if we let (x, y) approach (0, 0) along the parabola $y = x^2$, then

$$\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^4 + y^2} = \lim_{(x,y)\to(0,0)} \frac{x^2(x^2)}{x^4 + (x^2)^2}$$
$$= \lim_{(x,y)\to(0,0)} \frac{x^4}{2x^4} = \lim_{(x,y)\to(0,0)} \frac{1}{2} = \frac{1}{2}.$$

Thus not every path through (0, 0) leads to the same limiting value of $f(x, y)$, and hence by (16.4), the limit does not exist. ■



(i) Interior point of R



(ii) Boundary point of R

FIGURE 16.15

If R is a region in the xy-plane, then a point (a, b) is called an **interior point** of R if there exists a circle with center (a, b) that contains only points of R. An illustration of an interior point is shown in (i) of Figure 16.15. A point (a, b) is called a **boundary point** of R if *every* circle with center (a, b) contains points that are in R and points that are not in R, as illustrated in (ii) of Figure 16.15.

A region is called **closed** if it contains all of its boundary points. A region is **open** if it contains *none* of its boundary points; that is, every point of the region is an interior point. A region that contains some, but not all, of its boundary points is neither open nor closed. These concepts are analogous to closed intervals, open intervals, and half-open intervals of real numbers.

Suppose a function f of two variables is defined for all (x, y) in a region R, except possibly at (a, b). If (a, b) is an interior point, then Definition (16.3) can be used to investigate the limit of f as (x, y) approaches (a, b). *If (a, b) is a boundary point we shall use Definition* (16.3) *with the added restriction that (x, y) must be both in R and in the circle of radius δ.* Theorems on limits may then be extended to boundary points.

A function f of two variables is **continuous** at an interior point (a, b) of a region R if $f(a, b)$ exists, $f(x, y)$ has a limit as (x, y) approaches (a, b), and

$$\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b).$$

This notion can be extended to a boundary point (a, b) of R by applying the previous remarks on limits concerning boundary points. If R is in the domain D of f, then f is said to be **continuous on R** if it is continuous at every pair (a, b) in R. If f is continuous on R, then a small change in (x, y) produces a small change in $f(x, y)$. Referring to the graph S of f, if (x, y) is close to (a, b), then the point $(x, y, f(x, y))$ on S is close to $(a, b, f(a, b))$. Thus there are no holes or vertical steps in the graph of a continuous function.

Theorems on continuity that are analogous to those for functions of one variable may be proved. In particular, polynomial functions are continuous everywhere and rational functions are continuous except at points where the denominator vanishes.

The preceding discussion on limits and continuity can be extended to functions of three or more variables. For example, if f is a function of three variables we may state the following definition.

Definition (16.5)

Let a function f of three variables be defined throughout the interior of a sphere with center (a, b, c), except possibly at (a, b, c) itself. The **limit of $f(x, y, z)$ as (x, y, z) approaches (a, b, c) is L**, written

$$\lim_{(x, y, z) \to (a, b, c)} f(x, y, z) = L$$

means that for every $\varepsilon > 0$ there corresponds a $\delta > 0$ such that if

$$0 < \sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2} < \delta,$$

then $$|f(x, y, z) - L| < \varepsilon.$$

We may give a graphical interpretation for Definition (16.5) that is similar to that illustrated in Figure 16.13. The difference is that in place of the circle in the xy-plane, we use a sphere of radius δ with center at the point (a, b, c) in three dimensions (see Exercise 25).

An *interior point* or *boundary point* P of a three-dimensional region R is defined as in two dimensions, except that we use a *sphere* with center P instead of a circle. The limit definition (16.5) may be extended to the case where (a, b, c) is a boundary point by adding the restriction that (x, y, z) is both in R and in the sphere of radius δ.

A function f of three variables is *continuous* at an interior point (a, b, c) of a region if $f(a, b, c)$ exists, $f(x, y, z)$ has a limit as (x, y, z) approaches (a, b, c), and

$$\lim_{(x, y, z) \to (a, b, c)} f(x, y, z) = f(a, b, c).$$

Continuity at a boundary point is defined by using the extension of Definition (16.5) to boundary points.

There is no essential difference in defining limits for functions of four or more variables; however, in these cases we no longer give geometric interpretations (see Exercise 27).

In Section 16.5 we shall discuss composite functions of several variables. As a simple illustration, if f is a function of two variables and g is a function of one variable, then a function h of two variables may be defined by letting $h(x, y) = g(f(x, y))$, provided the range of f is in the domain of g.

Example 4 In each of the following, express $g(f(x, y))$ in terms of x and y, and find the domain of the resulting composite function.

(a) $f(x, y) = xe^y$, $g(t) = 3t^2 + t + 1$

(b) $f(x, y) = y - 4x^2$, $g(t) = \sin\sqrt{t}$

Solution In each case we substitute $f(x, y)$ for t. Thus:

(a) $g(f(x, y)) = g(xe^y) = 3x^2e^{2y} + xe^y + 1.$

(b) $g(f(x, y)) = g(y - 4x^2) = \sin\sqrt{y - 4x^2}.$

The domain in part (a) is $\mathbb{R} \times \mathbb{R}$, and the domain in part (b) consists of all ordered pairs (x, y) such that $y \geq 4x^2$. ■

The following analogue of (2.28) may be proved for functions of the type illustrated in Example 4.

Theorem (16.6)

> If a function f of two variables is continuous at (a, b), and a function g of one variable is continuous at $f(a, b)$, then the function h defined by $h(x, y) = g(f(x, y))$ is continuous at (a, b).

Theorem (16.6) is important because it provides a technique for establishing the continuity of certain functions by using previously proved results, as illustrated in the next example.

Example 5 If $h(x, y) = e^{x^2+5xy+y^3}$, show that h is continuous at every pair (a, b).

Solution If we let $f(x, y) = x^2 + 5xy + y^3$ and $g(t) = e^t$, it follows that $h(x, y) = g(f(x, y))$. Since f is a polynomial function, it is continuous at every pair (a, b). Moreover, g is continuous at every $t = f(a, b)$. Thus, by Theorem (16.6), h is continuous at (a, b). ■

16.2 Exercises

In Exercises 1–10 find the limits, if they exist.

1 $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2 - 2}{3 + xy}$

2 $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^3 - x^2y + xy^2 - y^3}{x^2 + y^2}$

3 $\displaystyle\lim_{(x,y)\to(0,0)} \frac{2x^2 - y^2}{x^2 + 2y^2}$

4 $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2 - 2xy + 5y^2}{3x^2 + 4y^2}$

5 $\displaystyle\lim_{(x,y)\to(0,0)} \frac{4x^2y}{x^3 + y^3}$

6 $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^4 - y^4}{x^2 + y^2}$

7 $\displaystyle\lim_{(x,y)\to(1,2)} \frac{xy - 2x - y + 2}{x^2 + y^2 - 2x - 4y + 5}$

8 $\displaystyle\lim_{(x,y)\to(0,0)} \frac{3xy}{5x^4 + 2y^4}$

9 $\displaystyle\lim_{(x,y)\to(0,0)} \frac{3x^3 - 2x^2y + 3y^2x - 2y^3}{x^2 + y^2}$

10 $\displaystyle\lim_{(x,y,z)\to(0,0,0)} \frac{xy + yz + xz}{x^2 + y^2 + z^2}$

In Exercises 11–16 discuss the continuity of the function f.

11 $f(x, y) = \ln(x + y - 1)$

12 $f(x, y) = xy/(x^2 - y^2)$

13 $f(x, y, z) = 1/(x^2 + y^2 - z^2)$

14 $f(x, y, z) = \sqrt{xy}\tan z$

15 $f(x, y) = \sqrt{x}e^{\sqrt{1-y^2}}$

16 $f(x, y) = \sqrt{25 - x^2 - y^2}$

In Exercises 17–20 find $h(x, y) = g(f(x, y))$ and use Theorem (16.6) to determine where h is continuous.

17 $f(x, y) = x^2 - y^2, g(t) = (t^2 - 4)/t$

18 $f(x, y) = 3x + 2y - 4, g(t) = \ln(t + 5)$

19 $f(x, y) = x + \tan y, g(z) = z^2 + 1$

20 $f(x, y) = y \ln x, g(w) = e^w$

21 If $f(x, y) = x^2 + 2y$, $g(t) = e^t$, and $h(t) = t^2 - 3t$, find $g(f(x, y))$, $h(f(x, y))$, and $f(g(t), h(t))$.

22 If $f(x, y, z) = 2x + ye^z$ and $g(t) = t^2$, find $g(f(x, y, z))$.

23 If $f(u, v) = uv - 3u + v$, $g(x, y) = x - 2y$, and $k(x, y) = 2x + y$, find $f(g(x, y), k(x, y))$.

24 If $f(x, y) = 2x + y$, find $f(f(x, y), f(x, y))$.

25 Give a graphical interpretation of Definition (16.5).

26 Prove that if the limit L in Definition (16.3) exists, then it is unique.

27 Extend Definition (16.3) to functions of four variables.

28 Prove that if f is a continuous function of two variables and $f(a, b) > 0$, then there is a circle C in the xy-plane with center (a, b) such that $f(x, y) > 0$ for all pairs (x, y) that are in the domain of f and within C.

29 Prove, directly from Definition (16.3), that

(a) $\lim_{(x, y)\to(a, b)} x = a.$ (b) $\lim_{(x, y)\to(a, b)} y = b.$

30 If $\lim_{(x, y)\to(a, b)} f(x, y) = L$ and c is any real number, prove, directly from Definition (16.3), that

$$\lim_{(x, y)\to(a, b)} cf(x, y) = cL.$$

16.3 Partial Derivatives

In this section we begin the study of derivatives of functions of several variables.

Definition (16.7)

If f is a function of two variables, then the **first partial derivatives of f with respect to x and y** are the functions f_x and f_y defined as follows:

$$f_x(x, y) = \lim_{h\to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

$$f_y(x, y) = \lim_{h\to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

provided the limits exist.

Note that in the definition of $f_x(x, y)$, the variable y is fixed. It is for this reason that the notation for limits of functions of one variable is used instead of that introduced in the previous section. Indeed, if we let $y = b$ and define g by $g(x) = f(x, b)$, then $g'(x) = f_x(x, b) = f_x(x, y)$. Thus, to find $f_x(x, y)$ we may regard y as a constant and differentiate $f(x, y)$ with respect to x in the usual way. Similarly, to find $f_y(x, y)$ the variable x is regarded as a constant and $f(x, y)$ is differentiated with respect to y.

Other common notations for partial derivatives are

$$f_x = \frac{\partial f}{\partial x} \quad \text{and} \quad f_y = \frac{\partial f}{\partial y}.$$

For brevity we often speak of $\partial f/\partial x$ or $\partial f/\partial y$ as *the partial of f with respect to x or y*, respectively.

If $w = f(x, y)$ we write

$$f_x(x, y) = \frac{\partial}{\partial x} f(x, y) = \frac{\partial w}{\partial x} = w_x$$

$$f_y(x, y) = \frac{\partial}{\partial y} f(x, y) = \frac{\partial w}{\partial y} = w_y.$$

Example 1 If $f(x, y) = x^3y^2 - 2x^2y + 3x$, find
(a) $f_x(x, y)$ and $f_y(x, y)$ (b) $f_x(2, -1)$ and $f_y(2, -1)$.

Solution
(a) Regarding y as a constant and differentiating with respect to x, we obtain

$$f_x(x, y) = 3x^2y^2 - 4xy + 3.$$

Regarding x as a constant and differentiating with respect to y gives us

$$f_y(x, y) = 2x^3y - 2x^2.$$

(b) Using the results of (a),

$$f_x(2, -1) = 3(2)^2(-1)^2 - 4(2)(-1) + 3 = 23$$

$$f_y(2, -1) = 2(2)^3(-1) - 2(2)^2 = -24.$$

■

Formulas such as the Product Rule, Quotient Rule, and Power Rule are true for partial derivatives. For example, if u and v are functions of x and y, and n is a real number,

$$\frac{\partial}{\partial x}(uv) = u\frac{\partial v}{\partial x} + v\frac{\partial u}{\partial x}, \qquad \frac{\partial}{\partial x}\left(\frac{u}{v}\right) = \frac{v\dfrac{\partial u}{\partial x} - u\dfrac{\partial v}{\partial x}}{v^2},$$

$$\frac{\partial}{\partial x}(u^n) = nu^{n-1}\frac{\partial u}{\partial x}, \qquad \frac{\partial}{\partial x}\cos u = -\sin u\frac{\partial u}{\partial x}.$$

Example 2 Find $\dfrac{\partial w}{\partial y}$ if $w = xy^2e^{xy}$.

Solution We may proceed as follows.

$$\begin{aligned}\frac{\partial w}{\partial y} &= xy^2\frac{\partial}{\partial y}(e^{xy}) + e^{xy}\frac{\partial}{\partial y}(xy^2)\\ &= xy^2(xe^{xy}) + e^{xy}(2xy) = (xy + 2)xye^{xy}\end{aligned}$$

■

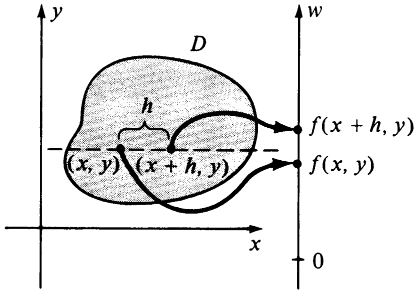


FIGURE 16.16

It is instructive to interpret Definition (16.7) by means of the illustration given at the beginning of the previous section, where $f(x, y)$ is the temperature of a flat metal plate at the point (x, y). Note that the points (x, y) and $(x + h, y)$ lie on a horizontal line, as indicated in Figure 16.16. In this case the difference $f(x + h, y) - f(x, y)$ is the change in temperature as a point moves from (x, y) to $(x + h, y)$, and the ratio

$$\frac{f(x + h, y) - f(x, y)}{h}$$

is the *average* change in temperature. For example, if the temperature change is 2 degrees and h is 4, then the average change in temperature is $\frac{2}{4}$, or $\frac{1}{2}$. Thus, *on the average*, the temperature changes at a rate of $\frac{1}{2}$ degree per unit change in distance. Taking the limit of the average change as h approaches 0 gives us the *rate of change* of temperature with respect to distance as the point (x, y) moves in a horizontal direction. Similarly, $f_y(x, y)$ measures the rate of change of $f(x, y)$ as (x, y) moves in a vertical direction.

As another illustration, suppose D represents the surface of a lake and $f(x, y)$ is the depth of the water under the point (x, y) on the surface. In this case $f_x(x, y)$ is the rate at which the depth changes as a point moves away from (x, y) parallel to the x-axis. Similarly, $f_y(x, y)$ is the rate of change of the depth in the direction of the y-axis.

There is an interesting geometric interpretation for Definition (16.7) in terms of the graph of f. In order to illustrate the meaning of $f_y(x, y)$, consider the points $M(x, y, 0)$ and $N(x, y + h, 0)$. The plane Γ, parallel to the yz-plane and passing through M and N, intersects S in a curve C. If we introduce a coordinate system on Γ, and if P and Q are the points on C that have projections M and N on the xy-plane, then as illustrated in Figure 16.17, $\overline{MP} = f(x, y)$, $\overline{NQ} = f(x, y + h)$, and $\overline{MN} = h$. It follows that the ratio $[f(x, y + h) - f(x, y)]/h$ is the slope of the secant line through P and Q in the plane Γ. The limit of this ratio as h approaches 0, namely $f_y(x, y)$, is the slope of the tangent line l to C at P. Similarly, if C' is the trace of S on the plane Γ', parallel to the xz-plane and passing through M, then $f_x(x, y)$ is the slope of the tangent line to C' at P.

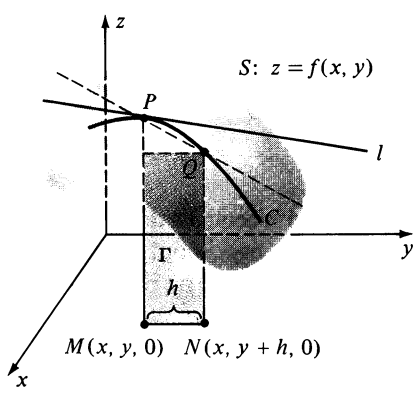


FIGURE 16.17

First partial derivatives of functions of three or more variables are defined in the same manner as that used in Definition (16.7). Specifically, all variables except one are held constant and differentiation takes place with respect to the remaining variable. Thus, given $f(x, y, z)$ we may find f_x, f_y, and f_z (or equivalently $\partial f/\partial x$, $\partial f/\partial y$, and $\partial f/\partial z$). For example,

$$f_z(x, y, z) = \lim_{h \to 0} \frac{f(x, y, z + h) - f(x, y, z)}{h}.$$

In a manner similar to our discussion in two dimensions, if $f(x, y, z)$ is the temperature at the point $P(x, y, z)$, then $f_x(x, y, z)$ is the rate of change of temperature with respect to distance along a line through P that is parallel to the x-axis. The partial derivatives $f_y(x, y, z)$ and $f_z(x, y, z)$ give the rates of change in the directions of the y- and z-axes, respectively.

Example 3 If $w = x^2y^3 \sin z + e^{xz}$, find $\partial w/\partial x$, $\partial w/\partial y$, and $\partial w/\partial z$.

Solution

$$\frac{\partial w}{\partial x} = 2xy^3 \sin z + ze^{xz}$$

$$\frac{\partial w}{\partial y} = 3x^2y^2 \sin z$$

$$\frac{\partial w}{\partial z} = x^2y^3 \cos z + xe^{xz} \qquad \blacksquare$$

If f is a function of two variables x and y, then f_x and f_y are also functions of two variables and we may consider *their* first partial derivatives. These are called the **second partial derivatives** of f and are denoted as follows.

$$\frac{\partial}{\partial x} f_x = (f_x)_x = f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2}$$

$$\frac{\partial}{\partial y} f_x = (f_x)_y = f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x}$$

$$\frac{\partial}{\partial x} f_y = (f_y)_x = f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y}$$

$$\frac{\partial}{\partial y} f_y = (f_y)_y = f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2}$$

If $w = f(x, y)$ we write

$$\frac{\partial^2}{\partial x^2} f(x, y) = f_{xx}(x, y) = \frac{\partial^2 w}{\partial x^2} = w_{xx}$$

and likewise for the other second partial derivatives.

It is customary to refer to f_{xy} and f_{yx} as the **mixed second partial derivatives of f** (or simply *mixed partials of f*). The next theorem brings out the fact that under suitable conditions, these mixed partials are equal. The proof may be found in texts on advanced calculus.

Theorem (16.8)

> Let f be a function of x and y. If f, f_x, f_y, f_{xy}, and f_{yx} are continuous on an open region R, then
>
> $$f_{xy} = f_{yx}$$
>
> throughout R.

The hypothesis of Theorem (16.8) is satisfied for most functions encountered in calculus and its applications. Similarly, if $w = f(x, y, z)$ and f

has continuous second partial derivatives, then it can be proved that the following equalities hold for mixed partials:

$$\frac{\partial^2 w}{\partial y\,\partial x} = \frac{\partial^2 w}{\partial x\,\partial y}; \qquad \frac{\partial^2 w}{\partial z\,\partial x} = \frac{\partial^2 w}{\partial x\,\partial z}; \qquad \frac{\partial^2 w}{\partial z\,\partial y} = \frac{\partial^2 w}{\partial y\,\partial z}.$$

Example 4 Find the second partial derivatives of f if

$$f(x, y) = x^3y^2 - 2x^2y + 3x.$$

Solution This function was considered in Example 1. Since the first partials of f are $f_x(x, y) = 3x^2y^2 - 4xy + 3$ and $f_y(x, y) = 2x^3y - 2x^2$,

$$f_{xx}(x, y) = \frac{\partial}{\partial x}(3x^2y^2 - 4xy + 3) = 6xy^2 - 4y$$

$$f_{xy}(x, y) = \frac{\partial}{\partial y}(3x^2y^2 - 4xy + 3) = 6x^2y - 4x$$

$$f_{yx}(x, y) = \frac{\partial}{\partial x}(2x^3y - 2x^2) = 6x^2y - 4x$$

$$f_{yy}(x, y) = \frac{\partial}{\partial y}(2x^3y - 2x^2) = 2x^3.$$

■

Third and higher partial derivatives are defined in like manner. For example,

$$\frac{\partial}{\partial x} f_{xx} = f_{xxx} = \frac{\partial}{\partial x}\left(\frac{\partial^2 f}{\partial x^2}\right) = \frac{\partial^3 f}{\partial x^3}$$

$$\frac{\partial}{\partial x} f_{xy} = f_{xyx} = \frac{\partial}{\partial x}\left(\frac{\partial^2 f}{\partial y\,\partial x}\right) = \frac{\partial^3 f}{\partial x\,\partial y\,\partial x},$$

and so on. If first, second, and third partial derivatives are continuous, then again it can be shown that the order of differentiation is immaterial, that is,

$$f_{xyx} = f_{yxx} = f_{xxy} \quad \text{and} \quad f_{yxy} = f_{xyy} = f_{yyx}.$$

Similar notations and results apply to functions of more than two variables. Of course, letters other than x and y may be used. To illustrate, if f is a function of r and s, then symbols such as

$$f_r(r, s), \quad f_s(r, s), \quad f_{rs}, \quad \frac{\partial f}{\partial r}, \quad \frac{\partial^2 f}{\partial r^2}$$

are employed for partial derivatives.

16.3 Exercises

In Exercises 1–18 find the first partial derivatives of f.

1 $f(x, y) = 2x^4y^3 - xy^2 + 3y + 1$

2 $f(x, y) = (x^3 - y^2)^2$

3 $f(r, s) = \sqrt{r^2 + s^2}$

4 $f(s, t) = t/s - s/t$

5 $f(x, y) = xe^y + y \sin x$

6 $f(x, y) = e^x \ln xy$

7 $f(t, v) = \ln \sqrt{(t + v)/(t - v)}$

8 $f(u, w) = \arctan (u/w)$

9 $f(x, y) = x \cos (x/y)$

10 $f(x, y) = \sqrt{4x^2 - y^2} \sec x$

11 $f(r, s, t) = r^2e^{2s} \cos t$

12 $f(x, y, t) = (x^2 - t^2)/(1 + \sin 3y)$

13 $f(x, y, z) = (y^2 + z^2)^x$

14 $f(r, s, v) = (2r + 3s)^{\cos v}$

15 $f(x, y, z) = xe^z - ye^x + ze^{-y}$

16 $f(r, s, v, p) = r^3 \tan s + \sqrt{s}e^{v^2} - v \cos 2p$

17 $f(q, v, w) = \sin^{-1} \sqrt{qv} + \sin vw$

18 $f(x, y, z) = xyze^{xyz}$

In Exercises 19–24 verify that $w_{xy} = w_{yx}$.

19 $w = xy^4 - 2x^2y^3 + 4x^2 - 3y$

20 $w = x^2/(x + y)$

21 $w = x^3e^{-2y} + y^{-2} \cos x$

22 $w = y^2e^{x^2} + (1/x^2y^3)$

23 $w = x^2 \cosh (z/y)$

24 $w = \sqrt{x^2 + y^2 + z^2}$

25 If $w = 3x^2y^3z + 2xy^4z^2 - yz$, find w_{xyz}.

26 If $w = u^4vt^2 - 3uv^2t^3$, find w_{tut}.

27 If $u = v \sec rt$, find u_{rvr}.

28 If $v = y \ln (x^2 + z^4)$, find v_{zzy}.

29 If $w = \sin xyz$ find $\partial^3 w/\partial z\, \partial y\, \partial x$.

30 If $w = x^2/(y^2 + z^2)$, find $\partial^3 w/\partial z\, \partial y^2$.

31 If $w = r^4s^3t - 3s^2e^{rt}$, verify that $w_{rrs} = w_{rsr} = w_{srr}$.

32 If $w = \tan uv + 2 \ln (u + v)$, verify that $w_{uvv} = w_{vuv} = w_{vvu}$.

A function f of x and y is **harmonic** if $\partial^2 f/\partial x^2 + \partial^2 f/\partial y^2 = 0$. Prove that the functions defined in Exercises 33–36 are harmonic.

33 $f(x, y) = \ln \sqrt{x^2 + y^2}$

34 $f(x, y) = \arctan (y/x)$

35 $f(x, y) = \cos x \sinh y + \sin x \cosh y$

36 $f(x, y) = e^{-x} \cos y + e^{-y} \cos x$

37 If $w = \cos (x - y) + \ln (x + y)$, show that $\partial^2 w/\partial x^2 - \partial^2 w/\partial y^2 = 0$.

38 If $w = (y - 2x)^3 - \sqrt{y - 2x}$, show that $w_{xx} - 4w_{yy} = 0$.

39 If $w = e^{-c^2t} \sin cx$, show that $w_{xx} = w_t$ for every real number c.

40 The Ideal Gas Law may be stated as $PV = knT$, where n is the number of moles of gas, V is the volume, T is the temperature, P is the pressure, and k is a constant. Show that

$$\frac{\partial V}{\partial T}\frac{\partial T}{\partial P}\frac{\partial P}{\partial V} = -1.$$

In Exercises 41 and 42 show that v satisfies the **wave equation**

$$\frac{\partial^2 v}{\partial t^2} = a^2 \frac{\partial^2 v}{\partial x^2}.$$

41 $v = (\sin akt)(\sin kx)$

42 $v = (x - at)^4 + \cos (x + at)$

Two functions u and v of x and y are said to satisfy the *Cauchy–Riemann equations* if $u_x = v_y$ and $u_y = -v_x$. Show that the pairs of functions in Exercises 43–46 satisfy these equations.

43 $u(x, y) = x^2 - y^2$, $v(x, y) = 2xy$

44 $u(x, y) = y/(x^2 + y^2)$, $v(x, y) = x/(x^2 + y^2)$

45 $u(x, y) = e^x \cos y$, $v(x, y) = e^x \sin y$

46 $u(x, y) = \cos x \cosh y + \sin x \sinh y$
$v(x, y) = \cos x \cosh y - \sin x \sinh y$

47 List all possible second partial derivatives of $w = f(x, y, z)$.

48 If $w = f(x, y, z, t, v)$, define w_t as a limit.

49 If $w = f(x, y, z)$ and f has continuous third partial derivatives, how many of them can be different?

50 If $w = f(x, y)$ and f has continuous fourth partial derivatives, how many of them can be different?

51 A flat metal plate is situated on an xy-plane such that the temperature T at the point (x, y) is given by $T = 10(x^2 + y^2)^2$, where T is in degrees and x, y are in centimeters. Find the rate of change of T with respect to distance at $P(1, 2)$ in the direction of (a) the x-axis; (b) the y-axis.

52 The surface of a certain lake is represented by a region D in the xy-plane such that the depth under the point corresponding to (x, y) is given by $f(x, y) = 300 - 2x^2 - 3y^2$, where x, y and $f(x, y)$ are in feet. If a boy in the water is at the point $(4, 9)$, find the rate at which the depth changes if he swims in the direction of (a) the x-axis; (b) the y-axis.

53 Suppose the electrical potential V at the point (x, y, z) is given by $V = 100/(x^2 + y^2 + z^2)$, where V is in volts and x, y, z are in inches. Find the rate of change of V with respect to distance at $P(2, -1, 1)$ in the direction of (a) the x-axis; (b) the y-axis; (c) the z-axis.

54 An object is situated in a rectangular coordinate system such that the temperature T at the point $P(x, y, z)$ is given by $T = 4x^2 - y^2 + 16z^2$, where T is in degrees and x, y, z are in centimeters. Find the rate of change of T at the point $P(4, -2, 1)$ in the direction of (a) the x-axis; (b) the y-axis; (c) the z-axis.

55 Let C be the trace of the paraboloid $z = 9 - x^2 - y^2$ on the plane $x = 1$. Find parametric equations of the tangent line l to C at the point $P(1, 2, 4)$. Sketch the paraboloid, C, and l. (*Hint*: See Figure 16.17.)

56 Let C be the trace of the graph of $z = \sqrt{36 - 9x^2 - 4y^2}$ on the plane $y = 2$. Find parametric equations of the tangent line l to C at the point $(1, 2, \sqrt{11})$. Sketch the surface, C, and l.

16.4 Increments and Differentials

If f is a function of two variables x and y, then the symbols Δx and Δy will denote increments of x and y. It should be observed that in the present situation Δy is an increment of the *independent* variable y and is not the same as that defined in (3.22), where y was a *dependent* variable. In terms of this increment notation, Definition (16.7) may be written

$$f_x(x, y) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$f_y(x, y) = \lim_{\Delta y \to 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}.$$

The increment of $w = f(x, y)$ is defined as follows.

Definition (16.9)

Let $w = f(x, y)$ and let Δx and Δy be increments of x and y, respectively. The **increment Δw of $w = f(x, y)$** is

$$\Delta w = f(x + \Delta x, y + \Delta y) - f(x, y).$$

Note that the increment Δw represents the change in the value of f if (x, y) changes to $(x + \Delta x, y + \Delta y)$.

Example 1 Let $w = f(x, y) = 3x^2 - xy$.

(a) Find Δw.

(b) Use Δw to calculate the change in $f(x, y)$ if (x, y) changes from $(1, 2)$ to $(1.01, 1.98)$.

Solution

(a) From Definition (16.9),

$$\begin{aligned}\Delta w &= [3(x + \Delta x)^2 - (x + \Delta x)(y + \Delta y)] - [3x^2 - xy]\\ &= [3x^2 + 6x\,\Delta x + 3(\Delta x)^2 - (xy + x\,\Delta y + y\,\Delta x + \Delta x\,\Delta y)] - 3x^2 + xy\\ &= 6x\,\Delta x + 3(\Delta x)^2 - x\,\Delta y - y\,\Delta x - \Delta x\,\Delta y.\end{aligned}$$

(b) To find the change in $f(x, y)$ we substitute $x = 1$, $y = 2$, $\Delta x = 0.01$, $\Delta y = -0.02$ in the formula for Δw, obtaining

$$\begin{aligned}\Delta w &= 6(1)(0.01) + 3(0.01)^2 - (1)(-0.02) - 2(0.01) - (0.01)(-0.02)\\ &= 0.0605.\end{aligned}$$

This number can also be found by calculating $f(1.01, 1.98) - f(1, 2)$. ■

Our first objective in this section is to obtain an alternative formula for the increment Δw in Definition (16.9). The reason for doing so will become apparent later, when we prove several important results that have many applications, both in mathematics and in the real world. To gain some insight into the type of formula we should expect, let us consider a differentiable function f of *one* variable x, and set $u = f(x)$. Let x_0 be a fixed value of x. If Δx is an increment of x, then by Definition (3.22),

$$\Delta u = f(x_0 + \Delta x) - f(x_0).$$

Since

$$\lim_{\Delta x \to 0} \frac{\Delta u}{\Delta x} = f'(x_0),$$

it follows that

$$\lim_{\Delta x \to 0} \left[\frac{\Delta u}{\Delta x} - f'(x_0)\right] = 0.$$

If we define ε by

$$\varepsilon = \frac{\Delta u}{\Delta x} - f'(x_0)$$

then ε is a function of Δx, and $\varepsilon \to 0$ as $\Delta x \to 0$. (Strictly speaking, we should use the symbol $\varepsilon(\Delta x)$ to denote the fact that ε depends on Δx.) It follows from the definition of ε that

$$\Delta u = f'(x_0)\,\Delta x + \varepsilon\,\Delta x.$$

This gives us an alternative formula for the increment of a function of *one* variable.

Suppose next that f is a function of *two* variables, and let $w = f(x, y)$. Suppose further that f_x and f_y exist at (x_0, y_0). Consider increments Δx, Δy, and, as in Definition (16.9), let

$$\Delta w = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0).$$

From the alternative formula for Δu obtained in the preceding paragraph, it is not unreasonable to expect that Δw may be written in the form

$$\Delta w = f_x(x_0, y_0)\,\Delta x + \varepsilon_1\,\Delta x + f_y(x_0, y_0)\,\Delta y + \varepsilon_2\,\Delta y$$

where the symbols ε_1 and ε_2 denote functions of Δx and Δy that have the limit 0 as $(\Delta x, \Delta y) \to (0, 0)$. The next theorem establishes this fact, provided that suitable restrictions are placed on f. As indicated previously, we shall use the abbreviations ε_1 and ε_2 in place of the more cumbersome notation $\varepsilon_1(\Delta x, \Delta y)$ and $\varepsilon_2(\Delta x, \Delta y)$.

Theorem (16.10)

Let $w = f(x, y)$, where the function f is defined on a rectangular region $R = \{(x, y): a < x < b, c < y < d\}$. Suppose f_x and f_y exist throughout R and are continuous at the pair (x_0, y_0) in R. If $(x_0 + \Delta x, y_0 + \Delta y)$ is in R and $\Delta w = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$, then

$$\Delta w = f_x(x_0, y_0)\,\Delta x + f_y(x_0, y_0)\,\Delta y + \varepsilon_1\,\Delta x + \varepsilon_2\,\Delta y$$

where ε_1 and ε_2 are functions of Δx and Δy that have the limit 0 as $(\Delta x, \Delta y) \to (0, 0)$.

Proof The graph of R and the pairs we wish to consider are illustrated in Figure 16.18.

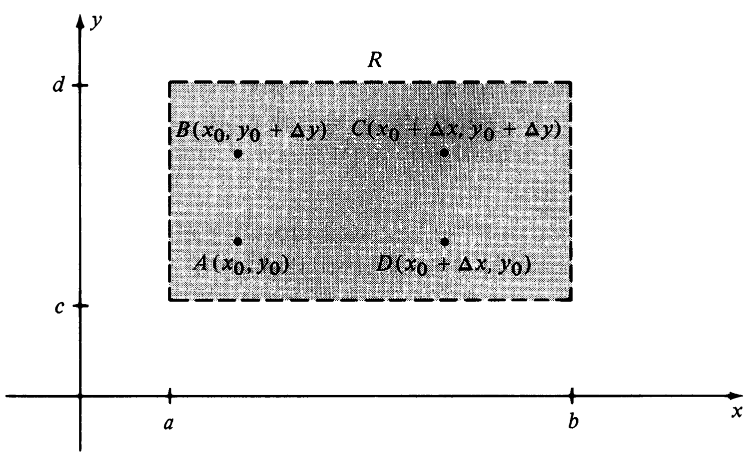


FIGURE 16.18

We first note that the increment Δw of (16.9) may be expressed as

$$\Delta w = [f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y)] \\ + [f(x_0, y_0 + \Delta y) - f(x_0, y_0)].$$

If we next define a function g of *one* variable x by $g(x) = f(x, y_0 + \Delta y)$, then $g'(x) = f_x(x, y_0 + \Delta y)$ for all x under consideration. Applying the Mean Value Theorem (4.12) to g we obtain

$$g(x_0 + \Delta x) - g(x_0) = g'(u)\,\Delta x$$

where u is between x_0 and $x_0 + \Delta x$. Using the definition of g, the last equation can be written

$$f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y) = f_x(u, y_0 + \Delta y)\,\Delta x.$$

Similarly, if we let $h(y) = f(x_0, y)$ and apply the Mean Value Theorem to the function h, we obtain

$$h(y_0 + \Delta y) - h(y_0) = h'(v)\,\Delta y$$

where v is between y_0 and $y_0 + \Delta y$. Thus,

$$f(x_0, y_0 + \Delta y) - f(x_0, y_0) = f_y(x_0, v)\,\Delta y.$$

Substitution in the expression for Δw obtained at the beginning of the proof gives us

$$\Delta w = f_x(u, y_0 + \Delta y)\,\Delta x + f_y(x_0, v)\,\Delta y.$$

Let us define ε_1 and ε_2 by

$$\varepsilon_1 = f_x(u, y_0 + \Delta y) - f_x(x_0, y_0)$$

$$\varepsilon_2 = f_y(x_0, v) - f_y(x_0, y_0).$$

Using the fact that f_x and f_y are continuous and noting that $u \to x_0$ and $v \to y_0$ as $\Delta x \to 0$ and $\Delta y \to 0$, respectively, it follows that ε_1 and ε_2 have the limit 0 as $(\Delta x, \Delta y) \to (0, 0)$. If we rewrite the preceding equations as

$$f_x(u, y_0 + \Delta y) = f_x(x_0, y_0) + \varepsilon_1$$

$$f_y(x_0, v) = f_y(x_0, y_0) + \varepsilon_2$$

and substitute into the last expression for Δw, we see that

$$\Delta w = [f_x(x_0, y_0) + \varepsilon_1]\,\Delta x + [f_y(x_0, y_0) + \varepsilon_2]\,\Delta y$$

which leads to the conclusion of the theorem. □

Example 2 If $w = 3x^2 - xy$, find expressions for ε_1 and ε_2 that satisfy the conclusion of Theorem 16.10.

Solution From the solution of Example 1,

$$\Delta w = 6x\,\Delta x + 3(\Delta x)^2 - x\,\Delta y - y\,\Delta x - \Delta x\,\Delta y$$

which may also be written

$$\Delta w = (6x - y)\,\Delta x + (-x)\,\Delta y + (3\,\Delta x)(\Delta x) + (-\Delta x)\,\Delta y.$$

If we define $\varepsilon_1 = 3\,\Delta x$ and $\varepsilon_2 = (-\Delta x)$, then the form in Theorem (16.10) is obtained. Note that ε_1 and ε_2 are not unique since we may also write

$$\Delta w = (6x - y)\,\Delta x + (-x)\,\Delta y + (3\,\Delta x - \Delta y)\,\Delta x + 0\,\Delta y$$

in which case $\varepsilon_1 = 3\,\Delta x - \Delta y$ and $\varepsilon_2 = 0$. ■

In Chapter 3 we defined differentials for functions of one variable. Recall that if $u = f(x)$ and Δx is an increment of x, then by (3.24) $dx = \Delta x$ and $du = f'(x)\,dx$. The next definition extends this concept to functions of two variables.

Definition (16.11)

Let $w = f(x, y)$, and let Δx and Δy be increments of x and y.

(i) The **differentials dx and dy of the independent variables** x and y are

$$dx = \Delta x \quad \text{and} \quad dy = \Delta y.$$

(ii) The **differential dw of the dependent variable** w is

$$dw = f_x(x, y)\,dx + f_y(x, y)\,dy = \frac{\partial w}{\partial x}\,dx + \frac{\partial w}{\partial y}\,dy.$$

If f satisfies the hypothesis of Theorem (16.10), then using the conclusion of that theorem, with the pair (x_0, y_0) replaced by (x, y), we see that

$$\Delta w - dw = \varepsilon_1\,\Delta x + \varepsilon_2\,\Delta y$$

where ε_1 and ε_2 approach 0 as $(\Delta x, \Delta y) \to (0, 0)$. It follows that if Δx and Δy are small, then $\Delta w - dw \approx 0$, that is, $dw \approx \Delta w$. This fact may be used to approximate the change in w corresponding to small changes in x and y.

Example 3 If $w = 3x^2 - xy$, find dw and use it to approximate the change in w if (x, y) changes from $(1, 2)$ to $(1.01, 1.98)$. How does this compare with the exact change in w?

Solution Applying Definition (16.11),

$$\begin{aligned} dw &= \frac{\partial w}{\partial x}\,dx + \frac{\partial w}{\partial y}\,dy \\ &= (6x - y)\,dx + (-x)\,dy. \end{aligned}$$

Substituting $x = 1, y = 2, dx = \Delta x = 0.01$, and $dy = \Delta y = -0.02$, we obtain

$$dw = (6 - 2)(0.01) + (-1)(-0.02) = 0.06.$$

It was shown in Example 1 that $\Delta w = 0.0605$. Hence the error involved in using dw is 0.0005. ■

Example 4 The radius and altitude of a right circular cylinder are measured as 3 in. and 8 in., respectively, with a possible error in measurement of ± 0.05 in. Use differentials to approximate the maximum error in the calculated volume of the cylinder.

Solution The volume V of a cylinder of radius r and altitude h is $V = \pi r^2 h$. Consequently,

$$dV = \frac{\partial V}{\partial r}\,dr + \frac{\partial V}{\partial h}\,dh = 2\pi rh\,dr + \pi r^2\,dh.$$

Substituting $r = 3$, $h = 8$, and $dr = dh = \pm 0.05$ gives us the following approximation to the maximum error:

$$dV = 48\pi(\pm 0.05) + 9\pi(\pm 0.05) = \pm 2.85\pi \approx \pm 8.95 \text{ in.}^3.$$ ■

For a function of one variable, the term *differentiable* means that the derivative exists. For functions of two variables we use the much stronger condition stated in the next definition.

Definition (16.12)

If $w = f(x, y)$, then f is **differentiable** at (x_0, y_0) provided Δw can be expressed in the form

$$\Delta w = f_x(x_0, y_0)\,\Delta x + f_y(x_0, y_0)\,\Delta y + \varepsilon_1\,\Delta x + \varepsilon_2\,\Delta y$$

where ε_1 and ε_2 have the limit 0 has $(\Delta x, \Delta y) \to (0, 0)$.

A function f of two variables is said to be **differentiable on a region *R*** if it is differentiable at each point of R. The next theorem follows directly from Theorem (16.10) and Definition (16.12). The terminology *rectangular region* means a region of the type described in (16.10).

Theorem (16.13)

If $w = f(x, y)$, and f_x and f_y are continuous on a rectangular region R, then f is differentiable on R.

The following result shows that a differentiable function is continuous, and is one of the reasons for using Definition (16.12) as the definition of differentiability.

Theorem (16.14)

> If a function f of two variables is differentiable at (x_0, y_0), then f is continuous at (x_0, y_0).

Proof We begin by writing the formula for Δw in Definition (16.12) as follows:

$$\Delta w = [f_x(x_0, y_0) + \varepsilon_1]\,\Delta x + [f_y(x_0, y_0) + \varepsilon_2]\,\Delta y.$$

If we let $x = x_0 + \Delta x$ and $y = y_0 + \Delta y$, then

$$\begin{aligned}\Delta w &= f(x, y) - f(x_0, y_0)\\ &= [f_x(x_0, y_0) + \varepsilon_1](x - x_0) + [f_y(x_0, y_0) + \varepsilon_2](y - y_0).\end{aligned}$$

It follows that

$$\lim_{(x, y)\to(x_0, y_0)} [f(x, y) - f(x_0, y_0)] = 0$$

or

$$\lim_{(x, y)\to(x_0, y_0)} f(x, y) = f(x_0, y_0).$$

Hence f is continuous at (x_0, y_0). □

Applying Theorem (16.13) gives us the following corollary.

Corollary (16.15)

> If f is a function of two variables, and if f_x and f_y are continuous on a rectangular region R, then f is continuous on R.

It can be shown by means of examples that the mere *existence* of f_x and f_y is not enough to ensure continuity of f (see Exercise 39). This brings out a difference from the single variable case, where the existence of f' implies continuity of f.

The discussion in this section can be extended to functions of more than two variables. For example, suppose $w = f(x, y, z)$ where f is defined on a suitable region R (such as a rectangular parallelepiped) and f_x, f_y, f_z exist in R and are continuous at (x, y, z). If x, y, and z are given increments Δx, Δy, and Δz, respectively, then the corresponding increment

$$\Delta w = f(x + \Delta x, y + \Delta y, z + \Delta z) - f(x, y, z)$$

can be written in the form

$$\Delta w = f_x(x, y, z)\,\Delta x + f_y(x, y, z)\,\Delta y + f_z(x, y, z)\,\Delta z + \varepsilon_1\,\Delta x + \varepsilon_2\Delta y + \varepsilon_3\Delta z$$

where ε_1, ε_2, and ε_3 are functions of Δx, Δy, and Δz that have the limit 0 as $(\Delta x, \Delta y, \Delta z) \to (0, 0, 0)$. Results that are analogous to (16.13)–(16.15) may be established for functions of three variables.

The following definition is an extension of (16.11) to functions of three variables.

Definition (16.16)

Let $w = f(x, y, z)$ and let Δx, Δy, and Δz be increments of x, y, and z, respectively.

(i) The **differentials of the independent variables x, y, and z** are

$$dx = \Delta x, \quad dy = \Delta y, \quad dz = \Delta z.$$

(ii) The **differential of the dependent variable w** is

$$dw = \frac{\partial w}{\partial x}\,dx + \frac{\partial w}{\partial y}\,dy + \frac{\partial w}{\partial z}\,dz.$$

We can use dw to approximate Δw if the increments of x, y, and z are small. Differentiability is defined as in (16.12). The extension to four or more variables is made in like manner.

Example 5 Suppose the dimensions (in inches) of a rectangular parallelepiped change from 9, 6, and 4 to 9.02, 5.97, and 4.01, respectively. Use differentials to approximate the change in volume. What is the exact change in volume?

Solution If we denote the dimensions by x, y, and z, then the volume is $V = xyz$ and, by Definition (16.16),

$$dV = yz\,dx + xz\,dy + xy\,dz.$$

Substituting $x = 9$, $y = 6$, $z = 4$, $dx = 0.02$, $dy = -0.03$, and $dz = 0.01$ gives us

$$\begin{aligned}\Delta V \approx dV &= 24(0.02) + 36(-0.03) + 54(0.01)\\ &= 0.48 - 1.08 + 0.54 = -0.06.\end{aligned}$$

Thus the volume decreases by approximately 0.06 in.^{3} The exact change in volume is

$$\Delta V = (9.02)(5.97)(4.01) - (9)(6)(4) = -0.063906. \qquad \blacksquare$$

16.4 Exercises

In Exercises 1–4 find values of ε_1 and ε_2 that satisfy Definition (16.12).

1 $f(x, y) = 4y^2 - 3xy + 2x$

2 $f(x, y) = (2x - y)^2$

3 $f(x, y) = x^3 + y^3$

4 $f(x, y) = 2x^2 - xy^2 + 3y$

In Exercises 5–16 find dw.

5 $w = x^3 - x^2y + 3y^2$

6 $w = 5x^2 + 4y - 3xy^3$

7 $w = x^2 \sin y + 2y^{3/2}$

8 $w = ye^{-2x} - 3x^4$

9 $w = x^2e^{xy} + (1/y^2)$

10 $w = \ln (x^2 + y^2) + x \tan^{-1} y$

11 $w = x^2 \ln (y^2 + z^2)$

12 $w = x^2y^3z + e^{-2z}$

13 $w = xyz/(x + y + z)$

14 $w = x^2e^{yz} + y \ln z$

15 $w = x^2z + 4yt^3 - xz^2t$

16 $w = x^2y^3zt^{-1}v^4$

17 Use differentials to approximate the change in

$$f(x, y) = x^2 - 3x^3y^2 + 4x - 2y^3 + 6$$

if (x, y) changes from $(-2, 3)$ to $(-2.02, 3.01)$.

18 Use differentials to approximate the change in

$$f(x, y) = x^2 - 2xy + 3y$$

if (x, y) changes from $(1, 2)$ to $(1.03, 1.99)$.

19 Use differentials to approximate the change in

$$f(x, y, z) = x^2z^3 - 3yz^2 + x^{-3} + 2y^{1/2}z$$

if (x, y, z) changes from $(1, 4, 2)$ to $(1.02, 3.97, 1.96)$.

20 Use differentials to approximate the change in

$$w = r^2 + 3sv + 2p^3$$

if r changes from 1 to 1.02, s from 2 to 1.99, v from 4 to 4.01, and p from 3 to 2.97.

21 Use differentials to approximate $\sqrt[3]{26.98}\sqrt{36.04}$.

22 Use differentials to approximate $(32.03)^{2/5}/(1.95)^4$.

23 The dimensions of a closed rectangular box are measured as 3 ft, 4 ft, and 5 ft with a possible error of $\frac{1}{16}$ in. Use differentials to approximate the maximum error in the calculated value of (a) the surface area of the box; (b) the volume of the box.

24 The two shortest sides of a right triangle are measured as 3 cm and 4 cm, respectively, with a possible error of 0.02 cm. Use differentials to approximate the maximum error in the calculated value of (a) the hypotenuse; (b) the area of the triangle.

25 An open cylindrical tin can has diameter 3 in. and altitude 4 in. Use differentials to approximate the amount of tin in the can if the tin is 0.015 in. thick.

26 The total resistance R of three resistances R_1, R_2, and R_3, connected in parallel (see figure) is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}.$$

If measurements of R_1, R_2, and R_3 are 100, 200, and 400 ohms, respectively, with a maximum error of 1% in each measurement, approximate the maximum error in the calculated value of R.

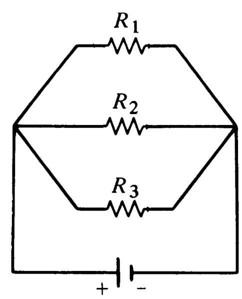


FIGURE FOR EXERCISE 26

27 The specific gravity of an object is given by $s = A/(A - W)$, where A and W are the weights (in pounds) of the object in air and water, respectively. If measurements are $A = 12$ lb and $W = 5$ lb with maximum errors of $\frac{1}{2}$ oz in air and 1 oz in water, what is the maximum error in the calculated value of s?

28 The pressure P, volume V, and temperature T of a confined gas are related by the *Ideal Gas Law* $PV = kT$, where k is a constant. If $P = 0.5$ lb/in.^{2} when $V = 64$ in.^{3} and $T = 80°$, approximate the change in P if V and T change to 70 in.^{3} and 76°, respectively.

29 In using the specific gravity formula $s = A/(A - W)$ (see Exercise 27), suppose that there are percentage errors of 2% and 4% in the measurements A and W, respectively. Express the maximum possible percentage error in the calculated value of s in terms of A and W.

30 In using the Ideal Gas Law $PV = kT$ (see Exercise 28), suppose that there are percentage errors of 0.8% and 0.5% in the measurements of T and P, respectively. Approximate the maximum percentage error in the calculated value of V.

31 The electrical resistance R of a wire is directly proportional to its length and inversely proportional to the square of its diameter. If the length is measured with a possible error of 1% and the diameter is measured with a possible error of 3%, what is the maximum percentage error in the calculated value of R?

32 It was shown in Section 6.8 that the flow of blood through an arteriole is given by $F = \pi PR^4/8vl$, where l is the length of the arteriole, R is the radius, P is the pressure difference between the two ends, and v is the viscosity of the blood. Suppose that v and l are constant.If the radius decreases by 2% and the pressure increases by 3%, use differentials to find the percentage change in the flow.

33 The temperature T at the point $P(x, y, z)$ in a rectangular coordinate system is given by $T = 8(2x^2 + 4y^2 + 9z^2)^{1/2}$. Use differentials to approximate the temperature difference between the points (6, 3, 2) and (6.1, 3.3, 1.98).

34 Approximate the change in area of an isosceles triangle if each of the two equal sides increases from 100 to 101 and the angle between them decreases from 120° to 119°.

35 If the cylinder described in Example 4 of this section has a closed top and bottom, use differentials to approximate the maximum error in the calculated total surface area.

36 Use differentials to approximate the change in surface area of the parallelepiped described in Example 5 of this section. What is the exact change?

In Exercises 37 and 38 prove that f is differentiable at every point in its domain.

37 $f(x, y) = (x^2 - y^2)/(x^2 + y^2)$

38 $f(x, y, z) = (x + y + z)/(x^2 + y^2 + z^2)$

39 Suppose $f(x, y) = \begin{cases} \dfrac{xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$

Prove that

(a) $f_x(0, 0)$ and $f_y(0, 0)$ exist. (*Hint*: Use (16.7).)

(b) f is not continuous at (0, 0).

(c) f is not differentiable at (0, 0).

40 Suppose

$$f(x, y, z) = \begin{cases} \dfrac{xyz}{x^3 + y^3 + z^3} & \text{if } (x, y, z) \neq (0, 0, 0) \\ 0 & \text{if } (x, y, z) = (0, 0, 0) \end{cases}$$

Prove that f_x, f_y, and f_z exist at (0, 0, 0), but that f is not differentiable at (0, 0, 0).

16.5 The Chain Rule

If f, g, and k are functions of two variables such that $w = f(u, v)$, $u = g(x, y)$, and $v = k(x, y)$, and if for each pair (x, y) in a subset D of $\mathbb{R} \times \mathbb{R}$ the corresponding pair (u, v) is in the domain of f, then $w = f(g(x, y), k(x, y))$ defines w as a (composite) function of x and y. For example, if

$$w = u^2 + u \sin v, \quad u = xe^{2y}, \quad \text{and} \quad v = xy$$

then
$$w = x^2e^{4y} + xe^{2y} \sin xy.$$

The next theorem provides formulas for expressing $\partial w/\partial x$ and $\partial w/\partial y$ in terms of the first partial derivatives of the functions g, k, and f. In the statement of the theorem it is assumed that domains are chosen such that the composite function is defined on a suitable domain D. Each of the formulas stated in Theorem (16.17) is referred to as a *Chain Rule*.

The Chain Rule (16.17)

If $w = f(u, v)$, $u = g(x, y)$, and $v = k(x, y)$ where f is differentiable and g and k have continuous first partial derivatives, then

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial w}{\partial v}\frac{\partial v}{\partial x}$$

$$\frac{\partial w}{\partial y} = \frac{\partial w}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial w}{\partial v}\frac{\partial v}{\partial y}.$$

Proof If x is given an increment Δx and y is held constant (that is, $\Delta y = 0$), then

(a) $$\Delta w = f(g(x + \Delta x, y), k(x + \Delta x, y)) - f(g(x, y), k(x, y)).$$

Also,

(b) $$\Delta u = g(x + \Delta x, y) - g(x, y)$$
$$\Delta v = k(x + \Delta x, y) - k(x, y)$$

and hence

$$g(x + \Delta x, y) = g(x, y) + \Delta u = u + \Delta u$$

$$k(x + \Delta x, y) = k(x, y) + \Delta v = v + \Delta v.$$

Substituting into equation (a),

$$\Delta w = f(u + \Delta u, v + \Delta v) - f(u, v).$$

Since f is differentiable, we may write

(c) $$\Delta w = \frac{\partial w}{\partial u}\Delta u + \frac{\partial w}{\partial v}\Delta v + \varepsilon_1 \Delta u + \varepsilon_2 \Delta v$$

where the symbols ε_1 and ε_2 denote functions of Δu and Δv which have the limit 0 as $(\Delta u, \Delta v) \to (0, 0)$. Moreover, we may assume that both ε_1 and ε_2 are 0 if $(\Delta u, \Delta v) = (0, 0)$, for if they are not, they can be replaced by functions μ_1 and μ_2 which have this property and are equal to ε_1 and ε_2 elsewhere. With this agreement, the functions ε_1 and ε_2 in (c) are continuous at (0, 0). Dividing both sides of that equation by Δx gives us

(d) $$\frac{\Delta w}{\Delta x} = \frac{\partial w}{\partial u}\frac{\Delta u}{\Delta x} + \frac{\partial w}{\partial v}\frac{\Delta v}{\Delta x} + \varepsilon_1\frac{\Delta u}{\Delta x} + \varepsilon_2\frac{\Delta v}{\Delta x}.$$

From equations (a) and (b),

$$\lim_{\Delta x \to 0}\frac{\Delta w}{\Delta x} = \frac{\partial w}{\partial x}, \quad \lim_{\Delta x \to 0}\frac{\Delta u}{\Delta x} = \frac{\partial u}{\partial x}, \quad \lim_{\Delta x \to 0}\frac{\Delta v}{\Delta x} = \frac{\partial v}{\partial x}.$$

If Δx approaches 0, then Δu and Δv also approach 0 and hence so do ε_1 and ε_2. Consequently, if we take the limit in equation (d) as $\Delta x \to 0$, we obtain

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial w}{\partial v}\frac{\partial v}{\partial x}.$$

The second formula in the statement of the theorem is established in similar fashion. □

It is of interest to note that (16.17) is a generalization of the Chain Rule (3.30) for functions of one variable. Specifically, if we let $w = f(u)$, $u = g(x)$, and $v = 0$, then the partial derivatives become ordinary derivatives, and the first formula in (16.17) takes on the form

$$\frac{dw}{dx} = \frac{dw}{du}\frac{du}{dx}$$

which is in agreement with (3.30).

Example 1 If $w = u^3 + v^2$, $u = xy^2$, and $v = x^2 \sin y$, use the Chain Rule to find $\partial w/\partial x$ and $\partial w/\partial y$.

Solution Note that w is a function of x and y. Specifically,

$$w = (xy^2)^3 + (x^2 \sin y)^2.$$

We could find $\partial w/\partial x$ and $\partial w/\partial y$ directly from this expression; however, our objective is to illustrate the Chain Rule. Thus, applying (16.17),

$$\begin{aligned}\frac{\partial w}{\partial x} &= (3u^2)y^2 + (2v)(2x \sin y)\\ &= 3(xy^2)^2y^2 + (2x^2 \sin y)(2x \sin y)\\ &= 3x^2y^6 + 4x^3 \sin^2 y.\end{aligned}$$

Similarly,

$$\begin{aligned}\frac{\partial w}{\partial y} &= (3u^2)(2xy) + (2v)(x^2 \cos y)\\ &= 3(xy^2)^2(2xy) + 2(x^2 \sin y)(x^2 \cos y)\\ &= 6x^3y^5 + 2x^4 \sin y \cos y.\end{aligned}$$ ■

Note that after applying the Chain Rule in Example 1 we substituted for u and v, thereby expressing $\partial w/\partial x$ and $\partial w/\partial y$ in terms of x and y. This was done to emphasize the fact that w is a (composite) function of the *two* variables x and y.

The Chain Rule can be extended to any number of variables, as stated in the following theorem.

General Chain Rule (16.18)

If w is a differentiable function of n variables $u_1, u_2, \ldots, u_n$ where each u_i is a function of m variables $x_1, x_2, \ldots, x_m$ having continuous first partial derivatives, and if w is a (composite) function of $x_1, x_2, \ldots, x_m$, then

$$\frac{\partial w}{\partial x_i} = \frac{\partial w}{\partial u_1}\frac{\partial u_1}{\partial x_i} + \frac{\partial w}{\partial u_2}\frac{\partial u_2}{\partial x_i} + \cdots + \frac{\partial w}{\partial u_n}\frac{\partial u_n}{\partial x_i}$$

for $i = 1, 2, \ldots, m$.

Example 2 If $w = r^2 + sv + t^3$, where $r = x^2 + y^2 + z^2$, $s = xyz$, $v = xe^y$, and $t = yz^2$, use the Chain Rule to find $\partial w/\partial z$.

Solution Note that w is a function of r, s, v, t and that each of these four variables is a function of x, y, and z. Applying Theorem (16.18),

$$\begin{aligned}\frac{\partial w}{\partial z} &= \frac{\partial w}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial w}{\partial s}\frac{\partial s}{\partial z} + \frac{\partial w}{\partial v}\frac{\partial v}{\partial z} + \frac{\partial w}{\partial t}\frac{\partial t}{\partial z} \\ &= (2r)(2z) + v(xy) + s(0) + (3t^2)(2yz) \\ &= 4z(x^2 + y^2 + z^2) + xe^y(xy) + 0 + 3(yz^2)^2(2yz) \\ &= 4z(x^2 + y^2 + z^2) + x^2ye^y + 6y^3z^5.\end{aligned}$$

■

The following special case of the Chain Rule follows directly from Theorem (16.18) by letting $x_i = t$ for each i.

Theorem (16.19)

If w is a function of $u_1, u_2, \ldots, u_n$ and each u_i is a function of *one* variable t, then w is a function of t and

$$\frac{dw}{dt} = \frac{\partial w}{\partial u_1}\frac{du_1}{dt} + \frac{\partial w}{\partial u_2}\frac{du_2}{dt} + \cdots + \frac{\partial w}{\partial u_n}\frac{du_n}{dt}.$$

In Theorem (16.19) the notations dw/dt and du_i/dt are used because w and each u_i is a function of *one* variable t. This situation could occur in applications where w is a function of several variables and each variable is a function of time t. In this case the derivative dw/dt would give us the rate of change of w with respect to time.

Example 3 If $w = x^2 + yz$ and $x = 3t^2 + 1$, $y = 2t - 4$, $z = t^3$, find dw/dt.

Solution The problem could be solved without the Chain Rule by writing

$$w = (3t^2 + 1)^2 + (2t - 4)t^3$$

and then finding dw/dt by single variable methods. However, if we apply Theorem (16.19), then

$$\begin{aligned}\frac{dw}{dt} &= \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt} \\ &= (2x)(6t) + z(2) + y(3t^2) \\ &= 2(3t^2 + 1)6t + t^3(2) + (2t - 4)3t^2 \\ &= 44t^3 - 12t^2 + 12t.\end{aligned}$$

■

Partial derivatives can be used to find derivatives of functions that are defined implicitly. Suppose, as in Section 3.7, an equation $F(x, y) = 0$ defines a differentiable function f of one variable x such that $F(x, f(x)) = 0$ for all x in the domain D of f. If we let

$$w = F(u, y), \quad \text{where } u = x \text{ and } y = f(x)$$

then applying Theorem (16.19) and using the fact that $w = F(x, f(x)) = 0$ for all x in D,

$$\frac{dw}{dx} = \frac{\partial w}{\partial u}\frac{du}{dx} + \frac{\partial w}{\partial y}\frac{dy}{dx} = 0,$$

that is,

$$\frac{\partial w}{\partial u}(1) + \frac{\partial w}{\partial y} f'(x) = 0.$$

If $\partial w/\partial y \neq 0$, then (since $u = x$)

$$f'(x) = -\frac{\partial w/\partial x}{\partial w/\partial y} = -\frac{F_x(x, y)}{F_y(x, y)}.$$

We may summarize this discussion as follows.

Theorem (16.20)

If an equation $F(x, y) = 0$ implicitly defines a differentiable function f of one variable x such that $y = f(x)$, then

$$\frac{dy}{dx} = -\frac{F_x(x, y)}{F_y(x, y)}.$$

Example 4 Find y' if $y = f(x)$ satisfies the equation

$$y^4 + 3y - 4x^3 - 5x - 1 = 0.$$

Solution If $F(x, y)$ is the expression on the left side of the equation, then by Theorem (16.20),

$$y' = -\frac{-12x^2 - 5}{4y^3 + 3} = \frac{12x^2 + 5}{4y^3 + 3}.$$

Compare this solution with that of Example 2 in Section 3.7, which was obtained using single-variable methods. ■

Given an equation such as

$$x^2 - 4y^3 + 2z - 7 = 0$$

we can solve for z, obtaining

$$z = \tfrac{1}{2}(-x^2 + 4y^3 + 7)$$

which is of the form $\qquad z = f(x, y).$

In analogy with the single variable case, we say that the function f of two variables x and y is defined *implicitly* by the given equation. The next theorem gives us formulas for finding f_x and f_y or, equivalently, $\partial z/\partial x$ and $\partial z/\partial y$ without actually solving the equation for z.

Theorem (16.21)

If an equation $F(x, y, z) = 0$ defines an implicit differentiable function f of two variables x and y such that $z = f(x, y)$ for all (x, y) in the domain of f, then

$$\frac{\partial z}{\partial x} = -\frac{F_x(x, y, z)}{F_z(x, y, z)}, \qquad \frac{\partial z}{\partial y} = -\frac{F_y(x, y, z)}{F_z(x, y, z)}.$$

Proof The statement that $F(x, y, z) = 0$ defines the function f, where $z = f(x, y)$, means that $F(x, y, f(x, y)) = 0$ for all (x, y) in the domain of f. If we let

$$w = F(u, v, z), \quad \text{where } u = x, v = y, z = f(x, y)$$

then by the Chain Rule and the fact that $w = 0$ for all (x, y) in D,

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial w}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial x} = 0.$$

Since $u = x$ and $v = y$, we have

$$\frac{\partial w}{\partial x}(1) + \frac{\partial w}{\partial y}(0) + \frac{\partial w}{\partial z}\frac{\partial z}{\partial x} = 0$$

and, if $\partial w/\partial z \neq 0$,

$$\frac{\partial z}{\partial x} = -\frac{\partial w/\partial x}{\partial w/\partial z} = -\frac{F_x(x, y, z)}{F_z(x, y, z)}.$$

The formula for $\partial z/\partial y$ may be obtained in similar fashion. □

Example 5 If $z = f(x, y)$ satisfies the equation

$$x^2z^2 + xy^2 - z^3 + 4yz - 5 = 0$$

find $\partial z/\partial x$ and $\partial z/\partial y$.

Solution If we let $F(x, y, z)$ denote the expression on the left of the given equation, then by Theorem (16.21),

$$\frac{\partial z}{\partial x} = -\frac{2xz^2 + y^2}{2x^2z - 3z^2 + 4y}$$

$$\frac{\partial z}{\partial y} = -\frac{2xy + 4z}{2x^2z - 3z^2 + 4y}.$$

■

16.5 Exercises

Use the Chain Rule to solve Exercises 1–14.

In Exercises 1 and 2 find $\partial w/\partial x$ and $\partial w/\partial y$.

1 $w = u \sin v$, $u = x^2 + y^2$, $v = xy$

2 $w = uv + v^2$, $u = x \sin y$, $v = y \sin x$

In Exercises 3 and 4 find $\partial w/\partial r$ and $\partial w/\partial s$.

3 $w = u^2 + 2uv$, $u = r \ln s$, $v = 2r + s$

4 $w = e^{tv}$, $t = r + s$, $v = rs$

In Exercises 5 and 6 find $\partial z/\partial x$ and $\partial z/\partial y$.

5 $z = r^3 + s + v^2$, $r = xe^y$, $s = ye^x$, $v = x^2y$

6 $z = pq + qw$, $p = 2x - y$, $q = x - 2y$, $w = -2x + 2y$

In Exercises 7 and 8 find $\partial r/\partial u$, $\partial r/\partial v$, and $\partial r/\partial t$.

7 $r = x \ln y$, $x = 3u + vt$, $y = uvt$

8 $r = w^2 \cos z$, $w = u^2vt$, $z = ut^2$

9 Find $\partial p/\partial r$ if $p = u^2 + 3v^2 - 4w^2$, $u = x - 3y + 2r - s$, $v = 2x + y - r + 2s$, $w = -x + 2y + r + s$.

10 Find $\partial s/\partial$ if $s = tr + ue^v$, $t = xy^2z$, $r = x^2yz$, $u = xyz^2$, $v = xyz$.

In Exercises 11–14 find dw/dt.

11 $w = x^3 - y^3$, $x = 1/(t + 1)$, $y = t/(t + 1)$

12 $w = \ln(u + v)$, $u = e^{-2t}$, $v = t^3 - t^2$

13 $w = r^2 - s \tan v$, $r = \sin^2 t$, $s = \cos t$, $v = 4t$

14 $w = x^2y^3z^4$, $x = 2t + 1$, $y = 3t - 2$, $z = 5t + 4$

In Exercises 15–18 find y' under the assumption that $y = f(x)$ satisfies the given equation.

15 $2x^3 + x^2y + y^3 = 1$

16 $x^4 + 2x^2y^2 - 3xy^3 + 2x = 0$

17 $6x + \sqrt{xy} = 3y - 4$

18 $x^{2/3} + y^{2/3} = 4$

In Exercises 19–22 find $\partial z/\partial x$ and $\partial z/\partial y$ under the assumption that $z = f(x, y)$ satisfies the given equation.

19 $2xz^3 - 3yz^2 + x^2y^2 + 4z = 0$

20 $xz^2 + 2x^2y - 4y^2z + 3y - 2 = 0$

21 $xe^{yz} - 2ye^{xz} + 3ze^{xy} = 1$

22 $yx^2 + z^2 + \cos xyz = 4$

23 If $w = f(x, y)$, $x = r\cos\theta$, and $y = r\sin\theta$ show that

$$\left(\frac{\partial w}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2 = \left(\frac{\partial w}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial w}{\partial \theta}\right)^2.$$

24 If $w = f(x, y)$ and $x = e^r\cos\theta$, $y = e^r\sin\theta$, show that

$$\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} = e^{-2r}\left(\frac{\partial^2 w}{\partial r^2} + \frac{\partial^2 w}{\partial \theta^2}\right).$$

25 If $w = f(x, y)$ and $x = r\cos\theta$, $y = r\sin\theta$, show that

$$\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} = \frac{\partial^2 w}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2 w}{\partial \theta^2} + \frac{1}{r}\frac{\partial w}{\partial r}.$$

26 If $v = f(x - at) + g(x + at)$ where f and g have second partial derivatives, show that v satisfies the wave equation

$$\frac{\partial^2 v}{\partial t^2} = a^2 \frac{\partial^2 v}{\partial x^2}.$$

(Compare with Exercise 42 of Section 16.3.)

27 If $w = \cos(x + y) + \cos(x - y)$, show, without using addition formulas, that $w_{xx} - w_{yy} = 0$.

28 If $w = f(x^2 + y^2)$, show that $y(\partial w/\partial x) - x(\partial w/\partial y) = 0$. (*Hint*: Let $u = x^2 + y^2$.)

29 If $w = f(u, v)$, $u = g(x, y)$, and $v = k(x, y)$, show that

$$\frac{\partial^2 w}{\partial x^2} = \frac{\partial^2 w}{\partial u^2}\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial^2 w}{\partial v\,\partial u} + \frac{\partial^2 w}{\partial u\,\partial v}\right)\frac{\partial u}{\partial x}\frac{\partial v}{\partial x} + \frac{\partial^2 w}{\partial v^2}\left(\frac{\partial v}{\partial x}\right)^2 + \frac{\partial w}{\partial u}\frac{\partial^2 u}{\partial x^2} + \frac{\partial w}{\partial v}\frac{\partial^2 v}{\partial x^2}.$$

30 Given w, u, and v as in Exercise 25, show that

$$\frac{\partial^2 w}{\partial y\,\partial x} = \frac{\partial^2 w}{\partial u^2}\frac{\partial u}{\partial x}\frac{\partial u}{\partial y} + \frac{\partial^2 w}{\partial v\,\partial u}\frac{\partial u}{\partial x}\frac{\partial v}{\partial y} + \frac{\partial^2 w}{\partial u\,\partial v}\frac{\partial u}{\partial y}\frac{\partial v}{\partial x} + \frac{\partial^2 w}{\partial v^2}\frac{\partial v}{\partial x}\frac{\partial v}{\partial y} + \frac{\partial w}{\partial u}\frac{\partial^2 u}{\partial y\,\partial x} + \frac{\partial w}{\partial v}\frac{\partial^2 v}{\partial y\,\partial x}.$$

31 If n resistances $R_1, R_2, \ldots, R_n$ are connected in parallel, then the total resistance R is given by

$$\frac{1}{R} = \sum_{i=1}^{n} \frac{1}{R_i}.$$

Prove that for $i = 1, 2, \ldots, n$,

$$\frac{\partial R}{\partial R_i} = \left(\frac{R}{R_i}\right)^2.$$

32 A function f of two variables is said to be **homogeneous of degree *n*** if $f(tx, ty) = t^n f(x, y)$ for every t such that (tx, ty) is in the domain of f. Show that for such functions, $xf_x(x, y) + yf_y(x, y) = nf(x, y)$. (*Hint*: Differentiate $f(tx, ty)$ with respect to t.)

33 The radius r and altitude h of a right circular cylinder are increasing at rates of 0.01 in./min and 0.02 in./min, respectively. Use the Chain Rule to find the rate at which the volume is increasing at the time when $r = 4$ in. and $h = 7$ in. At what rate is the curved surface area changing at this time?

34 The equal sides and included angle of an isosceles triangle are increasing at rates of 0.1 ft/hr and 2°/hr, respectively. Use the Chain Rule to find the rate at which the area of the triangle is increasing at the time when the length of the equal sides is 20 ft and the included angle is 60°.

35 The pressure P, volume V, and temperature T of a confined gas are related by the Ideal Gas Law $PV = kT$, where k is a constant. If P and V are changing at the rates dP/dt and dV/dt, respectively, use the Chain Rule to find a formula for dT/dt. Check your answer by using the product rule for functions of one variable.

36 If the base radius r and altitude h of a right circular cylinder are changing at the rates dr/dt and dh/dt, respectively, use the Chain Rule to find a formula for dV/dt, where V is the volume of the cylinder. Check your answer by using single variable methods.

37 A certain gas obeys the Ideal Gas Law $PV = 8T$. Suppose that the gas is being heated at a rate of 2°/min and the pressure is increasing at a rate of $\frac{1}{2}$ (lb/in^2)/min. If, at a certain instant, the temperature is 200° and the pressure is 10 lb/in.^{2}, find the rate at which the volume is changing.

38 Sand is leaking out of a hole in a container at a rate of 6 in.^{3}/min. As it leaks out it forms a pile in the shape of a right circular cone whose base radius is increasing at a rate of $\frac{1}{4}$ in./min. If, at the instant that 40 in.^{3} has leaked out, the radius is 5 in., find the rate at which the height of the pile is increasing.

39 Prove the following Mean Value Theorem for a function f of two variables x and y:

If f has first partial derivatives that are continuous on a rectangular region $R = \{(x, y): a < x < b,\ c < y < d\}$, and if $A(x_1, y_1)$ and $B(x_2, y_2)$ are points in R, then there is a point $P(x^*, y^*)$ on the line segment AB such that

$$f(x_2, y_2) - f(x_1, y_1) = f_x(x^*, y^*)(x_2 - x_1) + f_y(x^*, y^*)(y_2 - y_1).$$

40 Use Exercise 39 to prove the following extension of Theorem (4.33): If $f_x(x, y) = 0$ and $f_y(x, y) = 0$ for all (x, y) in a rectangular region R, then $f(x, y)$ is constant on R.

41 Extend Exercise 39 to a function of three variables.

42 Extend Exercise 40 to a function of three variables.

43 Suppose $u = f(x, y)$ and $v = g(x, y)$ satisfy the *Cauchy–Riemann equations* $u_x = v_y$ and $u_y = -v_x$. If $x = r\cos\theta$, $y = r\sin\theta$, show that

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta} \quad \text{and} \quad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}.$$

16.6 Directional Derivatives

Let us return to the illustration in which $w = f(x, y)$ is the temperature of a flat metal plate at the point $P(x, y)$. In Section 16.3 the partial derivatives $f_x(x, y)$ and $f_y(x, y)$ were interpreted as the rates of change of w with respect to distance at P in the horizontal or vertical directions, respectively. This can be generalized to any direction as follows. Let Q be any other point on the plate and let $R(x + \Delta x, y + \Delta y)$ be a point on the vector $\overrightarrow{PQ}$, as illustrated in Figure 16.19. The increment

$$\Delta w = f(x + \Delta x, y + \Delta y) - f(x, y)$$

is the difference in temperature between points P and R.

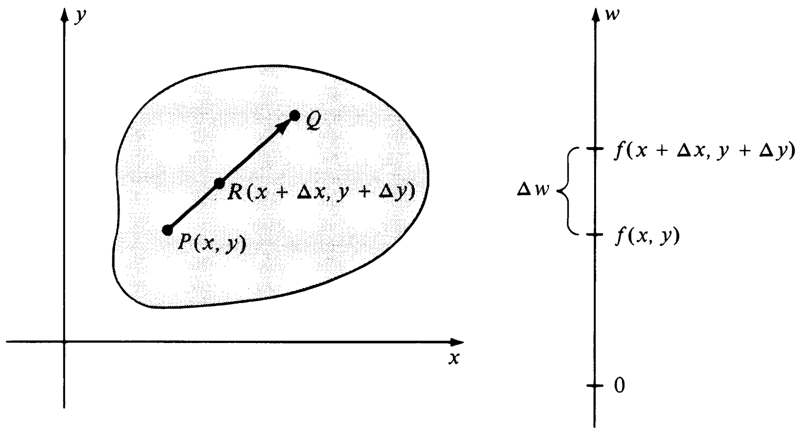


FIGURE 16.19

The ratio

$$\frac{\Delta w}{d(P, R)} = \frac{\Delta w}{\sqrt{(\Delta x)^2 + (\Delta y)^2}}$$

is the *average change in temperature* as a point moves along $\overrightarrow{PQ}$ from P to R. For example, if the temperature at P is 50° and the temperature at R is 51.5°, then $\Delta w = 1.5°$. If $d(P, R) = 3$ in., then the average rate of change in temperature as a point moves from P to R is

$$\frac{\Delta w}{d(P, R)} = 0.5°/\text{in.}$$

If the ratio $\Delta w/d(P, R)$ has a limit as R approaches P along $\overrightarrow{PQ}$, that is, as $(\Delta x, \Delta y) \to (0, 0)$, then the limit is called *the rate of change of w with respect to distance in the direction of the vector* $\overrightarrow{PQ}$.

If f is any function of two variables, then the preceding discussion may be used to motivate the concept of the *directional derivative* of f at $P(x, y)$ in any direction. Let us specify the direction from P by means of a unit vector $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ and consider the line l through P and parallel to $\mathbf{u}$, as illustrated in (i) of Figure 16.20.

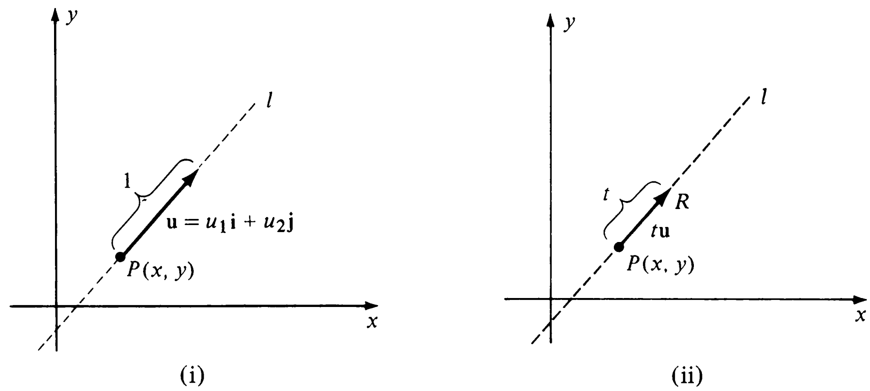


FIGURE 16.20

If R is another point on l and $\overrightarrow{PR}$ has the same direction as $\mathbf{u}$, then

$$\overrightarrow{PR} = t\mathbf{u} = (tu_1)\mathbf{i} + (tu_2)\mathbf{j}$$

where $t > 0$ (see (ii) of Figure 16.20). The coordinates of R are

$$(x + tu_1, y + tu_2)$$

and hence

$$\Delta w = f(x + tu_1, y + tu_2) - f(x, y).$$

Since $d(P, R) = |\overrightarrow{PR}| = t$, the average change in w corresponding to the displacement $\overrightarrow{PR}$ is

$$\frac{\Delta w}{d(P, R)} = \frac{f(x + tu_1, y + tu_2) - f(x, y)}{t}.$$

Taking the limit as $t \to 0^+$ gives us the rate of change of f at P in the direction of $\mathbf{u}$. This is the motivation for the next definition.

Definition (16.22)

If f is a function of x and y and if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ is a unit vector, then **the directional derivative of f at $P(x, y)$ in the direction of $\mathbf{u}$**, denoted by $D_\mathbf{u} f(x, y)$, is

$$D_\mathbf{u} f(x, y) = \lim_{t \to 0} \frac{f(x + tu_1, y + tu_2) - f(x, y)}{t}.$$

It is easy to see that the first partial derivatives of f are special cases of the directional derivative. As a matter of fact, if $\mathbf{u} = \mathbf{i}$, then $u_1 = 1$, $u_2 = 0$, and the limit in Definition (16.22) reduces to

$$D_{\mathbf{i}} f(x, y) = \lim_{t \to 0} \frac{f(x + t, y) - f(x, y)}{t} = f_x(x, y).$$

Similarly, if $\mathbf{u} = \mathbf{j}$, then $u_1 = 0$, $u_2 = 1$, and we obtain

$$D_{\mathbf{j}} f(x, y) = \lim_{t \to 0} \frac{f(x, y + t) - f(x, y)}{t} = f_y(x, y).$$

If $\mathbf{a}$ is any vector having the same direction as $\mathbf{u}$ we shall also refer to $D_{\mathbf{u}} f(x, y)$ as **the directional derivative of f in the direction of a**.

The following theorem provides a formula for finding directional derivatives.

Theorem (16.23)

If f is a differentiable function of two variables and $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ is a unit vector, then

$$D_{\mathbf{u}} f(x, y) = f_x(x, y)u_1 + f_y(x, y)u_2.$$

Proof If g is the function of one variable t defined by

$$g(t) = f(x + tu_1, y + tu_2),$$

then from Theorem (3.7) and Definition (16.22)

$$\begin{aligned} g'(0) &= \lim_{t \to 0} \frac{g(t) - g(0)}{t - 0} \\ &= \lim_{t \to 0} \frac{f(x + tu_1, y + tu_2) - f(x, y)}{t} \\ &= D_{\mathbf{u}} f(x, y). \end{aligned}$$

If we write

$$g(t) = f(r, v), \quad \text{where } r = x + tu_1, v = y + tu_2,$$

then from Theorem (16.19)

$$\begin{aligned} g'(t) &= f_r(r, v)\frac{du}{dt} + f_v(r, v)\frac{dv}{dt} \\ &= f_r(r, v)u_1 + f_v(r, v)u_2. \end{aligned}$$

As we saw in the first part of the proof, the directional derivative equals $g'(0)$. However, if $t = 0$ then $r = x$, $v = y$, and we obtain the formula given in the statement of the theorem. □

Example 1 If $f(x, y) = x^3y^2$, find the directional derivative of f at the point $P(-1, 2)$ in the direction of the vector $\mathbf{a} = 4\mathbf{i} - 3\mathbf{j}$.

Solution We wish to find $D_{\mathbf{u}}f(-1, 2)$, where $\mathbf{u}$ is a *unit* vector having the direction of $\mathbf{a}$. Since

$$\mathbf{u} = \frac{1}{|\mathbf{a}|}\mathbf{a} = \frac{1}{5}(4\mathbf{i} - 3\mathbf{j}) = \frac{4}{5}\mathbf{i} - \frac{3}{5}\mathbf{j}$$

and $$f_x(x, y) = 3x^2y^2 \quad \text{and} \quad f_y(x, y) = 2x^3y,$$

it follows from (16.23) that

$$D_{\mathbf{u}}f(x, y) = 3x^2y^2\left(\frac{4}{5}\right) + 2x^3y\left(-\frac{3}{5}\right).$$

Hence

$$\begin{aligned} D_{\mathbf{u}}f(-1, 2) &= 6(-1)^2(2)^2\left(\frac{4}{5}\right) + 2(-1)^3(2)\left(-\frac{3}{5}\right) \\ &= \frac{96}{5} + \frac{12}{5} = \frac{108}{5}. \end{aligned}$$ ■

We may use Theorem (16.23) to express a directional derivative as a dot product of two vectors as follows:

$$D_{\mathbf{u}}f(x, y) = \langle f_x(x, y), f_y(x, y)\rangle \cdot \langle u_1, u_2\rangle = \langle f_x(x, y), f_y(x, y)\rangle \cdot \mathbf{u}.$$

The vector in V_2 whose components are the first partial derivatives of $f(x, y)$ is designated by the following special name and notation.

Definition (16.24)

If f is a function of two variables, then the **gradient of f** (or of $f(x, y)$) is

$$\nabla f(x, y) = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}.$$

In applications, the gradient $\nabla f(x, y)$ is sometimes denoted by grad $f(x, y)$. From the previous discussion we have the following.

(16.25) $$D_{\mathbf{u}}f(x, y) = \nabla f(x, y) \cdot \mathbf{u}$$

Thus, *to find the directional derivative of f in the direction of the unit vector* $\mathbf{u}$, *we may dot the gradient of f with* $\mathbf{u}$. The symbol ∇ (called "**del**") is a *vector differential operator* and is symbolized by

$$\nabla = \mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y}.$$

It has properties similar to the operator d/dx (see Exercises 31–36). Standing alone it is meaningless; however, if it operates on $f(x, y)$ it produces the two-dimensional vector function given by Definition (16.24).

Example 2 If $f(x, y) = x^2 - 4xy$,

(a) find the gradient of f at the point $P(3, -2)$.

(b) use the gradient to find the directional derivative of f at $P(3, -2)$ in the direction from $P(3, -2)$ to $Q(4, 1)$.

(c) discuss the significance of the answer to part (b) if $f(x, y)$ is the temperature at (x, y).

Solution

(a) By Definition (16.24),

$$\nabla f(x, y) = (2x - 4y)\mathbf{i} - 4x\mathbf{j}.$$

Hence, at $P(3, -2)$,

$$\nabla f(3, -2) = (6 + 8)\mathbf{i} - 12\mathbf{j} = 14\mathbf{i} - 12\mathbf{j}.$$

(b) If we let $\mathbf{a} = \overrightarrow{PQ}$, then

$$\mathbf{a} = \langle 4 - 3, 1 - (-2)\rangle = \langle 1, 3\rangle = \mathbf{i} + 3\mathbf{j}.$$

A unit vector having the direction of $\overrightarrow{PQ}$ is

$$\mathbf{u} = \frac{1}{|\mathbf{a}|}\mathbf{a} = \frac{1}{\sqrt{10}}(\mathbf{i} + 3\mathbf{j}).$$

Applying (16.25) gives us

$$\begin{aligned} D_{\mathbf{u}} f(3, -2) &= \nabla f(3, -2) \cdot \mathbf{u} \\ &= (14\mathbf{i} - 12\mathbf{j}) \cdot \frac{1}{\sqrt{10}}(\mathbf{i} + 3\mathbf{j}) \\ &= \frac{1}{\sqrt{10}}(14 - 36) = -\frac{22}{\sqrt{10}} \approx -6.96. \end{aligned}$$

(c) If $f(x, y)$ is the temperature (in degrees) at (x, y), then the fact that $D_{\mathbf{u}} f(3, -2) \approx -6.96$ indicates that if a point moves in the direction of $\overrightarrow{PQ}$, the temperature at P is decreasing at approximately 6.96° per unit change in distance. ■

Let $P(x, y)$ be a fixed point, and consider the directional derivative $D_{\mathbf{u}} f(x, y)$ as $\mathbf{u} = \langle u_1, u_2\rangle$ varies. For certain unit vectors $\mathbf{u}$ the directional derivative may be positive (that is, $f(x, y)$ may increase), or negative ($f(x, y)$ may decrease), or it may be 0. In many applications it is important to find the direction in which $f(x, y)$ increases most rapidly and also to find the maximum rate of change. The next theorem provides this information.

Theorem (16.26) The maximum value of $D_{\mathbf{u}} f(x, y)$ at the point $P(x, y)$ occurs in the direction of $\nabla f(x, y)$. Moreover, this maximum value is $|\nabla f(x, y)|$.

Proof Let γ denote the angle between $\mathbf{u}$ and $\nabla f(x, y)$. Applying (16.25) and Theorem (14.22),

$$\begin{aligned} D_{\mathbf{u}} f(x, y) &= \nabla f(x, y) \cdot \mathbf{u} \\ &= |\nabla f(x, y)|\,|\mathbf{u}| \cos \gamma = |\nabla f(x, y)| \cos \gamma. \end{aligned}$$

The maximum value occurs if $\cos \gamma = 1$; that is, if $\gamma = 0$, in which case $\mathbf{u}$ has the same direction as $\nabla f(x, y)$. Moreover, in this direction we see that $D_{\mathbf{u}} f(x, y) = |\nabla f(x, y)|$. This completes the proof. □

In terms of rates of change, Theorem (16.26) tells us that $f(x, y)$ increases most rapidly in the direction of $\nabla f(x, y)$. It follows that $f(x, y)$ *decreases* most rapidly in the direction of $-\nabla f(x, y)$.

Example 3 If $f(x, y) = x^2 - 4xy$, find the direction in which $f(x, y)$ increases most rapidly at the point $P(3, -2)$. What is this maximum rate of increase?

Solution The function f was considered in Example 2, where we found that $\nabla f(3, -2) = 14\mathbf{i} - 12\mathbf{j}$. Hence, by Theorem (16.26), $f(x, y)$ increases most rapidly at $P(3, -2)$ in the direction of the vector $14\mathbf{i} - 12\mathbf{j}$. The maximum rate is

$$|14\mathbf{i} - 12\mathbf{j}| = \sqrt{196 + 144} = \sqrt{340} \approx 18.4.$$ ■

The directional derivative of a function f of three variables is defined in a manner similar to Definition (16.22). Specifically, if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ is a unit vector, then

$$D_{\mathbf{u}} f(x, y, z) = \lim_{t \to 0} \frac{f(x + tu_1, y + tu_2, z + tu_3) - f(x, y, z)}{t}.$$

As in the two variable case, $D_{\mathbf{u}} f(x, y, z)$ is the rate of change of f with respect to distance at $P(x, y, z)$ in the direction of $\mathbf{u}$. If f is a function of three variables, then in a manner analogous to Definition (16.24), the **gradient of f** (or of $f(x, y, z)$), denoted by $\nabla f(x, y, z)$ or grad $f(x, y, z)$, is defined as follows.

(16.27) $$\nabla f(x, y, z) = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k}$$

The next result is the three dimensional version of Theorem (16.23).

Theorem (16.28)

If f is a function of three variables and $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ is a unit vector, then

$$\begin{aligned} D_{\mathbf{u}} f(x, y, z) &= \nabla f(x, y, z) \cdot \mathbf{u} \\ &= f_x(x, y, z)u_1 + f_y(x, y, z)u_2 + f_z(x, y, z)u_3. \end{aligned}$$

It is easy to show, as in Theorem (16.26), that of all possible directional derivatives $D_{\mathbf{u}} f(x, y, z)$ at the point $P(x, y, z)$, the one in the direction of $\nabla f(x, y, z)$ has the largest value. Moreover, this maximum value is $|\nabla f(x, y, z)|$.

Example 4 Suppose a rectangular coordinate system is located in space such that the temperature T at the point (x, y, z) is given by the formula $T = 100/(x^2 + y^2 + z^2)$.

(a) Find the rate of change of T with respect to distance at the point $P(1, 3, -2)$ in the direction of the vector $\mathbf{a} = \mathbf{i} - \mathbf{j} + \mathbf{k}$.

(b) In what direction from P does T increase most rapidly? What is the maximum rate of change of T at P?

Solution

(a) By definition, the gradient of T is

$$\nabla T = \frac{\partial T}{\partial x}\mathbf{i} + \frac{\partial T}{\partial y}\mathbf{j} + \frac{\partial T}{\partial z}\mathbf{k}.$$

Since

$$\frac{\partial T}{\partial x} = \frac{-200x}{(x^2 + y^2 + z^2)^2}, \quad \frac{\partial T}{\partial y} = \frac{-200y}{(x^2 + y^2 + z^2)^2}, \quad \frac{\partial T}{\partial z} = \frac{-200z}{(x^2 + y^2 + z^2)^2}$$

we have

$$\nabla T = \frac{-200}{(x^2 + y^2 + z^2)^2}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}).$$

If we let $\nabla T]_P$ denote the value of ∇T at the point $P(1, 3, -2)$, then

$$\nabla T]_P = \frac{-200}{196}(\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}).$$

A unit vector $\mathbf{u}$ having the same direction as $\mathbf{a} = \mathbf{i} - \mathbf{j} + \mathbf{k}$ is

$$\mathbf{u} = \frac{1}{\sqrt{3}}(\mathbf{i} - \mathbf{j} + \mathbf{k}).$$

Using Theorem (16.28), the rate of change of T at P in the direction of $\mathbf{a}$ is

$$D_{\mathbf{u}}T]_P = \nabla T]_P \cdot \mathbf{u} = \frac{-200}{196}\frac{(1-3-2)}{\sqrt{3}} = \frac{200}{49\sqrt{3}} \approx 2.4.$$

If, for example, T is measured in degrees and distance is in inches, then T is increasing at a rate of 2.4 degrees per inch at P in the direction of $\mathbf{a}$.

(b) The maximum rate of change of T at P occurs in the direction of the gradient, that is, in the direction of the vector $-\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$. The maximum rate of change equals the magnitude of the gradient, that is,

$$|\nabla T]_P| = \frac{200}{196}\sqrt{1+9+4} \approx 3.8. \qquad \blacksquare$$

16.6 Exercises

In Exercises 1–6 find the gradient of f at the indicated point.

1 $f(x, y) = \sqrt{x^2 + y^2}$; $P(-4, 3)$

2 $f(x, y) = 7y - 5x$; $P(2, 6)$

3 $f(x, y) = e^{3x} \tan y$; $P(0, \pi/4)$

4 $f(x, y) = x \ln (x - y)$; $P(5, 4)$

5 $f(x, y, z) = yz^3 - 2x^2$; $P(2, -3, 1)$

6 $f(x, y, z) = xy^2e^z$; $P(2, -1, 0)$

In Exercises 7–20 find the directional derivative of f at the point P in the indicated direction.

7 $f(x, y) = x^2 - 5xy + 3y^2$; $P(3, -1)$, $\mathbf{u} = (\sqrt{2}/2)(\mathbf{i} + \mathbf{j})$

8 $f(x, y) = x^3 - 3x^2y - y^3$; $P(1, -2)$, $\mathbf{u} = \frac{1}{2}(-\mathbf{i} + \sqrt{3}\mathbf{j})$

9 $f(x, y) = \arctan y/x$; $P(4, -4)$, $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j}$

10 $f(x, y) = x^2 \ln y$; $P(5, 1)$, $\mathbf{a} = -\mathbf{i} + 4\mathbf{j}$

11 $f(x, y) = \sqrt{9x^2 - 4y^2 - 1}$; $P(3, -2)$, $\mathbf{a} = \mathbf{i} + 5\mathbf{j}$

12 $f(x, y) = (x - y)/(x + y)$; $P(2, -1)$, $\mathbf{a} = 3\mathbf{i} + 4\mathbf{j}$

13 $f(x, y) = x \cos^2 y$; $P(2, \pi/4)$, $\mathbf{a} = \langle 5, 1\rangle$

14 $f(x, y) = xe^{3y}$; $P(4, 0)$, $\mathbf{a} = \langle -1, 3\rangle$

15 $f(x, y, z) = xy^3z^2$; $P(2, -1, 4)$, $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$

16 $f(x, y, z) = x^2 + 3yz + 4xy$; $P(1, 0, -5)$, $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$

17 $f(x, y, z) = z^2e^{xy}$; $P(-1, 2, 3)$, $\mathbf{a} = 3\mathbf{i} + \mathbf{j} - 5\mathbf{k}$

18 $f(x, y, z) = \sqrt{xy} \sin z$; $P(4, 9, \pi/4)$, $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$

19 $f(x, y, z) = (x + y)(y + z)$; $P(5, 7, 1)$, $\mathbf{a} = \langle -3, 0, 1\rangle$

20 $f(x, y, z) = z^2 \tan^{-1} (x + y)$; $P(0, 0, 4)$, $\mathbf{a} = \langle 6, 0, 1\rangle$

In Exercises 21–24

(a) find the directional derivative of f at P in the direction from P to Q;

(b) find a unit vector in the direction in which f increases most rapidly at P and find the rate of change of f in that direction.

(c) find a unit vector in the direction in which f decreases most rapidly at P and find the rate of change of f in that direction.

21 $f(x, y) = x^2e^{-2y}$; $P(2, 0)$, $Q(-3, 1)$

22 $f(x, y) = \sin (2x - y)$; $P(-\pi/3, \pi/6)$, $Q(0, 0)$

23 $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$; $P(-2, 3, 1)$, $Q(0, -5, 4)$

24 $f(x, y, z) = x/y - y/z$; $P(0, -1, 2)$, $Q(3, 1, -4)$

25 A metal plate is situated on an xy-plane such that the temperature T is inversely proportional to the distance from the origin. If the temperature at $P(3, 4)$ is 100°, find the rate of change of T at P in the direction of the vector $\mathbf{i} + \mathbf{j}$. In what direction does T increase most rapidly at P? In what direction does T decrease most rapidly at P? In what direction is the rate of change 0?

26 The surface of a certain lake is represented by a region D in the xy-plane such that the depth (in feet) under the point corresponding to (x, y) is $f(x, y) = 300 - 2x^2 - 3y^2$. If a boy in the water is at the point (4, 9), in what direction should he swim in order for the depth to decrease most rapidly? In what direction will the depth remain the same?

27 The electrical potential V at the point $P(x, y, z)$ in a rectangular coordinate system is $V = x^2 + 4y^2 + 9z^2$. Find the rate of change of V at $P(2, -1, 3)$ in the

direction from P to the origin. Find the direction that produces the maximum rate of change of V at P. What is the maximum rate of change at P?

28 An object is situated in a rectangular coordinate system such that the temperature T at the point $P(x, y, z)$ is given by $T = 4x^2 - y^2 + 16z^2$. Find the rate of change of T at the point $P(4, -2, 1)$ in the direction of the vector $2\mathbf{i} + 6\mathbf{j} - 3\mathbf{k}$. In what direction does T increase most rapidly at P? What is this maximum rate of change? In what direction does T decrease most rapidly at P? What is this rate of change?

29 Prove Theorem (16.28).

30 Prove that $D_{\mathbf{k}} f(x, y, z) = f_z(x, y, z)$.

If $u = f(x, y)$ and $v = g(x, y)$ where f and g are differentiable, prove the identities in Exercises 31–36.

31 $\nabla(cu) = c\,\nabla u$, where c is a constant

32 $\nabla(u + v) = \nabla u + \nabla v$

33 $\nabla(uv) = u\,\nabla v + v\,\nabla u$

34 $\nabla\left(\dfrac{u}{v}\right) = \dfrac{v\,\nabla u - u\,\nabla v}{v^2}, \quad v \neq 0$

35 $\nabla u^n = nu^{n-1}\,\nabla u$ for every real number n

36 If $w = h(u)$, then $\nabla w = \dfrac{dw}{du}\nabla u$.

37 Let $\mathbf{u}$ be a unit vector and let θ be the angle, measured in the counterclockwise direction, from the positive x-axis to the position vector corresponding to $\mathbf{u}$.

(a) Show that

$$D_{\mathbf{u}} f(x, y) = f_x(x, y)\cos\theta + f_y(x, y)\sin\theta$$

(b) If $f(x, y) = x^2 + 2xy - y^2$ and $\theta = 5\pi/6$, find $D_{\mathbf{u}} f(2, -3)$.

38 Refer to Exercise 37. If $f(x, y) = (xy + y^2)^4$ and $\theta = \pi/3$, find $D_{\mathbf{u}} f(2, -1)$.

39 If f, f_x, and f_y are continuous and $\nabla f(x, y) = \mathbf{0}$ throughout a rectangular region $R = \{(x, y)\colon a < x < b,\ c < y < d\}$, prove that $f(x, y)$ is constant on R (cf. Exercise 40 of Section 16.5).

40 Suppose $w = f(x, y)$ and $x = g(t)$, $y = h(t)$, where all functions are differentiable. If $\mathbf{r}(t) = x\mathbf{i} + y\mathbf{j}$, prove that $dw/dt = \nabla w \cdot \mathbf{r}'(t)$.

16.7 Tangent Planes and Normal Lines to Surfaces

Suppose a surface S is the graph of an equation $F(x, y, z) = 0$, where F has continuous first partial derivatives. Let $P_0(x_0, y_0, z_0)$ be a point on S at which F_x, F_y, and F_z are not all zero. A **tangent line** to S at P_0 is, by definition, a tangent line l to any curve C that lies on S and contains P_0, as illustrated in Figure 16.21.

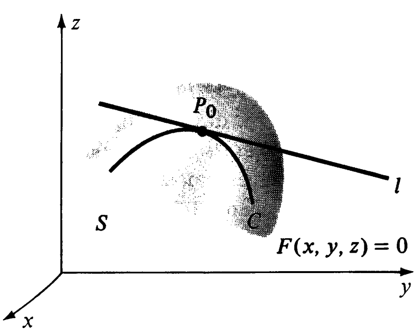


FIGURE 16.21

Suppose parametric equations for C are $x = f(t)$, $y = g(t)$, $z = h(t)$ and let t_0 be the value of t that corresponds to P_0. For each t the point $(f(t), g(t), h(t))$ on C is also on S and, therefore,

$$F(f(t), g(t), h(t)) = 0.$$

If we let

$$w = F(x, y, z), \quad \text{where } x = f(t),\ y = g(t),\ z = h(t),$$

then using Theorem (16.19) and the fact that $w = 0$ for all t,

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt} = 0,$$

that is,

$$F_x(x, y, z)f'(t) + F_y(x, y, z)g'(t) + F_z(x, y, z)h'(t) = 0.$$

In particular, at the point P_0,

$$F_x(x_0, y_0, z_0)f'(t_0) + F_y(x_0, y_0, z_0)g'(t_0) + F_z(x_0, y_0, z_0)h'(t_0) = 0.$$

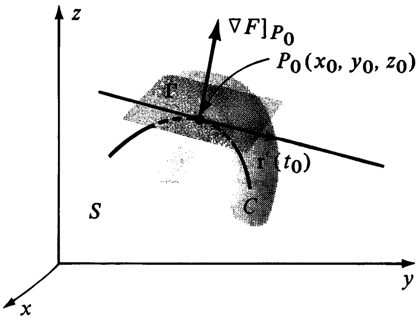


FIGURE 16.22

The preceding equation can be written compactly as

$$\nabla F]_{P_0} \cdot \mathbf{r}'(t_0) = 0,$$

where the symbol $\nabla F]_{P_0}$ denotes $\nabla F(x_0, y_0, z_0)$, and $\mathbf{r}(t) = \langle f(t), g(t), h(t)\rangle$. Since $\mathbf{r}'(t_0)$ is a tangent vector to C at P_0, this implies that *the vector* $\nabla F]_{P_0}$ *is orthogonal to every tangent line* l *to* S *at* P_0. The plane Γ through P_0 with normal vector $\nabla F]_{P_0}$ is called the **tangent plane to S at P_0**. We have shown that every tangent line l to S at P_0 lies in the tangent plane at P_0. A sketch illustrating these concepts appears in Figure 16.22. The following theorem summarizes our discussion.

Theorem (16.29)

Suppose $F(x, y, z)$ has continuous first partial derivatives and P_0 is a point on the graph S of $F(x, y, z) = 0$. If F_x, F_y, and F_z are not all 0 at P_0, then the vector $\nabla F]_{P_0}$ is normal to the tangent plane to S at P_0.

We shall refer to the vector $\nabla F]_{P_0}$ in Theorem (16.29) as a vector that is *normal to the surface* S at P_0. Applying Theorem (14.38) gives us the following corollary, where it is assumed that F_x, F_y, and F_z are not all 0 at (x_0, y_0, z_0).

Corollary (16.30)

An equation for the tangent plane to the graph of $F(x, y, z) = 0$ at the point (x_0, y_0, z_0) is

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0.$$

Example 1 Find an equation for the tangent plane to the ellipsoid given by $4x^2 + 9y^2 + z^2 - 49 = 0$ at the point $P_0(1, -2, 3)$.

Solution If $F(x, y, z)$ denotes the left side of the given equation, then

$$F_x(x, y, z) = 8x, \quad F_y(x, y, z) = 18y, \quad F_z(x, y, z) = 2z$$

and hence at P_0

$$F_x(1, -2, 3) = 8, \quad F_y(1, -2, 3) = -36, \quad F_z(1, -2, 3) = 6.$$

Applying Corollary (16.30) we obtain the equation

$$8(x - 1) - 36(y + 2) + 6(z - 3) = 0$$

which simplifies to

$$4x - 18y + 3z - 49 = 0.$$ ■

If $z = f(x, y)$ is an equation for S, then letting $F(x, y, z) = f(x, y) - z$, the equation in Corollary (16.30) becomes

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + (-1)(z - z_0) = 0.$$

We have proved the following result.

Theorem (16.31)

An equation for the tangent plane to the graph of $z = f(x, y)$ at the point (x_0, y_0, z_0) is

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

The line perpendicular to the tangent plane at a point P_0 on a surface S is called a **normal line** to S at P_0. If S is the graph of $F(x, y, z)$, then the normal line is determined by the vector $\nabla F]_{P_0}$.

Example 2 Find an equation of the normal line to the ellipsoid given by $4x^2 + 9y^2 + z^2 - 49 = 0$ at the point $P_0(1, -2, 3)$.

Solution This surface is the same as the one considered in Example 1. Consequently, the vector $\nabla F]_{P_0} = \langle 8, -36, 6 \rangle$ determines the desired normal line. Using Theorem (14.37) we obtain the following parametric equations for the normal line:

$$x = 1 + 8t, \quad y = -2 - 36t, \quad z = 3 + 6t$$

where t is a real number. A symmetric form for the line (see (14.41)) is

$$\frac{x - 1}{8} = \frac{y + 2}{-36} = \frac{z - 3}{6}. \qquad \blacksquare$$

Suppose f is a function of x and y, and S is the graph of the equation $z = f(x, y)$. The tangent plane Γ to S at $P_0(x_0, y_0, z_0)$ may be used to obtain a geometric interpretation for the differential

$$dz = f_x(x_0, y_0)\,\Delta x + f_y(x_0, y_0)\,\Delta y.$$

Since
$$\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0),$$

the point $P(x_0 + \Delta x, y_0 + \Delta y, z_0 + \Delta z)$ is on S. Let $D(x_0 + \Delta x, y_0 + \Delta y, z)$ be on the tangent plane Γ and consider the points $A(x_0 + \Delta x, y_0 + \Delta y, 0)$ and $B(x_0 + \Delta x, y_0 + \Delta y, z_0)$ as illustrated in Figure 16.23. Since D is on Γ its coordinates satisfy the equation in Theorem (16.31), that is,

$$\begin{aligned} z - z_0 &= f_x(x_0, y_0)(x_0 + \Delta x - x_0) + f_y(x_0, y_0)(y_0 + \Delta y - y_0) \\ &= f_x(x_0, y_0)\,\Delta x + f_y(x_0, y_0)\,\Delta y = dz. \end{aligned}$$

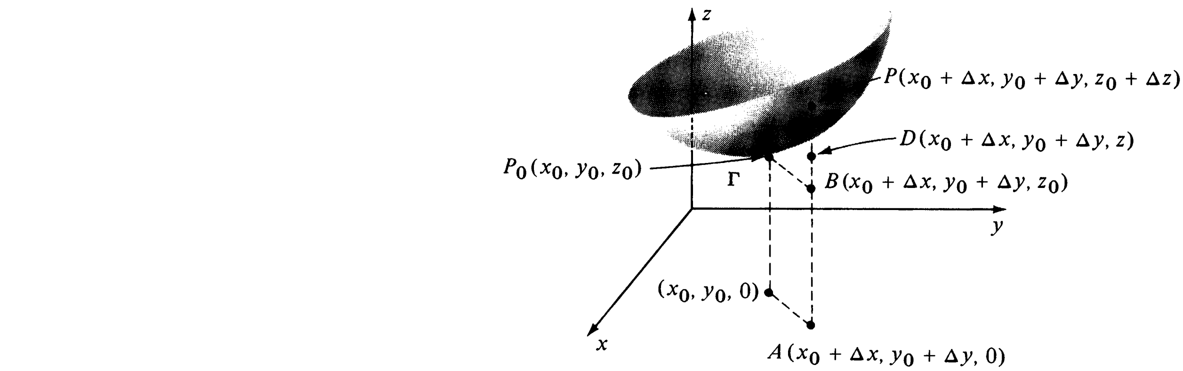


FIGURE 16.23

In terms of directed distances, we have shown that $dz = \overline{BD}$, that is, *dz is the directed distance from B to the point on the tangent plane that lies above (or below) B*. This is analogous to the single variable case $y = f(x)$ discussed in Section 3.5, where dy was interpreted as a distance to a point on a tangent line to the graph of f (see Figure 3.12). If follows that when dz is used as an approximation to z, it is assumed that the surface S almost coincides with its tangent plane at points close to P_0.

As an application of the discussion in this section, suppose a solid, liquid, or gas is situated in a three-dimensional coordinate system such that the temperature at the point $P(x, y, z)$ is $w = F(x, y, z)$. If $P_0(x_0, y_0, z_0)$ is a fixed point, then the graph of the equation

$$F(x, y, z) = F(x_0, y_0, z_0)$$

is the level (isothermal) surface S_0 that passes through P_0. For every point on S_0 the temperature is $w_0 = F(x_0, y_0, z_0)$. It is convenient to visualize all possible level surfaces that can be obtained by choosing different points. Several surfaces S_0, S_1, and S_2 through P_0, P_1, and P_2, respectively, corresponding to temperatures w_0, w_1, and w_2, are sketched in Figure 16.24. By Theorem (16.29), $\nabla F]_{P_0}$, $\nabla F]_{P_1}$, and $\nabla F]_{P_2}$ are normal vectors to S_0, S_1, and S_2 at the points P_0, P_1, and P_2, respectively.

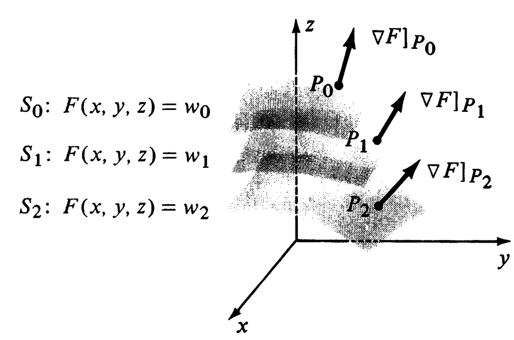


FIGURE 16.24

The preceding discussion may be summarized in the next theorem.

Theorem (16.32)

> If a function F of three variables, is differentiable at $P_0(x_0, y_0, z_0)$ and $\nabla F]_{P_0} \neq \mathbf{0}$, then $\nabla F]_{P_0}$ is a normal vector to the level surface of $F(x, y, z)$ that contains P_0.

Combining Theorem (16.32) with our work in the previous section gives us the fact that *the direction for the maximum rate of change of $F(x, y, z)$ at P_0 is along a normal vector to the level surface through P_0*, as illustrated by the isothermal surfaces in Figure 16.24. Another illustration of (16.32) occurs in electrical theory where $F(x, y, z)$ is the potential at $P(x, y, z)$. If a particle moves on one of the level (equipotential) surfaces, the potential remains constant. The potential changes most rapidly when a particle moves in a direction that is normal to an equipotential surface.

Example 3 If $F(x, y, z) = x^2 + y^2 + z$, sketch the level surface for F that passes through the point $P(1, 2, 4)$ and draw a vector with initial point P that corresponds to $\nabla F]_P$.

Solution The level surfaces for F are graphs of equations of the form $F(x, y, z) = c$, where c is a constant. In particular, the level surface that passes through $P(1, 2, 4)$ is the graph of

$$F(x, y, z) = F(1, 2, 4).$$

Using the formula for $F(x, y, z)$ gives us

$$x^2 + y^2 + z = (1)^2 + (2)^2 + 4 = 9$$

or equivalently,
$$z = 9 - x^2 - y^2.$$

The graph of this level surface is a paraboloid of revolution and is sketched in Figure 16.25. Since

$$\nabla F(x, y, z) = 2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$$

we see that

$$\nabla F]_P = 2(1)\mathbf{i} + 2(2)\mathbf{j} + \mathbf{k} = 2\mathbf{i} + 4\mathbf{j} + \mathbf{k}.$$

The vector with initial point P that represents $\nabla F]_P$ is shown in Figure 16.25. By Theorem (16.32), $\nabla F]_P$ is orthogonal to the level surface. ■

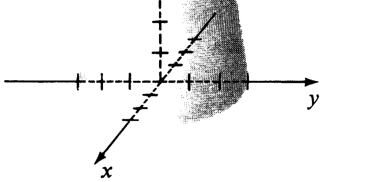


FIGURE 16.25

A result similar to Theorem (16.32) may be proved for functions of two variables. Specifically, given $f(x, y)$, we may show that the direction for the maximum rate of change of f at $P_0(x_0, y_0)$ is orthogonal to the level curve C through P_0; that is, orthogonal to the tangent line to C at P_0. For example, if $f(x, y) = 9 - x^2 - y^2$, the level curves are circles (see Figure 16.6) and hence the maximum rate of change of f takes place along lines through the origin. (Why?) This corresponds to going up or down the steepest part of the graph of f sketched in Figure 16.25.

Example 4 If $f(x, y) = x^2 + 2y^2$, sketch the level curve of f that passes through the point $P(3, 1)$ and draw a vector with initial point P that corresponds to $\nabla f]_P$.

Solution The level curve of f that passes through the point $P(3, 1)$ is given by $f(x, y) = f(3, 1)$, or

$$x^2 + 2y^2 = (3)^2 + 2(1)^2 = 11.$$

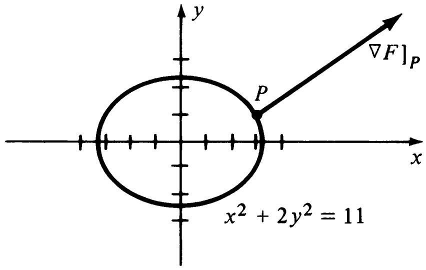


FIGURE 16.26

The graph is the ellipse sketched in Figure 16.26. The gradient of f is

$$\nabla f(x, y) = 2x\mathbf{i} + 4y\mathbf{j}$$

and hence $$\nabla f]_P = 2(3)\mathbf{i} + 4(1)\mathbf{j} = 6\mathbf{i} + 4\mathbf{j}.$$

The vector with initial point P that represents $\nabla f]_P$ is shown in Figure 16.26. Note that this vector is orthogonal to the level curve through P. ■

16.7 Exercises

In Exercises 1–10 find equations for the tangent plane and the normal line to the graph of the given equation at the indicated point P.

1 $4x^2 - y^2 + 3z^2 = 10$; $P(2, -3, 1)$

2 $9x^2 - 4y^2 - 25z^2 = 40$; $P(4, 1, -2)$

3 $z = 4x^2 + 9y^2$; $P(-2, -1, 25)$

4 $z = 4x^2 - y^2$; $P(5, -8, 36)$

5 $xy + 2yz - xz^2 + 10 = 0$; $P(-5, 5, 1)$

6 $x^3 - 2xy + z^3 + 7y + 6 = 0$; $P(1, 4, -3)$

7 $z = 2e^{-x} \cos y$; $P(0, \pi/3, 1)$

8 $z = \ln xy$; $P(\frac{1}{2}, 2, 0)$

9 $x = \ln (y/2z)$; $P(0, 2, 1)$

10 $xyz - 4xz^3 + y^3 = 10$; $P(-1, 2, 1)$

In Exercises 11–14 prove that an equation of the tangent plane to the given quadric surface at the point $P_0(x_0, y_0, z_0)$ may be written in the indicated form.

11 $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$; $\dfrac{xx_0}{a^2} + \dfrac{yy_0}{b^2} + \dfrac{zz_0}{c^2} = 1$

12 $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$; $\dfrac{xx_0}{a^2} - \dfrac{yy_0}{b^2} + \dfrac{zz_0}{c^2} = 1$

13 $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$; $\dfrac{xx_0}{a^2} - \dfrac{yy_0}{b^2} - \dfrac{zz_0}{c^2} = 1$

14 $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = cz$; $\dfrac{2xx_0}{a^2} + \dfrac{2yy_0}{b^2} = c(z + z_0)$

15 Find the points on the hyperboloid $x^2 - 2y^2 - 4z^2 = 16$ at which the tangent plane is parallel to the plane $4x - 2y + 4z = 5$.

16 Prove that the sum of the squares of the x-, y-, and z-intercepts of every tangent plane to the graph of the equation $x^{2/3} + y^{2/3} + z^{2/3} = a^{2/3}$ is a constant.

17 Prove that every normal line to a sphere passes through the center of the sphere.

18 Find the points on the paraboloid $z = 4x^2 + 9y^2$ at which the normal line is parallel to the line through $P(-2, 4, 3)$ and $Q(5, -1, 2)$.

19 Two surfaces are said to be **orthogonal** at a point of intersection $P(x, y, z)$ if their normal lines at P are orthogonal. Show that the graphs of $F(x, y, z) = 0$ and $G(x, y, z) = 0$ (where F and G have partial derivatives) are orthogonal at P if and only if

$$F_xG_x + F_yG_y + F_zG_z = 0.$$

20 Prove that the sphere $x^2 + y^2 + z^2 = a^2$ and the cone $x^2 + y^2 - z^2 = 0$ are orthogonal at every point of intersection (see Exercise 19).

In Exercises 21–24 sketch the level curve C for f that passes through the point P and draw a vector (with initial point P) corresponding to $\nabla f]_P$ (see Exercises 23–26 of Section 16.1).

21 $f(x, y) = y^2 - x^2$; $P(2, 1)$

22 $f(x, y) = 3x - 2y$; $P(-2, 1)$

23 $f(x, y) = x^2 - y$; $P(-3, 5)$

24 $f(x, y) = xy$; $P(3, 2)$

In Exercises 25–30 sketch the level surface S for F that passes through the point P and draw a vector (with initial point P) corresponding to $\nabla F]_P$ (see Exercises 33–38 of Section 16.1).

25 $F(x, y, z) = x^2 + y^2 + z^2$; $P(1, 5, 2)$

26 $F(x, y, z) = z - x^2 - y^2$; $P(2, -2, 1)$

27 $F(x, y, z) = x + 2y + 3z$; $P(3, 4, 1)$

28 $F(x, y, z) = x^2 + y^2 - z^2$; $P(3, -1, 1)$

29 $F(x, y, z) = x^2 + y^2$; $P(2, 0, 3)$

30 $F(x, y, z) = z$; $P(2, 3, 4)$

16.8 Extrema of Functions of Several Variables

Throughout the remainder of this chapter, the term **rectangular region** will be used for points in a coordinate plane that lie inside a rectangle whose sides are parallel to the coordinate axes. If we wish to include the points on the boundary, the term **closed rectangular region** will be used. A region R is **bounded** if it is a subregion of a closed rectangular region.

In a manner analogous to the single variable case, a function f of two variables is said to have a **local maximum** at (a, b) if there is a rectangular region R containing (a, b) such that $f(x, y) \leq f(a, b)$ for all other pairs (x, y) in R. Geometrically, if a surface S is the graph of f, then the local maxima correspond to the high points on S. If f_y exists, then as pointed out in Section 16.3, $f_y(a, b)$ is the slope of the tangent line to the trace C of S on the plane $x = a$ (see Figure 16.17). It follows that if $f(a, b)$ is a local maximum, then $f_y(a, b) = 0$. Similarly $f_x(a, b) = 0$.

The function f has a **local minimum** at (c, d) if there is a rectangular region R containing (c, d) such that $f(x, y) \geq f(c, d)$ for all other (x, y) in R. If f has first partial derivatives, they must be zero at (c, d). The local minima correspond to the low points on the graph of f.

It can be shown that if a function f of two variables is continuous on a closed and bounded region R, then f has an **absolute maximum** $f(a, b)$ and an **absolute minimum** $f(c, d)$ for some (a, b) and (c, d) in R. This means that

$$f(c, d) \leq f(x, y) \leq f(a, b)$$

for all (x, y) in R. The proof may be found in texts on advanced calculus.

The local maxima and minima are called the **local extrema** of f. The **extrema** include the absolute maximum and minimum (if they exist). If f has continuous first partial derivatives, then from the preceding discussion, the pairs that give rise to local extrema are solutions of *both* of the equations

$$f_x(x, y) = 0 \quad \text{and} \quad f_y(x, y) = 0.$$

As was the case for functions of one variable, local extrema can also occur at pairs where f_x or f_y does not exist. Since all of these pairs are crucial for finding local extrema, we shall give them a special name.

***Definition* (16.33)**

> Let f be a function of two variables. A pair (a, b) is a **critical point** of f if either
>
> (i) $f_x(a, b) = 0$ and $f_y(a, b) = 0$, or
>
> (ii) $f_x(a, b)$ or $f_y(a, b)$ does not exist.

If f has continuous first partial derivatives, and if (a, b) is a critical point, then the graph of f has a horizontal tangent plane at the point $(a, b, f(a, b))$. (Why?) When searching for local extrema of a function we usually begin by finding the critical points and then, in some way, we test each pair to determine whether it corresponds to a local maximum or minimum.

An absolute maximum or minimum of a function of two variables may occur at a boundary point of its domain R. The investigation of such **boundary extrema** usually requires a separate procedure, as was the case with endpoint extrema for functions of one variable. If there are no boundary points as, for example, when R is the entire xy-plane or the interior of a circle, then there can be no boundary extrema.

Example 1 Let $f(x, y) = 1 + x^2 + y^2$, where $0 \le x^2 + y^2 \le 4$. Find the extrema of f.

Solution The domain R of f is the region in the xy-plane consisting of the circle $x^2 + y^2 = 4$ and its interior. Since $f_x(x, y) = 2x$ and $f_y(x, y) = 2y$, the critical points are the (simultaneous) solutions of

$$2x = 0 \quad \text{and} \quad 2y = 0.$$

The only pair that satisfies both equations is $(0, 0)$ and hence $f(0, 0) = 1$ is the only possible local extremum. Moreover, since

$$f(x, y) = 1 + x^2 + y^2 > 1 \quad \text{if } (x, y) \ne (0, 0),$$

f has the local minimum 1 at $(0, 0)$. This fact may also be seen by referring to the graph of $z = f(x, y)$, sketched in Figure 16.27. It should be clear that the number $f(0, 0) = 1$ is also the absolute minimum of f.

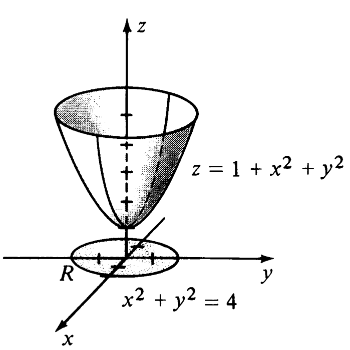


FIGURE 16.27

To find possible boundary extrema, we investigate points (a, b) that lie on the boundary of the domain of f, that is, on the circle $x^2 + y^2 = 4$. Referring to Figure 16.27, we see that any such point, for example $(0, 2)$, leads to the absolute maximum $f(0, 2) = 5$. ■

As illustrated in the next example, not every critical point may lead to an extremum.

Example 2 If $f(x, y) = y^2 - x^2$ and the domain of f is $\mathbb{R} \times \mathbb{R}$, find the extrema of f.

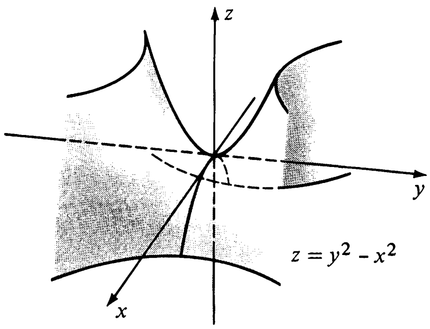


FIGURE 16.28

Solution Since $f_x(x, y) = -2x$ and $f_x(x, y) = 2y$, the only possible local extremum is $f(0,0) = 0$. (Why?) However, if $y \neq 0$, then $f(0, y) = y^2 > 0$; and if $x \neq 0$, then $f(x, 0) = -x^2 < 0$. Thus every rectangular region in the xy-plane containing (0, 0) contains pairs at which functional values are greater than $f(0, 0)$ as well as pairs at which values are less than $f(0, 0)$. Consequently f has no local extrema. This is also evident from the graph of $z = f(x, y)$ sketched in Figure 16.28. Because of the shape of the surface near (0, 0, 0), this point is called a **saddle point** of the graph. Since the domain of f has no boundary, there are no boundary extrema. ■

The following test, which we state without proof, is useful for more complicated functions. In a sense, it is the counterpart of the Second Derivative Test for functions of one variable. The interested reader may consult advanced calculus books for a proof.

Test for Extrema (16.34)

Let f be a function of two variables which has continuous second partial derivatives on a rectangular region Q and let

$$g(x, y) = f_{xx}(x, y)f_{yy}(x, y) - [f_{xy}(x, y)]^2$$

for all (x, y) in Q. If (a, b) is in Q and $f_x(a, b) = 0$, $f_y(a, b) = 0$, then the following statements hold.

(i) $f(a, b)$ is a local maximum of f if $g(a, b) > 0$ and $f_{xx}(a, b) < 0$.

(ii) $f(a, b)$ is a local minimum of f if $g(a, b) > 0$ and $f_{xx}(a, b) > 0$.

(iii) $f(a, b)$ is not an extremum of f if $g(a, b) < 0$.

Note that if $g(a, b) > 0$, then $f_{xx}(a, b)f_{yy}(a, b)$ is positive and hence $f_{xx}(a, b)$ and $f_{yy}(a, b)$ agree in sign. Thus, we may replace $f_{xx}(a, b)$ in (i) and (ii) by $f_{yy}(a, b)$. If $g(a, b) = 0$ the test gives no information and in this case it is necessary to use a more direct approach such as that illustrated in Examples 1 and 2.

Example 3 If $f(x, y) = x^2 - 4xy + y^3 + 4y$, find the local extrema of f.

Solution Since $f_x(x, y) = 2x - 4y$ and $f_y(x, y) = -4x + 3y^2 + 4$, the critical points are solutions of the equations

$$2x - 4y = 0 \quad \text{and} \quad -4x + 3y^2 + 4 = 0.$$

Solving these simultaneously we obtain the pairs (4, 2) and $(\frac{4}{3}, \frac{2}{3})$. (Verify this fact.) The second partial derivatives of f are

$$f_{xx}(x, y) = 2, \quad f_{xy}(x, y) = -4, \quad \text{and} \quad f_{yy}(x, y) = 6y$$

and, therefore, $g(x, y) = 12y - 16$. Since $g(\frac{4}{3}, \frac{2}{3}) = -8 < 0$, it follows from (iii) of (16.34) that $f(\frac{4}{3}, \frac{2}{3})$ is not an extremum of f. Since $g(4, 2) = 8 > 0$ and $f_{xx}(4, 2) = 2 > 0$, it follows from (ii) of (16.34) that f has a local minimum $f(4, 2) = 0$. ■

Example 4 If $f(x, y) = x^2 - 4xy + y^3 + 4y$, find the absolute extrema of f on the triangular region R that has vertices $(-1, -1)$, $(7, -1)$ and $(7, 7)$.

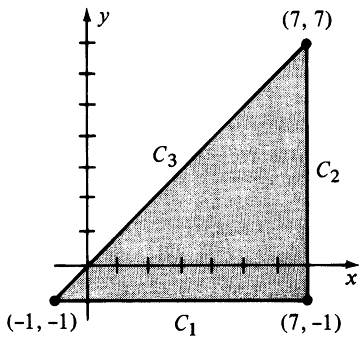


FIGURE 16.29

Solution The boundary of R consists of the line segments C_1, C_2, and C_3 shown in Figure 16.29. In Example 3 we found that f has a local minimum 0 at the point (4, 2), which is within R. Thus we need only check for boundary extrema.

On C_1 we have $y = -1$, and hence values of f are given by

$$f(x, -1) = x^2 + 4x - 1 - 4 = x^2 + 4x - 5.$$

This determines a function of one variable whose domain is the interval $[-1, 7]$. The first derivative is $2x + 4$, which equals 0 at $x = -2$, a number outside the interval $[-1, 7]$. Thus, there is no local extremum of $f(x, -1)$ on $[-1, 7]$. Indeed, $f(x, -1)$ is increasing throughout this interval and there is an endpoint minimum $f(-1, -1) = -8$ and an endpoint maximum $f(7, -1) = 72$.

On C_2 we have $x = 7$ and values of f are given by

$$f(7, y) = 49 - 28y + y^3 + 4y = y^3 - 24y + 49,$$

where $-1 \leq y \leq 7$. The first derivative of this function of y is $3y^2 - 24$, and hence there is a critical number if $3y^2 = 24$, or $y = \sqrt{8} = 2\sqrt{2}$. Using the second derivative $6y$ of $f(7, y)$ we see that $f(7, 2\sqrt{2})$ is a local minimum for f on the segment C_2. However, since $f(7, 2\sqrt{2}) = 49 - 32\sqrt{2} \approx 3.7$ and $f(-1, -1) = -8$, it is not an absolute minimum for f on R. The values of f at the endpoints of C_2 are

$$f(7, -1) = 72 \quad \text{and} \quad f(7, 7) = 224.$$

Finally, on C_3 we have $y = x$ and values of f are given by

$$f(x, x) = x^2 - 4x^2 + x^3 + 4x = x^3 - 3x^2 + 4x.$$

The first derivative $3x^2 - 6x + 4$ has no real roots and hence there are no critical numbers for $f(x, x)$. We have already calculated the endpoint values $f(-1, -1) = -8$ and $f(7, 7) = 224$. It follows that these are the absolute minimum and absolute maximum values for f on the triangular region R. ■

Example 5 A rectangular box with no top and having a volume of 12 ft^3 is to be constructed. The cost per square foot of the material to be used is \$4 for the bottom, \$3 for two of the opposite sides, and \$2 for the remaining pair of opposite sides. Find the dimensions of the box that will minimize the cost.

Solution Let x and y denote the dimensions (in feet) of the base, and z the altitude (in feet). Since there are two sides of area xz and two sides of area yz, the cost C (in dollars) of the material is given by

$$C = 4xy + 3(2xz) + 2(2yz)$$

where x, y, and z are all different from 0. Since $xyz = 12$, it follows that $z = 12/xy$. Substituting in the formula for C and simplifying, we obtain

$$C = 4xy + \frac{72}{y} + \frac{48}{x}.$$

There are no boundary points (Why?) and hence there are no boundary extrema. To determine the possible local extrema we must find the simultaneous solutions of the equations

$$C_x = 4y - \frac{48}{x^2} = 0 \quad \text{and} \quad C_y = 4x - \frac{72}{y^2} = 0.$$

These equations imply that

$$x^2y = 12 \quad \text{and} \quad xy^2 = 18.$$

We see from the first equation that $y = 12/x^2$. Substitution in the second equation gives us

$$x\left(\frac{144}{x^4}\right) = 18, \quad \text{or} \quad 144 = 18x^3.$$

Solving for x we obtain $x^3 = 8$, or $x = 2$. Since $y = 12/x^2$, the corresponding value of y is $\frac{12}{4}$, or 3. Theorem 16.34 can be used to show that these values of x and y determine a minimum value of C. Finally, using $z = 12/xy$ we obtain $z = 12/(2 \cdot 3) = 2$. Thus the minimum cost occurs if the dimensions of the base are 3 ft by 2 ft and the altitude is 2 ft. The longer sides should be made out of the \$2 material and the shorter sides out of the \$3 material. ■

The material in this section can be generalized to functions of more than two variables. For example, given $f(x, y, z)$ we define local maxima and minima in a manner analogous to the two-variable case. If f has first partial derivatives, then a local extremum can occur only at a point where f_x, f_y, and f_z are simultaneously 0. It is difficult to obtain tests for determining whether such a point corresponds to a maximum, a minimum, or neither. However, in applications it is often possible to determine this by analyzing the physical nature of the problem.

16.8 Exercises

In Exercises 1–16 find the extrema of f.

1 $f(x, y) = x^2 + 2xy + 3y^2$

2 $f(x, y) = x^2 - 3xy - y^2 + 2y - 6x$

3 $f(x, y) = x^3 + 3xy - y^3$

4 $f(x, y) = 4x^3 - 2x^2y + y^2$

5 $f(x, y) = x^2 + 4y^2 - x + 2y$

6 $f(x, y) = 5 + 4x - 2x^2 + 3y - y^2$

7 $f(x, y) = x^4 + y^3 + 32x - 9y$

8 $f(x, y) = \cos x + \cos y$

9 $f(x, y) = e^x \sin y$

10 $f(x, y) = x \sin y$

11 $f(x, y) = (4y + x^2y^2 + 8x)/xy$

12 $f(x, y) = x/(x + y)$

13 $f(x, y) = \sin x + \sin y + \sin (x + y)$; $0 \le x \le 2\pi, 0 \le y \le 2\pi$

14 $f(x, y) = (x + y + 1)^2/(x^2 + y^2 + 1)$

15 $f(x, y) = yx^2 + y^3 - 4x^2$

16 $f(x, y) = x^2 - 6x \cos y + 9$

In Exercises 17–22 find the absolute maximum and minimum values of f if the domain is the indicated region R. (Refer to Exercises 1–6 for local extrema.)

17 $f(x, y) = x^2 + 2xy + 3y^2$; $R = \{(x, y): -2 \le x \le 4, -1 \le y \le 3\}$.

(*Hint*: To determine boundary extrema, consider the following functions of *one* variable: $f(x, -1)$, $f(x, 3)$, $f(-2, y)$, and $f(-4, y)$.)

18 $f(x, y) = x^2 - 3xy - y^2 + 2y - 6x$; $R = \{(x, y): |x| \le 3, |y| \le 2\}$

19 $f(x, y) = x^3 + 3xy - y^3$; R the triangular region with vertices $(1, 2)$, $(1, -2)$ and $(-1, -2)$.

20 $f(x, y) = 4x^3 - 2x^2y + y^2$; R the region bounded by the graphs of $y = x^2$ and $y = 9$.

21 $f(x, y) = x^2 + 4y^2 - x + 2y$; R the region bounded by the ellipse $x^2 + 4y^2 = 1$.

22 $f(x, y) = 5 + 4x - 2x^2 + 3y - y^2$; R the triangle bounded by the lines $y = x$, $y = -x$, and $y = 2$.

23 Find the shortest distance from the point $P(2, 1, -1)$ to the plane $4x - 3y + z = 5$.

24 Find the shortest distance between the planes $2x + 3y - z = 2$ and $2x + 3y - z = 4$.

25 Find the points on the graph of $xy^3z^2 = 16$ that are closest to the origin.

26 Find three positive real numbers whose sum is 1000 and whose product is a maximum.

27 If an open rectangular box is to have a fixed volume V, what relative dimensions will make the surface area a minimum?

28 If an open rectangular box is to have a fixed surface area A, what relative dimensions will make the volume a maximum?

29 Find the dimensions of the rectangular parallelepiped of maximum volume with faces parallel to the coordinate planes that can be inscribed in the ellipsoid $16x^2 + 4y^2 + 9z^2 = 144$.

30 Generalize Exercise 29 to any ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$.

31 Find the dimensions of the rectangular parallelepiped of maximum volume that has three of its faces in the coordinate planes, one vertex at the origin, and another vertex in the first octant on the plane $4x + 3y + z = 12$.

32 Generalize Exercise 31 to any plane $x/a + y/b + z/c = 1$ where a, b, and c are positive real numbers.

33 A company plans to manufacture closed rectangular boxes that have a volume of 8 ft^3. If the material for the top and bottom costs twice as much as the material for the sides, find the dimensions that will minimize the cost.

34 A window has the shape of a rectangle surmounted by an isosceles triangle, as illustrated in the figure. If the perimeter of the window is 12 ft, what values of x, y, and θ will maximize the total area?

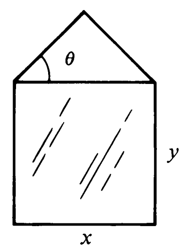


FIGURE FOR EXERCISE 34

35 Suppose a post office will not mail a rectangular box if the sum of its length and girth (the perimeter of a cross section that is perpendicular to the length) is more than 84 in. Find the dimensions of the box of maximum volume that can be mailed.

36 Find a vector in three dimensions having magnitude 8 such that the sum of its components is as large as possible.

37 In scientific experiments, corresponding values of two quantities x and y are often tabulated as follows:

x values	x_1	x_2	$\cdots$	x_n
y values	y_1	y_2	$\cdots$	y_n

Plotting the points (x_i, y_i) may lead the investigator to conjecture that x and y are related *linearly*, that is, $y = mx + b$ for some m and b. Thus it is desirable to find a line l having that equation which "best fits" the data, as illustrated in the figure on the next page. Statisticians call l a **linear regression line**.

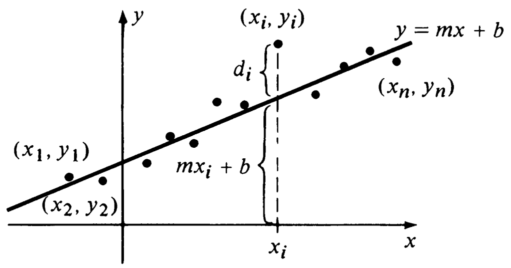


FIGURE FOR EXERCISE 27

One technique for finding l is to employ the **method of least squares**. To use this method consider, for each i, the *vertical deviation* $d_i = y_i - (mx_i + b)$ of the point (x_i, y_i) from the line $y = mx + b$ (see figure). Values of m and b are then determined which minimize the sum of the squares $\sum_{i=1}^{n} d_i^2$ (the squares d_i^2 are used because some of the d_i may be negative). Substituting for d_i produces the following function f of m and b:

$$f(m, b) = \sum_{i=1}^{n} (y_i - mx_i - b)^2.$$

Show that the line $y = mx + b$ of best fit occurs if

$$\left(\sum_{i=1}^{n} x_i\right)m + nb = \sum_{i=1}^{n} y_i$$

and
$$\left(\sum_{i=1}^{n} x_i^2\right)m + \left(\sum_{i=1}^{n} x_i\right)b = \sum_{i=1}^{n} x_i y_i.$$

Thus the line can be found by solving this system of two equations for the two unknowns m and b.

38 If the equations in Exercise 37 are true, show that the sum $\sum_{i=1}^{n} d_i$ of the deviations is 0. (This means that the positive and negative deviations cancel one another, and it is one reason for using $\sum_{i=1}^{n} d_i^2$ in the method of least squares.)

In Exercises 39 and 40, use the method of least squares (see Exercise 37) to find a line $y = mx + b$ that best fits the given data.

39

x values	1	4	7
y values	3	5	6

40

x values	1	4	6	8
y values	1	3	2	4

41 The following table lists the relationship between semester averages and scores on the final examination for ten students in a mathematics class.

Semester average	40	55	62	68	72	76	80	86	90	94
Final examination	30	45	65	72	60	82	76	92	88	98

Fit these data to a straight line and use it to estimate the final examination grade of a student with a class average of 70.

42 In studying the stress-strain diagram of an elastic material (see page 281), an engineer finds that part of the curve appears to be linear. Experimental values are listed in the following table, where the stress is measured in pounds and the strain in inches.

Stress	2	2.2	2.4	2.6	2.8	3.0
Strain	0.10	0.30	0.40	0.60	0.70	0.90

Fit these data to a straight line and estimate the strain when the stress is 2.5 lb.

16.9 Lagrange Multipliers

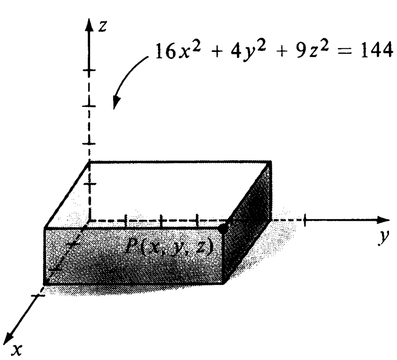


FIGURE 16.30

The definitions of extrema given in the preceding section can be extended to a function f of any number of variables. In many applications it is necessary to find the extrema of f when the variables are restricted in some manner. As an illustration, suppose we wish to find the volume of the largest rectangular box with faces parallel to the coordinate planes that can be inscribed in the ellipsoid $16x^2 + 4y^2 + 9z^2 = 144$. Note that by symmetry it is sufficient to examine the part in the first octant illustrated in Figure 16.30.

If $P(x, y, z)$ is the vertex indicated in the figure, then the volume V of the box is $V = 8xyz$. The problem is to find the maximum value of V subject to the **constraint** (or **side condition**)

$$16x^2 + 4y^2 + 9z^2 - 144 = 0.$$

Solving this equation for z and substituting in the formula for V, we obtain

$$V = (8xy)\tfrac{1}{3}\sqrt{144 - 16x^2 - 4y^2}.$$

The extrema may then be found using (16.34); however, this method is cumbersome because of the manipulations involved in finding partial derivatives and critical points. Another disadvantage of this technique is that for similar problems it may be impossible to solve for z. For these reasons it is sometimes simpler to employ the method of *Lagrange multipliers* discussed in this section.*

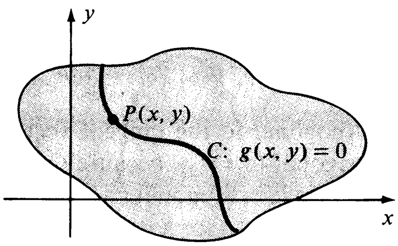


FIGURE 16.31

As another illustration, let $f(x, y)$ be the temperature at the point $P(x, y)$ on the flat metal plate illustrated in Figure 16.31, and let C be a curve that has the rectangular equation $g(x, y) = 0$. Suppose we wish to find the points on C at which the temperature attains its largest or smallest value. This amounts to finding the extrema of $f(x, y)$ *subject to the constraint* $g(x, y) = 0$. One technique for accomplishing this is to use the following result.

Lagrange's Theorem (16.35)

Let $f(x, y)$ and $g(x, y)$ have continuous first partial derivatives, and suppose f has an extremum $f(x_0, y_0)$ when (x, y) is subject to the constraint $g(x, y) = 0$. If $\nabla g(x_0, y_0) \neq \mathbf{0}$, then there is a real number λ such that

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0).$$

Proof The graph of $g(x, y) = 0$ is a curve C in the xy-plane. It can be shown that under the given conditions, C has a smooth parametric representation

$$x = k(t), \quad y = h(t),$$

where t is in some interval I. Let

$$\mathbf{r}(t) = x\mathbf{i} + y\mathbf{j} = k(t)\mathbf{i} + h(t)\mathbf{k}$$

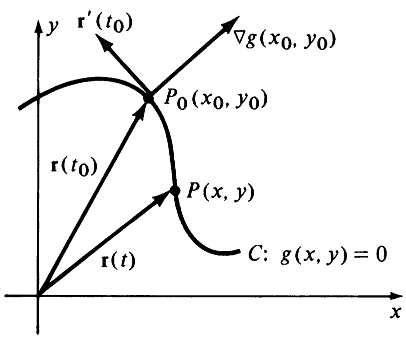


FIGURE 16.32

be the position vector of the point $P(x, y)$ on C (see Figure 16.32) and let the point $P_0(x_0, y_0)$ correspond to $t = t_0$; that is,

$$\mathbf{r}(t_0) = x_0\mathbf{i} + y_0\mathbf{j} = k(t_0)\mathbf{i} + h(t_0)\mathbf{j}.$$

If we define a function F of one variable t by

$$F(t) = f(k(t), h(t))$$

then as t varies we obtain functional values $f(x, y)$ which correspond to (x, y) on C; that is, $f(x, y)$ is subject to the constraint $g(x, y) = 0$. Since

* This method was discovered by the French mathematician Joseph Lagrange (1736–1813).

$f(x_0, y_0)$ is an extremum of f under these conditions, it follows that $F(t_0) = f(k(t_0), h(t_0))$ is an extremum of $F(t)$. Thus $F'(t_0) = 0$. If we regard F as a composite function, then by Theorem (16.19),

$$F'(t) = f_x(x, y)\frac{dx}{dt} + f_y(x, y)\frac{dy}{dt}$$

and, therefore,

$$0 = F'(t_0) = f_x(x_0, y_0)k'(t_0) + f_y(x_0, y_0)h'(t_0) = \nabla f(x_0, y_0) \cdot \mathbf{r}'(t_0).$$

This shows that the vector $\nabla f(x_0, y_0)$ is orthogonal to the tangent vector $\mathbf{r}'(t_0)$ to C. However, $\nabla g(x_0, y_0)$ is also orthogonal to $\mathbf{r}'(t_0)$, because C is a level curve for g. Since $\nabla f(x_0, y_0)$ and $\nabla g(x_0, y_0)$ are orthogonal to the same vector they are parallel, that is $\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0)$ for some λ. □

The number λ which appears in Theorem (16.35) is called a **Lagrange multiplier**. When applying this method we begin by considering the equations

$$\nabla f(x, y) = \lambda \nabla g(x, y) \quad \text{and} \quad g(x, y) = 0$$

or, equivalently,

(16.36)
$$f_x(x, y) = \lambda g_x(x, y), \quad f_y(x, y) = \lambda g_y(x, y) \quad \text{and} \quad g(x, y) = 0.$$

If f has an extremum at (x_0, y_0) subject to the constraint $g(x, y) = 0$, then (x_0, y_0) must be a solution of each equation in (16.36). Thus, to determine the extrema find the pairs (a_i, b_i) that, along with a suitable λ, satisfy all three equations in (16.36). If f has a maximum (or minimum) value it will be the largest (or smallest) of the corresponding functional values $f(a_i, b_i)$.

Example 1 Find the extrema of $f(x, y) = xy$ if (x, y) is restricted to the ellipse $4x^2 + y^2 = 4$.

Solution In this example the constraint is $g(x, y) = 4x^2 + y^2 - 4 = 0$. Setting $\nabla f(x, y) = \lambda \nabla g(x, y)$, we obtain

$$y\mathbf{i} + x\mathbf{j} = \lambda(8x\mathbf{i} + 2y\mathbf{j})$$

and equations (16.36) take on the form

$$y = 8x\lambda, \quad x = 2y\lambda \quad \text{and} \quad 4x^2 + y^2 - 4 = 0.$$

There are many ways to solve this system. One is to begin by writing

$$x = 2y\lambda = 2(8x\lambda)\lambda = 16x\lambda^2.$$

This leads to

$$x - 16x\lambda^2 = 0, \quad \text{or} \quad x(1 - 16\lambda^2) = 0$$

and hence either $x = 0$ or $\lambda = \pm\frac{1}{4}$. If $x = 0$, then using $4x^2 + y^2 - 4 = 0$ we obtain $y = \pm 2$. Thus the points $(0, \pm 2)$ are possibilities for extrema of $f(x, y)$. If $\lambda = \pm\frac{1}{4}$, then $y = 8x\lambda = 8x(\pm\frac{1}{4})$ or $y = \pm 2x$. Again using the fact that $4x^2 + y^2 - 4 = 0$ we get

$$4x^2 + 4x^2 - 4 = 0, \quad 8x^2 = 4 \quad \text{or} \quad x = \pm 1/\sqrt{2} = \pm\sqrt{2}/2.$$

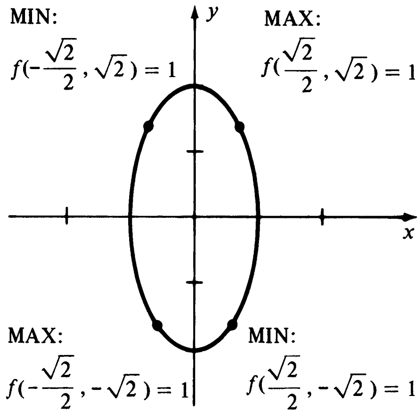


FIGURE 16.33

The corresponding y-values satisfy

$$4(\tfrac{1}{2}) + y^2 - 4 = 0, \quad y^2 = 2 \quad \text{or} \quad y = \pm\sqrt{2}.$$

This gives us the points $(\sqrt{2}/2, \pm\sqrt{2})$ and $(-\sqrt{2}/2, \pm\sqrt{2})$. The value of f at each of the points we have found is listed below.

(x, y)	$(0, 2)$	$(0, -2)$	$(\sqrt{2}/2, \sqrt{2})$	$(\sqrt{2}/2, -\sqrt{2})$	$(-\sqrt{2}/2, \sqrt{2})$	$(-\sqrt{2}/2, -\sqrt{2})$
$f(x, y)$	0	0	1	-1	-1	1

It follows that $f(x, y)$ takes on a maximum value 1 at either $(\sqrt{2}/2, \sqrt{2})$ or $(-\sqrt{2}/2, -\sqrt{2})$ and a minimum value -1 at $(\sqrt{2}/2, -\sqrt{2})$ or $(-\sqrt{2}/2, \sqrt{2})$. These facts are indicated on the ellipse in Figure 16.33. ■

Lagrange's Theorem (16.35) can be extended to functions of more than two variables. For example, to determine the extrema of $f(x, y, z)$ subject to the constraint $g(x, y, z) = 0$ we begin by finding the (simultaneous) solutions of the equations

(16.37) $$\nabla f(x, y, z) = \lambda \nabla g(x, y, z) \quad \text{and} \quad g(x, y, z) = 0$$

or, equivalently, of

(16.38) $$f_x = \lambda g_x, \quad f_y = \lambda g_y, \quad f_z = \lambda g_z, \quad g = 0$$

where we have simplified the notation by deleting the functional value symbols in $f_x(x, y, z)$, $g_x(x, y, z)$, etc. A similar situation exists for functions of four or more variables.

The next example was discussed briefly at the beginning of this section.

Example 2 Find the volume of the largest rectangular box with faces parallel to the coordinate planes that can be inscribed in the ellipsoid $16x^2 + 4y^2 + 9z^2 = 144$.

Solution A typical position for the box is illustrated in Figure 16.30. We wish to maximize

$$V = f(x, y, z) = 8xyz$$

subject to the constraint

$$g(x, y, z) = 16x^2 + 4y^2 + 9z^2 - 144 = 0.$$

As in (16.37), let us begin by considering $\nabla f(x, y, z) = \lambda \nabla g(x, y, z)$, or

$$8yz\mathbf{i} + 8xz\mathbf{j} + 8xy\mathbf{k} = \lambda(32x\mathbf{i} + 8y\mathbf{j} + 18z\mathbf{k}).$$

This, together with $g(x, y, z) = 0$ gives us the following system of four equations:

$$8yz = 32x\lambda, \quad 8xz = 8y\lambda, \quad 8xy = 18z\lambda, \quad 16x^2 + 4y^2 + 9z^2 - 144 = 0.$$

Multiplying the first equation by x, the second by y, and the third by z and adding gives us

$$24xyz = 32x^2\lambda + 8y^2\lambda + 18z^2\lambda = 2\lambda(16x^2 + 4y^2 + 9z^2).$$

This, together with the fact that $16x^2 + 4y^2 + 9z^2 = 144$ implies that

$$24xyz = 2\lambda(144) \quad \text{or} \quad xyz = 12\lambda.$$

The last equation may be used to find x, y, and z. For example, multiplying both sides of the equation $8yz = 32x\lambda$ by x, we obtain

$$\begin{aligned} 8xyz &= 32x^2\lambda \\ 8(12\lambda) &= 32x^2\lambda \\ 96\lambda - 32x^2\lambda &= 0 \\ 32\lambda(3 - x^2) &= 0. \end{aligned}$$

Consequently, either $\lambda = 0$ or $x = \sqrt{3}$. We may reject $\lambda = 0$ since in this case $xyz = 0$ and hence $V = 8xyz = 0$. Thus the only possibility for x is $\sqrt{3}$.

Similarly, multiplying both sides of the equation $8xz = 8y\lambda$ by y leads to

$$\begin{aligned} 8xyz &= 8y^2\lambda \\ 8(12\lambda) &= 8y^2\lambda \\ 96\lambda - 8y^2\lambda &= 0 \\ 8\lambda(12 - y^2) &= 0 \end{aligned}$$

and hence $y = \sqrt{12} = 2\sqrt{3}$.

Finally, multiplying the equation $8xy = 18z\lambda$ by z and using the same technique results in $z = 4\sqrt{3}/3$. It follows that the desired volume is

$$V = 8(\sqrt{3})(2\sqrt{3})\left(\frac{4\sqrt{3}}{3}\right) = 64\sqrt{3}. \qquad \blacksquare$$

Example 3 If $f(x, y, z) = 4x^2 + y^2 + 5z^2$, find the point on the plane $2x + 3y + 4z = 12$ at which $f(x, y, z)$ has its least value.

Solution We wish to find the minimum of $f(x, y, z) = 4x^2 + y^2 + 5z^2$ subject to the constraint $g(x, y, z) = 2x + 3y + 4z - 12 = 0$. If, as in (16.37),

$$\nabla(4x^2 + y^2 + 5z^2) = \lambda\nabla(2x + 3y + 4z - 12)$$

then

$$8x = 2\lambda, \quad 2y = 3\lambda, \quad \text{and} \quad 10z = 4\lambda.$$

Solving the last three equations for λ gives us

$$\lambda = 4x = \tfrac{2}{3}y = \tfrac{5}{2}z.$$

These conditions imply that

$$y = 6x \quad \text{and} \quad z = \tfrac{8}{5}x.$$

Substituting into the equation $2x + 3y + 4z - 12 = 0$, we obtain

$$2x + 18x + \tfrac{32}{5}x - 12 = 0$$

or $x = \frac{5}{11}$. Hence $y = 6(\frac{5}{11}) = \frac{30}{11}$ and $z = (\frac{8}{5})(\frac{5}{11}) = \frac{8}{11}$. It follows that the minimum value occurs at the point $(\frac{5}{11}, \frac{30}{11}, \frac{8}{11})$. ■

For some applications there may be more than one constraint. In particular, consider the problem of finding the extrema of $f(x, y, z)$ subject to the *two* constraints

$$g(x, y, z) = 0 \quad \text{and} \quad h(x, y, z) = 0.$$

It can be shown that if f has an extremum subject to these constraints, then the following condition must be satisfied for some real numbers λ and μ:

$$\nabla f(x, y, z) = \lambda\nabla g(x, y, z) + \mu\nabla h(x, y, z).$$

A specific illustration of this situation is given in the next example. This method can also be extended to functions of more than three variables and other numbers of side conditions.

Example 4 Let C denote the first octant arc of the curve in which the paraboloid $2z = 16 - x^2 - y^2$ and the plane $x + y = 4$ intersect. Find the points on C that are closest to, and farthest from, the origin. Find the minimum and maximum distances from the origin to C.

Solution The arc C is sketched in Figure 16.34. If $P(x, y, z)$ is an arbitrary point on C, then we wish to find the largest and smallest values of $d(O, P) = \sqrt{x^2 + y^2 + z^2}$. These may be found by determining the triples (x, y, z) that give the extrema of the radicand

$$f(x, y, z) = x^2 + y^2 + z^2$$

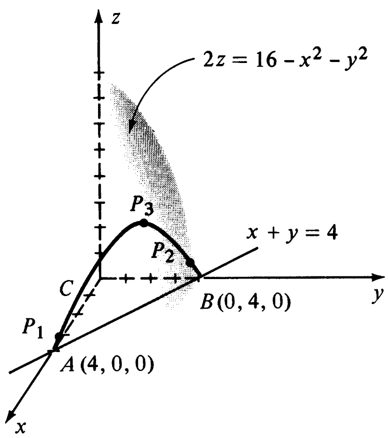


FIGURE 16.34

subject to the constraints

$$g(x, y, z) = x^2 + y^2 + 2z - 16 = 0$$

and
$$h(x, y, z) = x + y - 4 = 0.$$

As in our discussion, consider

$$\nabla(x^2 + y^2 + z^2) = \lambda\nabla(x^2 + y^2 + 2z - 16) + \mu\nabla(x + y - 4).$$

This leads to the following system of equations:

$$2x = 2x\lambda + \mu$$

$$2y = 2y\lambda + \mu$$

$$2z = 2\lambda$$

$$x^2 + y^2 + 2z - 16 = 0$$

$$x + y - 4 = 0$$

From the first two equations we obtain

$$2x - 2y = (2x\lambda + \mu) - (2y\lambda + \mu) = 2x\lambda - 2y\lambda$$

or
$$2(x - y) - 2\lambda(x - y) = 0$$

and therefore
$$2(x - y)(1 - \lambda) = 0.$$

Consequently, either $\lambda = 1$ or $x = y$.

If $\lambda = 1$, it follows that $2z = 2\lambda = 2(1)$ or $z = 1$. The first constraint $x^2 + y^2 + 2z - 16 = 0$ then gives us $x^2 + y^2 - 14 = 0$. Solving this equation simultaneously with $x + y - 4 = 0$ we find that either

$$x = 2 + \sqrt{3}, y = 2 - \sqrt{3} \quad \text{or} \quad x = 2 - \sqrt{3}, y = 2 + \sqrt{3}.$$

Thus, points on C that may lead to extrema are

$$P_1(2 + \sqrt{3}, 2 - \sqrt{3}, 1) \quad \text{and} \quad P_2(2 - \sqrt{3}, 2 + \sqrt{3}, 1).$$

The corresponding distances from O are

$$d(O, P_1) = \sqrt{15} = d(O, P_2).$$

If $x = y$, then using the constraint $x + y - 4 = 0$ we get $x + x - 4 = 0$, $2x = 4$, or $x = 2$. This gives us $P_3(2, 2, 4)$ and $d(O, P_3) = 2\sqrt{6}$.

Referring to Figure 16.34, we may now make the following observations. As a point moves continuously along C from $A(4, 0, 0)$ to $B(0, 4, 0)$, its distance from the origin starts at $d(O, A) = 4$, decreases to a minimum value $\sqrt{15}$ at P_1 and then increases to a maximum value $2\sqrt{6}$ at P_3. The distance then decreases to $\sqrt{15}$ at P_2 and again increases to 4 at B.

As a check on the solution, note that parametric equations for C are

$$x = 4 - t, \quad y = t, \quad z = 4t - t^2$$

where $0 \le t \le 4$. In this case

$$f(x, y, z) = (4 - t)^2 + t^2 + (4t - t^2)^2$$

and the extrema of f may be found using single variable methods. It is left to the reader to show that the same points are obtained. ■

16.9 Exercises

In Exercises 1–10 use Lagrange multipliers to find the local extrema of the given function f subject to the stated constraints.

1 $f(x, y) = y^2 - 4xy + 4x^2$; $x^2 + y^2 = 1$

2 $f(x, y) = 2x^2 + xy - y^2 + y$; $2x + 3y = 1$

3 $f(x, y, z) = x + y + z$; $x^2 + y^2 + z^2 = 25$

4 $f(x, y, z) = x^2 + y^2 + z^2$; $x + y + z = 25$

5 $f(x, y, z) = x^2 + y^2 + z^2$; $x - y + z = 1$

6 $f(x, y, z) = x + 2y - 3z$; $z = 4x^2 + y^2$

7 $f(x, y, z) = x^2 + y^2 + z^2$; $x - y = 1$; $y^2 - z^2 = 1$

8 $f(x, y, z) = z - x^2 - y^2$; $x + y + z = 1$; $x^2 + y^2 = 4$

9 $f(x, y, z, t) = xyzt$; $x - z = 2$; $y^2 + t = 4$

10 $f(x, y, z, t) = x^2 + y^2 + z^2 + t^2$; $x + y + 2z = 1$; $2x - z + t = 2$; $y + 3z + 2t = -1$

11 Find the point on the sphere $x^2 + y^2 + z^2 = 9$ that is closest to the point (2, 3, 4).

12 Let C denote the line of intersection of the planes $3x + 2y + z = 6$ and $x - 4y + 2z = 8$. Find the point on C that is closest to the origin.

13 A closed rectangular box having a volume of 2 ft^3 is to be constructed. If the cost per square foot of the material for the sides, bottom, and top is \$1.00, \$2.00, and \$1.50, respectively, what dimensions will minimize the cost?

14 Find the volume of the largest rectangular box that has three of its vertices on the positive x-, y-, and z-axes, respectively, and a fourth vertex on the plane $2x + 3y + 4z = 12$.

15 A container with a closed top is to be constructed in the shape of a right circular cylinder. If the surface area has a fixed value S, what relative dimensions will maximize the volume?

16 Prove that the triangle of maximum area having a given perimeter p is equilateral. (*Hint*: If the sides are x, y, z and if $s = p/2$, then the area is $[s(s - x)(s - y)(s - z)]^{1/2}$.)

17–26 Work Exercises 23–32 of Section 16.8 using Lagrange multipliers.

27 Prove that the product of the sines of the angles of a triangle is greatest when the triangle is equilateral.

28 A rectangular box with an open top is to have a fixed volume V. What dimensions will minimize the surface area?

29 The strength of a rectangular beam varies as the product of its width and the square of its depth. Find the dimensions of the strongest rectangular beam that can be cut from a cylindrical log whose cross sections are elliptical with major and minor axes of lengths 24 in. and 16 in., respectively.

30 If x units of capital and y units of labor are required to manufacture $f(x, y)$ units of a certain commodity, then the *Cobb–Douglas production function* is defined by $f(x, y) = kx^a y^b$, where k is a constant and a and b are positive numbers such that $a + b = 1$. Suppose that $f(x, y) = x^{1/5}y^{4/5}$ and that each unit of capital costs C dollars and each unit of labor costs L dollars. If the total amount available for these costs is M dollars, so that $xC + yL = M$, how many units of capital and labor will maximize production?

16.10 Review: Concepts

Define or discuss each of the following.

1 Function of more than one variable
2 Level curves
3 Level surfaces
4 Limits of functions of more than one variable
5 Continuous function
6 First partial derivatives
7 Higher partial derivatives
8 Differentials
9 Differentiable function
10 The Chain Rule
11 Implicit differentiation
12 Directional derivative
13 Gradient of a function
14 Tangent plane to a surface
15 Extrema of functions of two variables
16 Lagrange multipliers

Review: Exercises

In Exercises 1–4 determine the domain of the function f.

1 $f(x, y) = \sqrt{36 - 4x^2 + 9y^2}$

2 $f(x, y) = \ln xy$

3 $f(x, y, z) = (z^2 - x^2 - y^2)^{-3/2}$

4 $f(x, y, z) = \sec z/(x - y)$

In Exercises 5–10 find the first partial derivatives of f.

5 $f(x, y) = x^3 \cos y - y^2 + 4x$

6 $f(r, s) = r^2 e^{rs}$

7 $f(x, y, z) = (x^2 + y^2)/(y^2 + z^2)$

8 $f(u, v, t) = u \ln (v/t)$

9 $f(x, y, z, t) = x^2 z\sqrt{2y + t}$

10 $f(v, w) = v^2 \cos w + w^2 \cos v$

In Exercises 11 and 12 find the second partial derivatives of f.

11 $f(x, y) = x^3y^2 - 3xy^3 + x^4 - 3y + 2$

12 $f(x, y, z) = x^2 e^{y^2 - z^2}$

13 If $u = (x^2 + y^2 + z^2)^{-1/2}$ prove that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0.$$

14 (a) Find dw if $w = y^3 \tan^{-1} x^2 + 2x - y$.
(b) Find dw if $w = x^2 \sin yz$.

15 Find Δw and dw if $w = x^2 + 3xy - y^2$. Use Δw to find the exact change and dw to find the approximate change in w if (x, y) changes from $(-1, 2)$ to $(-1.1, 2.1)$.

16 In using Ohm's Law $R = E/I$, suppose that there are percentage errors of 3% and 2% in the measurements of E and I, respectively. Use differentials to estimate the maximum percentage error in the calculated value of R.

Use the Chain Rule to find the solutions in Exercises 17–19.

17 If $s = uv + vw - uw$, $u = 2x + 3y$, $v = 4x - y$, and $w = -x + 2y$, find $\partial s/\partial x$ and $\partial s/\partial y$.

18 If $z = ye^x$, $x = r + st$, $y = 2r + 3s - t$, find $\partial z/\partial r$, $\partial z/\partial s$, and $\partial z/\partial t$.

19 If $w = x \sin yz$, $x = 3e^{-t}$, $y = t^2$, $z = 3t$, find dw/dt.

20 Find the directional derivative of $f(x, y) = 3x^2 - y^2 + 5xy$ at the point $P(2, -1)$ in the direction of the vector $\mathbf{a} = -3\mathbf{i} - 4\mathbf{j}$. What is the maximum rate of increase of $f(x, y)$ at P?

21 Suppose that the temperature at the point (x, y, z) is given by $T(x, y, z) = 3x^2 + 2y^2 - 4z$. Find the rate of change of T at the point $P(-1, -3, 2)$ in the direction from P to the point $Q(-4, 1, -2)$.

22 A curve C is given parametrically by $x = t$, $y = t^2$, $z = t^3$. If $f(x, y, z) = y^2 + xz$, find $D_{\mathbf{u}}f(2, 4, 8)$, where $\mathbf{u}$ is a unit tangent vector to C at $P(2, 4, 8)$.

23 Find equations of the tangent plane and normal line to the graph of $7z = 4x^2 - 2y^2$ at the point $P(-2, -1, 2)$.

24 Show that every plane tangent to the cone

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 0$$

passes through the origin.

25 If $y = f(x)$ satisfies $x^3 - 4xy^3 - 3y + x - 2 = 0$, use partial derivatives to find $f'(x)$.

26 If $z = f(x, y)$ satisfies $x^2y + z\cos y - xz^3 = 0$, find $\partial z/\partial x$ and $\partial z/\partial y$.

27 Find the extrema of f if $f(x, y) = x^2 + 3y - y^3$.

28 The material for the bottom of a rectangular box costs twice as much per square inch as that for the sides and top. What relative dimensions will minimize the cost if the volume is fixed?

29 Given $f(x, y) = x^2/4 + y^2/25$, sketch several level curves associated with f and represent $\nabla f]_P$ by a vector for a point P on each curve.

30 Given $F(x, y, z) = z + 4x^2 + 9y^2$, sketch several level surfaces associated with F and represent $\nabla F]_P$ by a vector for a point P on each surface.

31 Find the local extrema of $f(x, y, z) = xyz$ subject to the constraint $x^2 + 4y^2 + 2z^2 = 8$.

32 Use Lagrange multipliers to find the local extrema of $f(x, y, z) = 4x^2 + y^2 + z^2$ subject to the constraints $2x - y + z = 4$ and $x + 2y - z = 1$. Check your answer using single variable methods.

33 Find the points on the graph of $1/x + 2/y + 3/z = 1$ that are closest to the origin.

34 A hopper in a grain elevator has the shape of a right circular cone of radius 2 ft, surmounted by a right circular cylinder. If the volume is 100 ft^3, find the altitudes h and k of the cylinder and cone, respectively, that will minimize the curved surface area.